DIGITAL DISASTERS: THE ECONOMIC COSTS OF SUBMARINE CABLE BREAKS*

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Abstract

Submarine cables carry more than 95% of international data traffic and are a critical engine of growth, financial deepening and cross-border trade. This paper measures the economic costs of their disruption. Using a panel of 164 economies over 2008–2019 and exploiting the plausibly exogenous timing and repair duration of accidental cable breaks, a dynamic difference-in-differences design identifies a persistent 4 to 6% fall in GDP per capita within four to five years. A monthly quasi-experiment on the Africa Coast to Europe cable over 2016–2019, comparing landing countries hit by a break with countries left connected on the same system, shows night-time lights dimming immediately and persistently after the break. GDP losses run through reduced trade flows, exporter participation, foreign investment and productivity, and tighter access to finance, while proximate economies gain 0.5 to about 1% in GDP as activity is reallocated within the region.

Keywords: Submarine cables, connectivity disruptions, economic growth, trade, spillovers, digital infrastructure.

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1 Introduction

Modern economies rely on cross-border data flows to coordinate trade, finance, and production. Submarine cables (SMCs) are the first-mile infrastructure for these flows, carrying more than 95% of international data traffic (Telegeography 2023) and thus forming the physical backbone connecting national economies to global markets. A growing literature shows that the arrival of submarine cables fosters economic growth, integration in global trade, and financial development, particularly in developing economies (Hjort and Poulsen 2019; Simione and Li 2021; D'Andrea and Limodio 2024; Imbruno et al. 2025). Yet SMCs fail regularly, with 100 to 200 faults recorded each year, mostly caused by fishing gear and vessel anchoring. Where international connectivity rests on only a handful of submarine routes and landing stations, cable failures can simultaneously disrupt payments, logistics, and digital services for days or even weeks, with consequences potentially cascading across sectors and amplifying across time (ITU et al. 2026). The European Commission, among other authorities, now designates submarine cables as critical infrastructure, issuing in February 2024 a recommendation on their security and resilience.[1] Its economic incidence, however, falls stronger elsewhere, as low- and middle-income economies are particularly exposed to this hazard. Despite this growing attention, the economic costs of submarine cable disruptions have yet to be measured.

In 2024, cable breaks were the leading cause of global internet shutdowns by countries affected, and the second most frequent after government-ordered shutdowns.[2] The macroeconomic consequences of these failures remain poorly understood. The literature lacks causal evidence on SMC failures specifically: existing assessments of connectivity disruptions are largely accounting exercises and consider other disruption causes, such as government shutdowns. Deloitte (2016) estimates that connectivity disruptions can destroy hundreds of millions of dollars in GDP within days, while The Brookings Institution (West 2016) estimated global GDP losses from government-induced internet shutdowns at $2.4 billion annually, with some country-level losses reaching nearly $1 billion. A joint ITU–UNDRR–Sciences Po report (ITU et al. 2026) underscores the knowledge gap, noting that the economic impacts of digital disruptions remain poorly understood and that existing risk frameworks lack models reflecting the depth of modern societies' digital dependence.

Connectivity disruptions can depress economic activity well beyond digitally-intensive sectors. When international connectivity is impaired, firms face immediate information and financial frictions. Orders cannot be verified, payments cannot be executed, and the data-intensive services embedded in modern trade—finance, logistics, transport, and ICT support—are delayed or disrupted. These frictions propagate through supply chains and trade networks as firms reliant on just-in-time logistics and cross-border coordination incur order, payment and delivery delays, and are likely amplified today by the increasing digitalization of economies and "servicification" of production and merchandise trade (Baldwin 2016).

Theory and empirical evidence link information frictions to trade volumes. Dasgupta and Mondria (2018) show that when the cost of acquiring information about foreign markets rises, rationally inattentive importers reduce the attention they allocate to those markets, contracting trade flows even absent any change in other trade costs. A related literature models trade as a costly search and matching process between buyers and sellers, in which information frictions leave a share of potential matches unrealized and impede trade transactions (Allen and Arkolakis 2014; Krolikowski and McCallum 2021; Eaton et al. 2026). Exploiting the staggered rollout of broadband in Norway, Akerman et al. (2022) find that lower information frictions enlarge the set of trading partners available

[1] See, for instance, the European Commission Recommendation 2024/779 of 26 February 2024 on secure and resilient submarine cable infrastructure (link).

[2] https://radar.cloudflare.com/year-in-review/2024

to exporters and importers. Closest to the present design, Steinwender (2018) considers temporary failures of the transatlantic telegraph cable as exogenous variation in information frictions, showing that each additional day of information lag widened the US–UK cotton price gap by 23%. Delays in merchandise delivery also translate into substantial trade costs, as each additional day reduces bilateral product trade by more than 1% on average (Djankov et al. 2010) and is equivalent to an ad-valorem tariff of 0.6 to 2.1% (Hummels and Schaur 2013). Because customs and port procedures, domestic payment systems, cloud-hosted business software, banking, and public administrations now run on the same international bandwidth as cross-border exchange, the loss from a break is likely to reach well beyond trade and affect the economy as a whole.

The paper provides the first quasi-experimental estimates of the economic costs of submarine cable failures. Using a new cross-country dataset spanning 2008–2019, I exploit plausibly exogenous variation in accidental disruption timing and duration in a dynamic event-study difference-in-differences framework. An average disruption, requiring around 8 cable repair days, is followed by a decline in GDP per capita that accumulates over the following years. Within four to five years, GDP per capita falls by 4–6%, depending on the fixed-effect specification. These losses are confined to countries with sparse cable networks and concentrated cable ownership, and are partly or fully cushioned by alternative routes, Internet exchange points, or a more diversified set of cable owners. I also find spillovers to proximate non-disrupted economies in the same region, whose GDP per capita rises by 0.5 to about 1% after a disruption, depending on the specification, as merchandise and bandwidth-intensive trade flows, exporter participation, foreign direct investment, and productivity gains shift toward them. Overall, even short-lived disruptions to international digital infrastructure have persistent effects on economic trajectories, which targeted investment in network redundancy can largely prevent.

Dynamic effects are estimated using the event-study framework of de Chaisemartin and d'Haultfoeuille (2024), which accommodates staggered, repeated, non-binary and non-absorbing treatments. Identification exploits the fact that cable failures are primarily triggered by maritime hazards and occur within a dense network architecture that makes disruption timing and location plausibly orthogonal to macroeconomic conditions.[3] Exposure follows the physical geography of the network (Imbs and Pauwels 2025; Porcellacchia et al. 2026). Most breaks occur where cables cross busy shipping lanes, including lanes hundreds of miles from the choke points that the ships are heading for. For instance, the cables serving West Africa head north along the Iberian coast to reach European landing stations, while those serving East Africa, South or East Asia pass through the Gulf of Aden on the way to Suez. Those choke points carry the highest risk of a fault, because anchoring and fishing in those crowded waters are the leading cause of breaks (Clare et al. 2023), and a single fault there severs many systems at once. The peripheral economies are where faults do lasting damage, because of the few alternative routes to cushion them.

Therefore, exposure is defined from the perspective of what each country experiences, since a single break can simultaneously impair connectivity among distant countries. A first-stage analysis based on high-frequency connectivity data establishes that coded disruptions translate into a measured loss of international connectivity. Drawing on routed-prefix counts from the RIPE routing-information service, compiled independently of the disruption coding, I find that the share of a country's networks reachable from abroad holds steady before a disruption and falls on its recorded onset. In the

[3]The few failures caused by human interventions (e.g. sabotage) or by natural disasters, which could threaten identification assumptions, are hence not coded as disruption events. The International Cable Protection Committee (ICPC) argues that accurate charting of submarine cable routes on nautical charts is critical for maritime safety and cable protection, because most cable faults are accidental and largely caused by fishing activity and vessel anchoring (on the order of 70–80% of faults). The ICPC also notes that removing cables from charts would do little to deter deliberate damage, since actors intent on sabotage can locate cables regardless, while de-charting would materially increase accidental incidents (link).

disruption month, the largest daily shortfall reaches about 10 percent of a country's routed address space, which then returns to its pre-event level within about three weeks.

Following Rambachan and Roth (2023), I show that medium-run GDP losses remain significant under a linear extension of the pre-break trend and under moderate departures from it. An infra-annual analysis using night-time light radiance confirms the magnitude and persistence of the GDP response to short-lived connectivity losses, and the absence of pre-trend. Breaks on the Africa Coast to Europe (ACE) cable between 2016 and 2019 affected some of its landing countries while leaving other countries connected to the same system unaffected. Following breaks on the ACE cable, night-time light radiance falls by 15.5% in economic capitals over the fifteen following months, and by 28.2% in secondary and intermediary cities. The capital response is immediate, persistent and rapidly stabilizes, whereas the response in secondary and intermediary cities starts later and accentuates over time, suggesting that the shock propagates from primary to secondary economic centres. Additionally, to address endogeneity concerns, I test for non-random exposure following Borusyak and Hull (2023), and I document that the cross-country incidence of disruptions is uncorrelated with pre-treatment growth. I also control for a wide range of confounding factors, exclude countries and regions one by one.

The causal literature on cross-border connectivity has so far measured what its arrival delivers. Broadband and cable rollout raised growth, productivity, and the reallocation toward services across developing economies (Hjort and Poulsen 2019; Simione and Li 2021; D'Andrea and Limodio 2024), and access through shared cable systems fostered trade and widened firms' participation in foreign markets (Herman and Oliver 2023; Imbruno et al. 2025). This paper addresses connectivity's withdrawal. Economies have organised trade, payments and production around continuous connectivity, and when a route fails they have no domestic substitute. These losses compound with disruption duration, since an unrerouted fault holds trade-time frictions in place for the days or weeks a deep-water repair requires (Djankov et al. 2010; Hummels and Schaur 2013). They also persist, since a cable break is a critical-infrastructure outage without a backstop. An electricity shortage can be partly buffered by on-site generation (Fried and Lagakos 2023), storage, or imports from a neighbouring grid. Lost international bandwidth has no comparable local substitute[4], because it is produced abroad and can only be replaced by traffic rerouted over other international cables.

Therefore, submarine cable failures warrant treatment as *digital disasters*. They are sudden, exogenous shocks that disable critical infrastructure access, and propagate economy-wide with persistence, with estimated welfare losses in the same magnitude class as those documented for climate-attributable extreme weather events (Carleton and Hsiang 2016; Newman and Noy 2023), financial and socio-political shocks (Cerra and Saxena 2008), or inter-state wars (Federle et al. 2026). Unlike those shocks, their effects primarily operate through disruptions to market access, production linkages, and cross-border financial flows rather than physical capital destruction, precluding the reconstruction-driven recovery that typically follows such events. Effects concentrate in countries with limited rerouting capacity where the absence of redundancy amplifies transmission and persistence.

The remainder of the paper is organized as follows. Section 2 reviews the institutional background, the transmission mechanisms, and the related literature. Section 3 sets out the event-study estimator, describes the data, presents a first stage linking coded disruptions to independently measured losses of routed connectivity. Section 4 reports baseline GDP per capita effects, heterogeneity by network redundancy, and regional spillovers. Section 5 checks the robustness of results, investigating night-time light monthly response to disruptions on the ACE network, testing for non-random exposure,

[4]Satellite links, including low-orbit broadband, have far less capacity than a submarine cable and higher latency and cost.

controlling for confounding factors, and probing sample composition. Section 6 studies the transmission channels through trade, productivity, foreign investment, and finance. Section 7 concludes with policy implications.

2 Background

2.1 Cable disruptions as a critical and plausibly exogenous connectivity shock

As broadband access expands, exposure rises where international capacity hinges on a small number of undersea routes and landing points. European policymakers explicitly treat this layer as critical infrastructure. Since 2022, the EU's NIS2 cybersecurity directive brings digital-infrastructure entities operating submarine cables into scope, while the Critical Entities Resilience (CER) directive sets a cross-EU framework to harden such assets.[5] In 2025, the Commission went further with an EU Action Plan on Cable Security to coordinate protection and resilience measures across borders. This policy arc reflects the reality depicted in this paper: subsea cable networks are now critical and their failure can have cascading effects on the economy (Bueger et al. 2022).

Occurrence

SMC faults are common. The global network spans 1.8 million km of telecom cables; 100–200 damage events are recorded annually, concentrated in shallow water (<100 m) where accidental human interactions dominate, that is, fishing (nearly half of documented faults) and anchoring. While suspicions of sabotaged cables make the headlines, those episodes remain marginal compared to accidental damages[6], which explains why the ICPC advocates for charting cables on nautical charts to prevent their occurrence.[7] In other words, the modal break is the by-product of routine seabed activity, and its timing is random relative to a country's macroeconomic conditions (Clare 2021; Telegeography 2025b). However, cable faults do not systematically lead to internet disruptions in connected countries.

Transmission

Because international connectivity follows shared trunk routes, where a fault occurs is not where it is materially felt. Figure 1 maps a December 2008 incident on the SEA-ME-WE and FLAG systems off the Egyptian coast, and shows that countries reported as experiencing multi-day outages or major slowdowns stretch across the Middle East and South Asia and reach as far as Zambia and the Maldives.[8] What determines a country's exposure is more its position in the physical cable network than its geographical proximity to the break (Imbs and Pauwels 2025). Economies adjacent to the maritime choke points where many systems converge, and peripheral economies that rely on a handful of landing cables, are the most exposed, because a single fault on a shared route can remove a large share

[5] Available here.

[6] Telegeography (2025a) invokes *Hanlon's razor* in a piece explaining the causes of undersea cable breaks: "*Never attribute to malice that which is adequately explained by stupidity.*" Talking about recent incidents in the Baltic Sea, a Washington Post article stressed that faults were not intentional but "*accidents caused by inexperienced crews serving aboard poorly maintained vessels.*" Nevertheless, as robustness check, I adjust the disruption exposure's definition by dropping events associated with suspected or confirmed episodes of sabotage, but also to natural disasters that could directly affect growth through physical and human damages.

[7] See ICPC viewpoint: "Charting Submarine Cables Is Critical for Maritime Safety & Infrastructure Protection", International Cable Protection Committee, Updated Wednesday 25 June 2025.

[8] Other episodes show the same pattern. In June 2017 a MainOne fault about 3,000 km off Portugal degraded service in Nigeria and along the West African seaboard until repair (report). In March 2024 an undersea landslide off West Africa cut WACS, ACE, SAT-3 and MainOne at once, degrading connectivity in 10–13 countries despite the offshore damage (report 1, report 2, report 3). Further SubTel Forum alerts document similar multi-day disruptions in Vietnam, Pakistan, and Southern Africa.

of their international capacity. This geographic basis of exposure is what allows cable disruptions to serve as a connectivity shock orthogonal to growth.

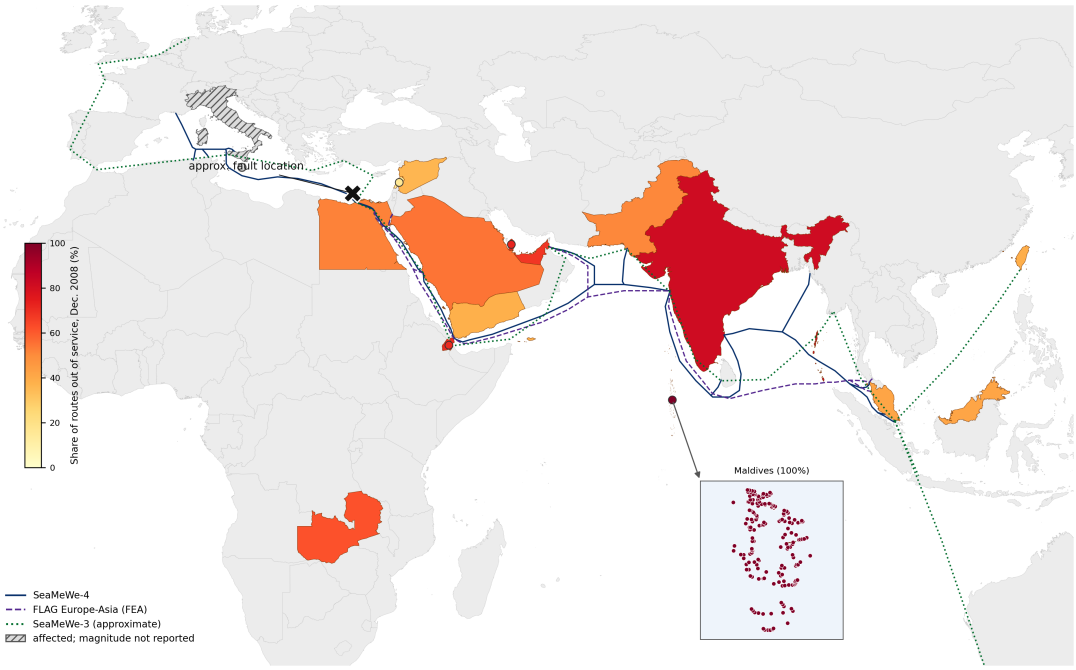


Fig. 1: Internet disruption events related to the SEA-ME-WE and FLAG systems failure, December 2008

Notes: Affected countries are those recorded in the *SubTel Forum*-based database for the December 2008 SEA-ME-WE and FLAG failures. Colour reports the share of routes out of service, as documented by SubmarineNetworks. Malta and Italy, recorded as affected (see Wikipedia page) but without a documented outage magnitude, are shown hatched. Small countries (Qatar, Lebanon, Djibouti, Malta) are marked with a dot for legibility, and the Maldives, the most affected with all routes out of service, is enlarged in the inset. The cross marks the approximate location of the cable faults, in the Mediterranean off the Egyptian coast near the Alexandria landing stations.

Whether, where and to what extent a submarine-cable fault will map into internet disruptions is hence difficult to predict *ex ante* because a physical break does not translate mechanically into who loses connectivity. Depending on how networks are engineered (redundant links, capacity, and contractual circuits), the same fault may make some online destinations unreachable, or instead prompt traffic to be diverted onto alternative routes with heterogeneous effects across networks and locations.[9] Evidence also indicates that impacts can extend beyond traffic that directly used the failed span. In fact, when traffic is shifted onto backups, congestion on those backups can degrade performance even for users whose traffic would not normally traverse the broken cable (Chan et al. 2011). Finally, as noted by the Regional Internet Registry for Europe, Middle East and Central Asia (RIPE), one cannot directly infer which end-to-end paths actually traversed the damaged cables and that comprehensive public data on "who uses which cable, when" are lacking, so the set of affected economies is largely revealed only *ex post*.[10]

Duration

Around 99% of cable systems are privately owned and thus not "flagged" to any single state, making repair times primarily an issue of international coordination. Repairing a damaged system indeed

[9] See RIPE blog: "Mediterranean Fibre Cable Cut - a RIPE NCC Analysis", Eds: Rene Wilhelm, Chris Buckridge, RIPE NCC Science Group.

[10] See RIPE blog: "Does the Internet Route Around Damage? - Baltic Sea Cable Cuts", Emile Aben, RIPE Labs, 20 Nov 2024.

follows a well-established sequence, independent from country's policies and context.[11] Operators localize the fault, then a specialized repair ship mobilizes, grapnels and recovers the cable, trims damaged sections, splices in new lengths, tests, and re-buries where required (Telegeography 2025b). Today, maintenance relies on regional contracts and zones (e.g., ACMA for the Atlantic, MECMA for the Mediterranean/Red Sea), with a handful of ships covering vast areas, highlighting capacity limits when multiple faults coincide: among the 77 cable ships, only 22 are designed for cable repair.[12] Industry analyses warn that much of the global repair fleet is ageing, and that meeting demand and avoiding longer queues will require US$3 billion for 15 replacement and 5 additional ships to sustain service levels as cable-km rise toward 2040.[13] Therefore, cable repair duration depends on geophysical factors and international logistics, another reason repair duration is treated as a plausibly exogenous "dose" in the event-study and dose–response estimates.

2.2 Related Literature

Early work on the ICT–growth nexus established that telecommunications and broadband networks generate large growth effects, functioning as enabling technologies with network externalities that spread across sectors (Röller and Waverman 2001; Czernich et al. 2011).

Building on these contributions, more recent work has turned specifically to submarine cables, the backbone of international connectivity, and extended the analysis to developing economies (Hjort and Poulsen 2019). First, recent studies have highlighted their contribution to Internet accessibility, especially in Africa (Hjort and Poulsen 2019; Cariolle 2021; Cariolle et al. 2025). Regarding their socio-economic impacts, several recent studies document trade effects of submarine cable deployments. Cariolle and da Piedade (2023) show that improved cable connectedness significantly promotes export upgrading in developing countries. Imbruno et al. (2025) find that bilateral connectivity through shared cable systems facilitates exporters' access to foreign markets, particularly for high-productivity firms. Haltenhof (2019) reports that bilateral connectivity stimulates services trade flows, with elasticities highest in data-intensive sectors, while Herman and Oliver (2023) documents complementary effects in both goods and services trade.

A final strand of research extends the emphasis from trade to domestic and microeconomic activity. Eichengreen et al. (2023) show that the availability of international bandwidth facilitates cross-border financial transactions, lowering frictions in payments and settlements that are critical for trade and capital flows. Complementarily, D'Andrea and Limodio (2024) find that high-speed internet via cables promoted financial technology adoption, expanded interbank market activity, and increased firm access to credit and sales. These results underscore how cable connectivity affects not only cross-border flows but also the functioning of domestic financial systems, amplifying growth potential. In this regard, Simione and Li (2021) use the staggered arrival of submarine cables in Sub-Saharan Africa as a quasi-experiment and find that resulting higher internet penetration rates significantly boost per-capita GDP growth and productivity, while also shifting employment and output shares toward the services sector. This macro evidence is corroborated at the firm level by Hjort and Poulsen (2019), who provide the first causal evidence on how fast internet arrival in Africa, through gradual SMC laying over the last decades, boosts firm productivity, employment, and export performance.

A distinct strand of recent work characterizes the supply-side economics and network structure of the cable market itself. Caoui and Steck (2025) model the global internet backbone as a dynamic

[11] Cable repair ships are regular civilian vessels flying a national flag, but they operate under international law and large regional maintenance contracts that cover vast ocean areas.
[12] See this animated piece from *The Verge*. See also the maintenance map in Appendix OA.1.1.
[13] See Telegeography's blog: "It's Going to Take $3 Billion to Ensure Submarine Cable Repair Ships Can Keep the World Connected", Telegeography staff, Jun 30, 2025.

entry game and show that buyer demand for route diversification, which hedges against disruption risk, is a primary determinant of cable entry and long-run market structure. Distortions from operators' inability to internalize the social value of diversity are as large as business-stealing distortions. Jeon and Rysman (2025) characterize cross-border data flows as international trade and show that cable capacity is a binding constraint on country-to-country data exchanges, with elasticities of data trade to capacity highest in low-provision settings. Imbs and Pauwels (2025) build a new dataset on global cable topology, redundancy, and capacity over 2010–2025 and show that network vulnerability is persistent, with fragility concentrated along the MENA–South Asia corridor and correlating with network centrality. Porcellacchia et al. (2026) embed the network in a traffic-assignment model with congestion and rerouting and derive sufficient statistics for the welfare cost of a disruption, measured as the loss of real income. They find that removing any single cable is close to costless, while coordinated cuts at the maritime choke points, where many systems converge, are far more damaging. None of these papers estimates, ex post, the realized macroeconomic costs that countries bear when disruptions occur.

This body of evidence on trade and growth effects of SMC connectivity has a direct implication for disruptions: when submarine cables fail, the same mechanisms that stimulate and organize economic activity in normal times operate in reverse, depressing flows through higher transaction and information costs, unreliable transport and logistics, and impaired cross-border coordination. However, the effects of connectivity shocks may be asymmetric as the economic damage from disruptions likely exceeds the proportional benefits of connectivity access due to heightened digital dependence. Modern economies have slowly adjusted around continuous connectivity, with just-in-time supply chains, real-time financial systems, and cloud-dependent operations creating vulnerabilities that amplify disruption costs beyond simple connectivity gains in reverse.

The closest evidence to this study's is provided by Steinwender (2018), who exploits the 1866 establishment of the transatlantic telegraph, and within the post-telegraph period its temporary technical failures, to identify a widening of the cross-market price gap and a contraction in trade in a single commodity market over a one-year window. That window captures the contemporaneous effect but cannot track a medium-run output path, and the contrast with the persistence documented here is consistent with the broader role digital connectivity now plays across production, trade, payments, and finance and with the deeper integration of economies, so that a present-day break propagates further and unwinds more slowly than a telegraph-era information lag. This paper thus scales that evidence from a single traded product to macroeconomic aggregates across 164 economies, and frames it in the current digital era.

2.3 Mechanisms

International connectivity is an enabling intermediate input for domestic production, consumption, trade, and finance (Deloitte 2016; Hjort and Tian 2025). Submarine-cable outages restrain international and domestic connectivity access and therefore act as a supply and demand shock that simultaneously lowers operational efficiency and raises market frictions. Notably, these shocks can generate (i) *direct* losses in the affected country due to higher coordination costs, financial and trade frictions; and (ii) *regional spillovers* as trade flows, contracts, and investment are diverted toward substitute locations.

2.3.1 Direct effects

Operational disruptions, payments frictions, and immediate productivity losses

Connectivity loss immediately impairs firms in their daily operations, by disrupting internal communication and monitoring, digital workflows, cloud-based tools, and remote coordination. Moreover, even when production and sales can continue, firms may be unable to issue invoices, receive electronic payments, reconcile accounts, or execute domestic and cross-border settlements. These impairments delay cash receipts while wages, rent, and input bills remain due, lengthening firms' cash-conversion cycles when uncertainty rises. In working-capital frameworks, longer payment lags tighten liquidity and lead firms to cut employment and production (Barrot and Nanda 2020). This channel therefore predicts an immediate contraction in activity and productivity through operational downtime and cash-flow stress.

Supply-chain disorganization and trade frictions

Beyond within-firm disorganisation, outages raise the costs of coordinating and executing trade activities. International shipments rely on continuous information exchange across firms, banks, insurers, freight forwarders, ports, and customs authorities. By slowing documentation, compliance, tracking, and release of goods, degraded connectivity acts like a time barrier to trade, as each additional day of delay reduces bilateral trade (Djankov et al. 2010) and is equivalent to a sizable ad-valorem cost (Hummels and Schaur 2013). Because trade is increasingly "servicified", bundling finance, insurance, transport intermediation, and digital logistics (Baldwin 2016), connectivity failures that impair these service inputs translate directly into weaker import and export performance. Outages also raise the cost of finding and monitoring trading partners. Information frictions leave potential buyer-seller matches unrealized (Allen and Arkolakis 2014; Krolikowski and McCallum 2021; Eaton et al. 2026), and rationally inattentive importers shift their attention away from markets that become costlier to monitor (Dasgupta and Mondria 2018). Steinwender (2018) provides quasi-experimental evidence of this information channel, showing that temporary technical failures of the transatlantic telegraph cable raised the US–UK cotton price gap by 23% per additional day of information lag. Studies of broadband rollout document the reverse effect. Akerman et al. (2022) find that lower information frictions enlarge the set of trading partners available to Norwegian firms, and Malgouyres et al. (2021) show that broadband expansion in France increased firm imports by around 25%, mainly through the product and sourcing-country extensive margins. When inputs are relationship-specific and difficult to substitute quickly, disruptions propagate along production networks as buyers cannot seamlessly switch suppliers (Barrot and Sauvagnat 2016), amplifying the initial shock beyond the directly exposed firms.

Amplification through financial frictions

Connectivity disruptions can amplify real effects through two complementary financial channels. First, as banking and credit provision are information- and communication-intensive, improved connectivity can expand credit supply and enhance monitoring and competition in lending (D'Andrea and Limodio 2024; D'Andrea et al. 2025). The reverse implication is that disruptions can temporarily impair screening/monitoring, raise operating costs, and worsen credit terms or availability. Second, firms' short-term liquidity needs rise during disruptions when payment frictions and bank intermediation problems make liquidity scarcer. The interaction amplifies production and trade frictions, as constrained firms are less able to bridge temporary disruptions, which can turn transitory shocks into persistent losses of jobs and market shares (Barrot and Nanda 2020; Barrot and Sauvagnat 2016). Over longer horizons, if disruptions raise operating and financing wedges differentially across firms and

sectors, they may depress investment and slow aggregate productivity growth through reallocation and pro-competitive channels (Varela 2018).

2.3.2 Regional spillovers

Regional spillovers may arise because higher trade, financial, information and production costs in the affected economy shift activity toward close substitutes. In spatial equilibrium and trade-based geography frameworks, changes in trade or interaction costs reallocate production, spending, and factor demands across locations, generating indirect effects beyond treated units (Allen and Arkolakis 2014; Ahlfeldt et al. 2015; Donaldson and Hornbeck 2016). In this context, a connectivity failure can induce trade and investment diversion, as contracts, customer support, outsourcing tasks, logistics intermediation, and other digitally enabled business services are rerouted toward nearby countries that remain reliably connected.

Unlike many natural disasters, cable outages do not destroy physical capital (except for cable owners and operators). Persistent effects are therefore more naturally interpreted as arising from coordination and liquidity frictions, relationship disruption, and the reallocation of contracts and investment. Evidence from the natural-disaster literature confirms that large shocks generate trade diversion. Exporters cut sales to affected destinations by reallocating toward unaffected ones (London and Gigout 2023), while importers substitute toward alternative suppliers (Freund et al. 2022). This substitution is likely stronger for coordination-intensive digital services, where regional alternatives are manifold.

The disaster literature also provides evidence that localized shocks can reallocate mobile capital toward unaffected areas. Friedt and Toner-Rodgers (2022) document large and persistent reductions in FDI in disaster-hit regions within India and sizable positive spillovers into otherwise unaffected regions. They show that a substantial share of the lost investment reflects relocation within the country, and the effects persist for several years. In this setting, if connectivity disruptions impair the whole economy, they can lower expected returns to operating in the affected country by raising operational and financing costs, discouraging FDI and accelerating reallocation toward regional alternatives. Foreign investors may therefore postpone, scale down, or relocate projects toward proximate regional alternatives that can deliver services and coordinate cross-border activity more reliably.

Such shifts can also shape aggregate productivity. If improved connectivity disproportionately reduces operating costs or financing constraints for outward-oriented and more productive firms (Imbruno et al. 2025), those same firms shrink, lose contracts, or delay expansion after connectivity drops; and possibly move their activity in less exposed markets hosting potentially less productive. At the same time, weaker entry and competitive pressure can slow selection and reallocation toward the most efficient firms, with adverse implications for TFP growth, consistent with the competition-and-reallocation mechanisms emphasized by Varela (2018). As a result, exposed economies can experience persistent competitiveness losses, while nearby non-disrupted economies can attract investment and exhibit higher productivity as activity relocates within the region.

3 Empirical framework

In this paper, I estimate the dynamic treatment effects of Internet disruptions caused by submarine cable failures on economic growth and its components, using a country–year panel of 164 countries covering 2008–2019. I use the event-study estimator of de Chaisemartin and d'Haultfœuille (hereafter CDH), designed to study staggered non-binary and non-absorbing treatment with possible

heterogeneous dynamic effects on macro-economic outcomes (de Chaisemartin and d'Haultfoeuille 2024).

3.1 Econometric modelling

Standard two-way fixed effects event-study estimators are biased under heterogeneous treatment effects with staggered adoption (Goodman-Bacon 2021; Sun and Abraham 2021; Callaway and Sant'Anna 2021; de Chaisemartin and d'Haultfoeuille 2024). Because SMC disruptions are staggered, non-binary, and non-absorbing (countries may experience multiple disruptions of varying intensity, with no mechanical persistence across years), I rely on the estimator of de Chaisemartin and d'Haultfoeuille (2024) (hereafter DID_M), which accommodates this setting with heterogeneity-robust Difference-in-Difference (DiD) and nonnegative aggregation weights.

Let y_{it} denote the macroeconomic outcome (e.g. log GDP per capita) for country i in year t, and let D_{it} denote treatment intensity in country i and year t (the number of SMC-induced connectivity disruptions or SMC repair days). Let F_i be the first year in which D_{it} departs from its initial baseline level D_{i1} (which is zero for almost all countries), and define event time as $k = t - F_i$. I estimate

$$y_{it} = \alpha_i + \lambda_t + \gamma_i t + \delta_{r(i),t} + \delta_{c(i),t} + \sum_{k=-K_0}^{K_1} \beta_k \, \text{Event}_{it}^k + \varepsilon_{it}, \tag{1}$$

where α_i and λ_t are country and year fixed effects, $\gamma_i t$ are country-specific linear trends, and $\delta_{r(i),t}$ and $\delta_{c(i),t}$ are region-by-year and income-group-by-year fixed effects, with $r(i)$ denoting country i's World Bank region and $c(i)$ its income group. The pre- and post-event horizons are set to $K_0 = 3$ and $K_1 = 6$, respectively. For never-treated units, $\text{Event}_{it}^k = 0$ by convention. The event-time–by–intensity regressor is

$$\text{Event}_{it}^k \equiv D_{it} \, \mathbf{1}\{t - F_i = k\}, \qquad k = -K_0, \ldots, K_1,$$

so Event_{it}^k equals the number of disruptions (or repair days) that country i experiences k years after its first departure from zero exposure, and is zero otherwise. The coefficients β_k trace out the effect of a one-unit increase in treatment intensity at event time k.[14] Identification relies on a conditional parallel-trends assumption for outcomes in the absence of changes in D_{it}.

3.2 Data

The empirical analysis uses an annual country panel for 2008–2020 that combines macroeconomic variables from the World Bank's Open Database, Global Productivity Database, Exporter Dynamics Database, the Bank for International Settlements Data Portal, with SMC infrastructure data from Telegeography and an original database on cable-related internet disruption episodes. The data coding protocol applied to build cable disruption data is explained in Online Appendix OA.1.1. The monthly evidence from ACE cable breaks in Section 5.2 draws on a separate high-frequency panel built from VIIRS night-time-light imagery around African cities, whose construction is detailed in Online Appendix A.2.3.

[14] By normalization, $\beta_{-1} = 0$. Because treatment is non-binary, the `normalized` option in `did_multiplegt_dyn` Stata command is used, which rescales each local effect by the size of the underlying treatment change before aggregation. The β_k can therefore be interpreted as the average effect, k years after first exposure, of a one-unit increase in treatment intensity (one additional disruption or one extra repair day), relative to a status-quo path in which the country remained at zero exposure throughout. See de Chaisemartin and d'Haultfoeuille (2024) for details.

3.2.1 Connectivity disruptions and SMC failures

Definition

To study the output effects of SMC-related connectivity shocks, I use two key variables: (i) the annual number of international connectivity disruptions caused by cable failures, a *treatment-incidence* measure; and (ii) the duration of SMC repair in days, a *treatment-intensity* measure. The treatment is defined at the country–year level as an episode in which a submarine–cable failure generates a measurable loss of international connectivity for that country. Countries that are topologically connected to the faulted cable but for which traffic is rerouted without noticeable service degradation are coded as *not treated*. Therefore, cable failure attribution follows the principle of observed material impact, not physical proximity to the event. Under this approach, a single submarine cable failure affecting multiple countries is recorded as several country-specific events, with affected nations sometimes located far from the break site.

The disruption database was hand-compiled from *SubTel Forum* —the SMC industry's main technical platform— which systematically documents national-level disruptions caused by cable failures.[15] Unlike Internet disruptions driven by internal policy factors, such as government-ordered shutdowns, submarine cable failures constitute plausibly exogenous shocks that primarily impair cross-border data flows, making them well suited for identifying the effects of international connectivity losses.[16] In fact, most SMC failures stem from incidents caused by maritime activities (fishing, anchoring) (Carter 2009; Clare et al. 2023), but the dataset also records less frequent causes such as natural hazards (seismic events, landslides, storms, flooding), sabotage, and other technical causes (e.g. power failures).[17] This taxonomy allows me to exclude failures plausibly linked to policy or to natural shocks that might affect economic activity independently from connectivity.

Timing and duration

The timing and duration of each event are read from the dates of the SubTel Forum posts that report a fault and its subsequent maintenance and restoration updates, with repair duration measured as the span between the first post flagging a fault and the post recording its resolution. The platform posts updates regularly, and for most events these posts date the onset of the disruption and the completion of the repair precisely, corroborated by operator notifications, social media or press releases. Where such corroboration is unavailable, I date the fault and its resolution by the SubTel Forum posts themselves, which bound the true dates within the posting lag. For a few events the repair-completion date was not documented, leaving the repair duration missing.

Descriptive statistics

Summary statistics of these variables are reported in Appendix Table A.1.1.2, and their time and geographic distributions in Figure 2. They reveal that submarine cable failures are relatively infrequent but highly variable events with sizable repair burdens. At the country-year level (Appendix Table A.1.1.2, Panel A), the average country experiences about 0.07 cable disruptions annually, with physical breaks (0.057) more common than technical outages (0.016). The standard deviations (0.30 and 0.26) indicate substantial cross-country heterogeneity. Among the 71 countries ever treated (Panel

[15] Data are collected from the SubTel Forum's cable faults and maintenance desk, complemented with Akamai "State of Internet Connectivity" reports and cross-checked with keyword-based web searches.

[16] Concerns about non-random exposure to these disruptions are addressed through pre-trend analysis and placebo-permutation tests in the spirit of Borusyak and Hull (2023).

[17] Natural hazards represent the second most common cause, accounting for less than 20% of failures (Clare et al. 2023), but often generate multiple deep-sea breaks where repairs are more complex. Clare et al. (2023) emphasize that climate change and sea-level rise will however intensify these risks.

B), the average rises to 1.14 disruptions a year, and a typical disruption is associated with about 8 repair days. In global annual totals over 2008–2020 (Panel C), the sample averages 20 disruptions and 202 repair days per year; excluding 2020 (Panel D) reduces these to 16 disruptions and 156 repair days, confirming that 2020 was an unusually intense year.

Aggregating failures by region in Figure 2 (right graphs), Sub-Saharan Africa (SSA) accounts for the largest totals of both Internet disruptions and cable repair days, followed by the Middle East & North Africa (MENA), East Asia & Pacific (EAP), and South Asia (SA). Europe & Central Asia (ECA) and Latin America & the Caribbean (LAC) register much smaller tolls. The contrast between the number of disruptions and the stock of repair days is informative, as a few episodes are long-lived and queue repair capacity. So events' duration also drives exposure. The annual series in Figure 2's left graphs show lumpy time patterns. The count of disruptions rises from 2019, with the largest shift occurring in 2020.[18] Repair-day exposure exhibits much larger variance, with a pronounced spike in 2020. This reinforces the choice to use repair duration as the complementary "dose" variable in the empirical analysis.

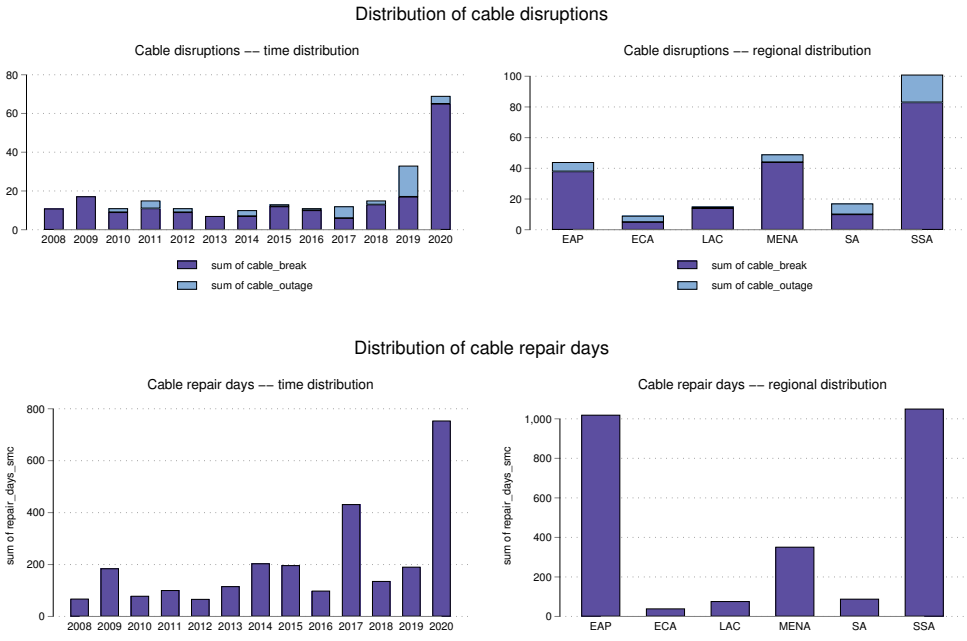


Fig. 2: Internet disruptions and repair days, regional and time distributions, 2008-2020

The geography of submarine cable failures

The country-level incidence of disruptions concentrates in locations defined by the physical layout of the cable network. Figure 3 maps the cumulative count of SMC-related disruptions per country over 2008–2019 against critical world maritime choke points. The highest counts fall on two types of location: economies on the periphery of the network that depend on a small number of landing cables, and countries adjacent to the maritime choke points where many systems converge — the Red Sea and Bab-el-Mandeb corridor feeding the Suez landings, the Strait of Malacca, the Mediterranean

[18] The baseline estimation sample drops this outlier year, but results are robust to its inclusion in the sample. See robustness Section 5.

approaches off North Africa, and the West African coast. Isolated island economies of the Indian Ocean, the Pacific, and the Caribbean, shown in the insets, record disruptions out of proportion to their size for the same reason. Most faults originate in fishing and anchoring over shallow continental shelves (Clare et al. 2023), so that a country's position relative to these shipping-dense seabeds governs how often its international routes are severed.

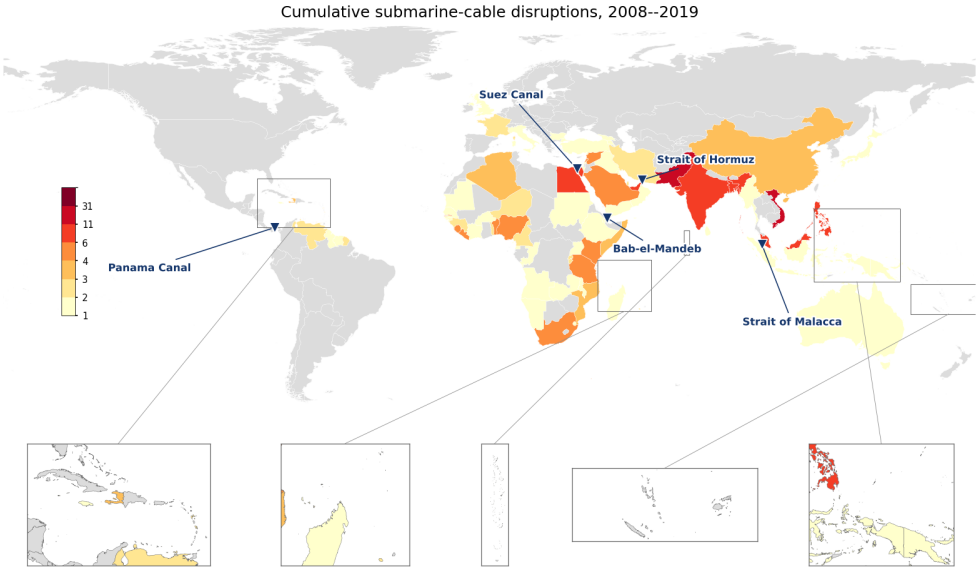


Fig. 3: Geographic distribution of SMC-related internet disruptions, cumulative 2008–2019.

Notes: Shading reports the cumulative count of submarine-cable-related internet disruptions recorded per country over 2008–2019 in the *SubTel Forum*-based database. Navy triangles mark the principal maritime choke points where many cable systems share crowded seabeds (Panama and Suez Canals, Bab-el-Mandeb, and the Straits of Hormuz and Malacca). Insets zoom on isolated island economies of the Indian Ocean, the Pacific, and the Caribbean, whose disruption counts are hard to read at world scale.

3.2.2 GDP growth and other economic outcomes

I study macroeconomic aggregates that plausibly respond to connectivity shocks. For each following aggregate I focus on levels in constant 2015 USD (log): GDP, household final consumption, government final consumption, fixed gross capital formation, exports and imports of goods and services, and Foreign Direct Investment (FDI).

Because submarine cables are the main digital infrastructure for cross-border data flows, I place particular emphasis on trade, productivity, and financial channels. Therefore, in addition to total trade in goods and services, I study disaggregated trade flows, such as merchandise trade and services trade, but also exchanges of commercial services, ICT services, financial services or transport services.[19] To investigate the mechanisms behind macroeconomic adjustments, I also study the productivity effects of cable disruptions, using the World Bank Global Productivity Database, and financial frictions, using Credit and Locational Banking Statistics from the Bank of International Settlements. Finally, I use the country-level exporter data from the World Bank's Exporter Dynamics Database to connect empirical evidence from macroeconomic aggregates to those from observed import and export extensive margins.

[19]These services-trade series are available only in current USD; I obtain series in constant dollars using implicit deflators, which are the ratio of the exports (or imports) of goods and services expressed in current USD over exports (imports) in constant USD.

The outcomes above are annual and national. I complement them with a monthly measure of local economic activity, night-time light radiance, built from the VIIRS Day/Night Band stray-light-corrected monthly composite (NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG) extracted through Google Earth Engine.[20] Online Appendix A.2.3 presents the data construction steps in full.

3.3 Estimation sample

Summary statistics of cable failures and main macroeconomic outcomes for the estimation sample are reported in Appendix Tables A.1.1.3 and A.1.1.5, respectively.

3.3.1 Sampling restrictions

Two groups of countries have been intentionally dropped from the sample. Sixteen *non-sovereign territories* are excluded, since their national accounts and their international connectivity follow those of the metropolitan state: American Samoa, Bermuda, the Channel Islands, Curaçao, the Faroe Islands, Gibraltar, Guam, the Isle of Man, Saint Martin, the Northern Mariana Islands, New Caledonia, French Polynesia, Sint Maarten, the Turks and Caicos Islands, the British Virgin Islands, and the United States Virgin Islands.

Thirteen countries recording a *state-based conflict at war intensity*, above 1,000 battle-related deaths in two or more years between 2008 and 2020, are excluded: Afghanistan, the Democratic Republic of the Congo, Ethiopia, Iraq, Libya, Nigeria, Pakistan, Sri Lanka, Somalia, Sudan, Syria, Ukraine, and Yemen. [21]

Vietnam is associated to an abnormally high number of disruptions and is consequently held out separately. In fact, it records 26 disruptions and 467 repair days over 2013–2019, of which 11 events and 259 days fall in 2017 alone, a concentration that reflects repeated faults on the AAG (Asia-America Gateway) cable traced to a design flaw. As shown previously, the year 2020 is also an outlier in terms of disruptions count and repair duration, and thereby, is excluded from the baseline estimation sample. Robustness estimations reported in Appendix Section OA.3 add Vietnam and 2020 back to the estimation sample.

3.3.2 Estimation sample's composition

The estimation sample retained in baseline estimations consists of 1,804 observations covering 164 countries and 12 years (2008-2019). The estimation sample distribution by region and income group is given in Table 1. It includes 517 observations from high–income countries (HICs) (28.66%), 935 from middle–income economies (MICs) (51.83%), and 352 from low–income countries (LICs) (19.51%). Regional representation is likewise diverse: Europe and Central Asia (ECA) accounts for the greatest sample share (32.32%), followed by Sub–Saharan Africa (SSA) (23.78%), Latin America and the Caribbean (LAC) (21.95%), East Asia and Pacific (EAP) (14.63%), the Middle-East and North Africa (MENA) (4.27%), South Asia (SA) (1.83%), and North America (1.22%). Among treated countries, SMC disruptions are more recorded among MICs, more located in SSA, while ECA is over-represented in the group of never-treated countries.[22].

[20] For each city I average radiance within ten kilometres of the centroid, weight it by the population living inside that ring, remove the calendar-month component, and take logs. The panel covers nineteen countries observed from January 2016 to December 2019, represented by their economic capital and by the remaining cities above 20,000 inhabitants.

[21] Identified from the UCDP/PRIO Armed Conflict Dataset v24.1. We chose a two-year rather than a one-year threshold, since the later would capture isolated flare-ups, such as Marawi in 2017 for the Philippines, and thus lead to drop too many countries.

[22] Additional analysis shows that results are robust to successively removing regions from the sample, including removing SSA (Section 5.4.1)

Table 1: Estimation sample composition

	Observations	Percent
Panel A. Estimation sample		
By income group		
High income (HIC)	517	28.66
Middle income (MIC)	935	51.83
Low income (LIC)	352	19.51
By region		
Europe and Central Asia	583	32.32
Sub–Saharan Africa (SSA)	429	23.78
Latin America & Caribbean (LAC)	396	21.95
East Asia and Pacific	264	14.63
Middle East & North Africa (MENA)	77	4.27
South Asia	33	1.83
North America	22	1.22
Total	**1,804**	**100.0**
Panel B. Switchers		
By income group		
High income (HIC)	77	14.29
Middle income (MIC)	253	46.94
Low income (LIC)	209	38.78
By region		
Sub–Saharan Africa (SSA)	286	53.06
Europe and Central Asia	77	14.29
East Asia and Pacific	66	12.24
Latin America & Caribbean (LAC)	66	12.24
Middle East & North Africa (MENA)	33	6.12
South Asia	11	2.04
Total	**539**	**100.0**

3.4 First-stage analysis

Identification assumes that a SubTel Forum–coded disruption maps into a loss of international connectivity. I test this against routing data built independently of the disruption coding. The connectivity measure is the count of IPv4 address blocks (routed prefixes) that a country originates and that remain visible in the global routing table, from the Routing Information Service (RIS) of the Réseaux IP Européens (RIPE). A fall in this count means part of a country's address space has become unreachable, which is the operational definition of a connectivity loss.

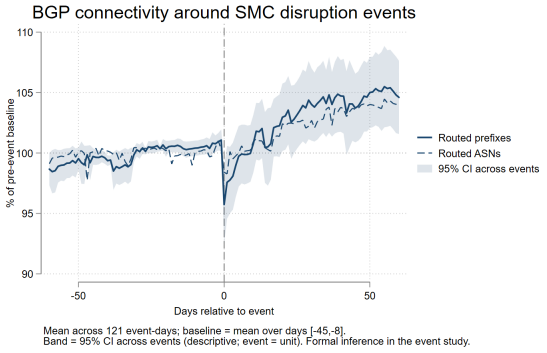
I use two designs, reported together in Figure 4. The first is a daily event study around the multi-country episodes whose onset date is fixed from the SubTel Forum timestamps and contemporaneous operator and press reports, covering 17 episodes, 62 countries, 107 country-events.[23] Because disruption events are spread across distant calendar windows, from January 2008 to January 2020, the daily and weekly analyses (panels a and b) stack them in event time, aligning each country on its own verified onset date, with event fixed effects and standard errors clustered by event.[24] Two daily outcomes reflecting a country's routed connectivity are drawn from the global BGP routing table (Routing Information Service, RIPE). First, *routed prefixes*, which are the number of distinct IPv4 address blocks the country originates that are visible in the global routing table. A fall means part of the address space is withdrawn and becomes unreachable, reflecting partial withdrawals of address space. Second, *routed ASNs*, which is the count of distinct networks registered in the country that remain reachable in the same routing table, each identified by its Autonomous System Number (ASN).[25] Routed prefixes drop on the event date, and recover within three weeks (panel a). Panel b

[23] See Appendix Table A.2.1.1

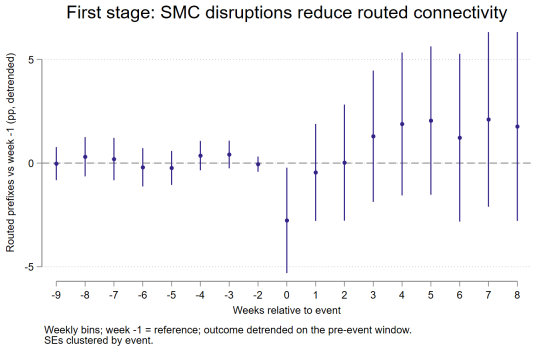
[24] The DID_M estimator does not apply to this design, since it compares disrupted countries with not-yet-disrupted ones within the same calendar period

[25] An autonomous system is one operator's network, identified by a number assigned by a regional internet registry, which announces its own routes to the rest of the internet. Internet service providers, mobile operators, hosting and content providers,

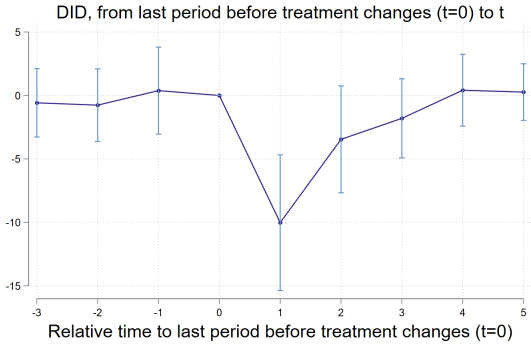
shows that routed prefixes are flat over the nine weeks before onset, fall 2.8 percentage points below the pre-break week in the week of the break, and are back to baseline the following week.



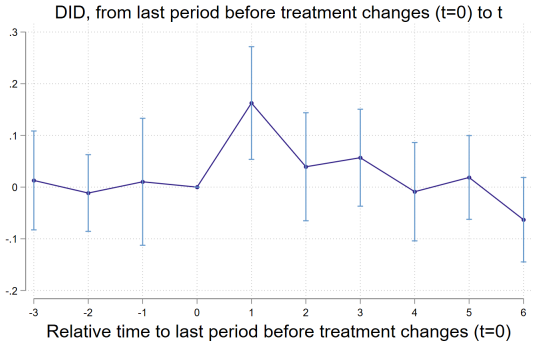
(a) Daily analysis (TWFE): mean routed prefixes around onset



(b) Weekly analysis (TWFE): detrended routed prefixes



(c) Monthly analysis (CDH): worst-month drawdown



(d) Monthly analysis (CDH): outage episodes

Fig. 4: First stage: submarine-cable disruptions reduce routed connectivity.

Notes: Outcomes measure a country's routed connectivity in the global BGP routing table (Routing Information Service, RIPE). Routed prefixes = the number of distinct IPv4 address blocks the country originates that are visible in the global routing table. Routed ASNs = the number of distinct networks registered in the country that remain reachable in the same routing table, each identified by its Autonomous System Number (ASN). **(a)** cross-event mean routed prefixes in event time, as a percentage of the pre-event baseline (mean over days $[-45, -8]$), 95% band across the 107 country-events (17 episodes) of Online Appendix Table A.2.1.1. **(b)** event-study coefficients in weekly bins, outcome detrended by an episode-specific linear pre-trend, week -1 the reference, SE clustered by country-event. **(c)–(d)** dynamic effects from de Chaisemartin and d'Haultfoeuille (2024), SE clustered by country. *Worst-month drawdown* = deepest daily shortfall of routed prefixes relative to their trailing 60-day baseline within the month (percentage points, negative). *Outage episode* = a run of consecutive days with routed prefixes more than 10% below that baseline (gaps ≤ 3 days merged).

The second design is a country-month panel of 172 economies over 2008–2020, using four monthly outcomes built from RIPE RIS data on routed prefixes: i) the worst-month drawdown, ii) outage days, iii) outage episodes, and iv) severity. All four rest on a country's daily drawdown, the percentage shortfall of its routed prefixes relative to their trailing 60-day baseline. The *worst-month drawdown* is the deepest daily drawdown in the month. *Outage days* count the days on which the drawdown exceeds 10%, and *outage episodes* count the distinct runs of such days, merging runs separated by three days or less. *Severity* is the sum of daily drawdowns over outage days, and thus combines depth and duration. Panels (c) and (d) of Figure 4 plot the dynamic effects of disruptions on the worst-month drawdown and outage episodes, estimated with the $\mathbf{DID}_M$ estimator on the full SubTel Forum event dataset. Appendix Tables A.2.1.2 and A.2.1.3 report the TWFE and CDH estimates for

and some large enterprises each hold one or more. The ASN count falls only when an entire network stops being announced, and is therefore the coarser and more severe signal.

all four outcomes, respectively. In the month of a disruption, the worst-month drawdown deepens by 10 percentage points, and outage episodes rise by 0.16. Both effects are significant in the disruption month, and fade in the following months. Outage duration and severity remain high through the second month after the disruption (Appendix Tables A.2.1.2 and A.2.1.3). Outage days rise by 0.85 after one month and 1.87 after two, and severity increases by about 53 in both months. Connectivity thus drops immediately, while the time spent below baseline accumulates over two months before reverting.

4 Main Results

This section presents baseline evidence on the economic costs of cable failures. First, I test whether exposure to cable disruptions is followed by movements in GDP per capita, and whether the magnitude of these effects increases with the duration of disruptions. Second, I turn to heterogeneity tests, exploring whether digital infrastructure characteristics condition the transmission of shocks to the whole economy. Third, I study regional GDP spillovers induced by breaks for regional non affected neighbours.

4.1 Baseline estimations

Figure 5 reports the response of GDP per capita to a cable disruption, with distinct fixed effects (FEs) calibrations. Detailed estimates are reported in Appendix A.2.2. The upper panel (a) reports estimates without group-by-year FEs, while the middle and lower panels (b) and (c) report those with income-group-by-year and region-by-year FEs, respectively. Left-hand graphs report the dynamic effects of the cable disruption variable, while the right-hand sides' report the cable repair duration variable's.

Baseline treatment-path diagnostics in Appendix Table A.2.2.1 show that the cable-disruption count generates seven distinct treatment paths, covering all 22 switchers with a fully-observed six-period window out of 58 switchers in the baseline estimation sample. Three paths account for 82% of these switchers: a one-off break in the first treated year with no further disruption over the following six years (55%, path 1), a break in the first year followed by renewed disruption in years five and six (14%, path 2), and two disruptions concentrated in the first treated year alone (14%, path 3). The remaining four paths, each covering a single switcher, combine breaks at more scattered horizons.

Across specifications, placebo leads are flat, consistent with the parallel-trends assumption, while the GDP response becomes increasingly negative over the post-disruption horizon $[t, t+5]$. Our preferred specification includes income-group $\times$ year fixed effects, which compares each disrupted country with countries in the same income group. The sup-t test rejects the joint nullity of the treatment effects at the 2% level in this setting, against 5 and 6% level in the two others (Appendix Table A.2.2.3). Across all calibrations, GDP falls by approximately 1% on impact, with the estimated loss widening to 2–3% two years after the disruption. The decline then reaches its trough four to five years after the event, lying between -4.8% to -5.9%.[26] Replacing the cable disruption number with the number of cable repair days confirms previous growth patterns and adds interpretability to the results (Figure 5, right panels; Appendix A.2.2). One additional cable repair day depresses GDP per capita by about -0.1% to -0.2% over [t; t+2], reaching -0.48% to -0.66% over [t+4;t+5]. With an average duration of repair 8 days per disruption, this estimate converges with estimates using the disruption episode variable. In magnitude, estimated losses are sizeable, of the order documented for

[26] The medium-run estimates rest on 22 switchers and 413 contributing observations at $t+5$ in the specification with income-group$\times$year fixed effects, and on 22 switchers and 666 contributing observations at $t+5$ in the specification without group$\times$year fixed effects.

power outages (Fried and Lagakos 2023), banking crises (Cerra and Saxena 2008), or inter-state wars (Federle et al. 2026).

4.2 Heterogeneity analysis

Four documented features of a country's international connectivity infrastructure can limit the output loss from a cable failure. First, more SMC landings provide spare capacity. Second, a broader set of connected partners, counted as distinct foreign landing countries and systems, widens the set of routes available when one path fails. Third, Internet Exchange Points keep regional and domestic traffic inside the country or the region, which lowers the share of traffic depending on international links.[27] Fourth, ownership diversity, measured by the number of distinct commercial owners among the cables landed in a country, shapes how easily traffic can be rerouted after a fault. When owners are more diverse, operators whose cables are unaffected can absorb rerouted traffic through their own capacity and backup arrangements.[28] Where paths are few and local routing capacity is limited, one fault removes most of the substitution available to a country's traffic.

To identify heterogeneous effects of SMC failures depending on network redundancy, I split countries at the median of each connectivity variable in the 2008 cross-section, fix each country's group at that baseline assignment, and re-estimate the event studies (Figure 6, panels (a)–(d)). The 2008 medians are one landed cable, two connected partners and two distinct cable owners. For IXPs the split is between countries with no exchange point and countries with at least one.[29]

Among countries with at most one landed cable, GDP per capita falls 7.3% below trend four years after a disruption, against 2.5% where more cables land (Figure 6, Panel (a)). Among countries with at most two connected partners, it falls 3.8% below trend two years after a disruption and 7.6% four years after, against 1.2% where more partners are connected (Panel (b)). In both splits the below-median group is significantly different from zero from three years onward, while the above-median group is not significant at any horizon. Interestingly, the number of connected partners separates the two groups more distinctly than the number of landed cables, suggesting that route diversity matters more than cable density for digital resilience.[30] The IXP split gives the widest of the four gaps, as GDP per capita in countries with no IXP in 2008 falls 10.5% below trend four years after a disruption, against 1.7% where at least one is present (Panel (c)). Last, ownership diversity gives a gap of the same size as connectedness (Panel (d)). GDP per capita in countries with at most two distinct cable owners falls 7.5% below trend four years after a disruption, against 7.6% in the low-connectedness group, and is significantly different from zero from three years onward. Where more than two owners are present, the shortfall reaches 1.2% and is insignificant at every horizon.

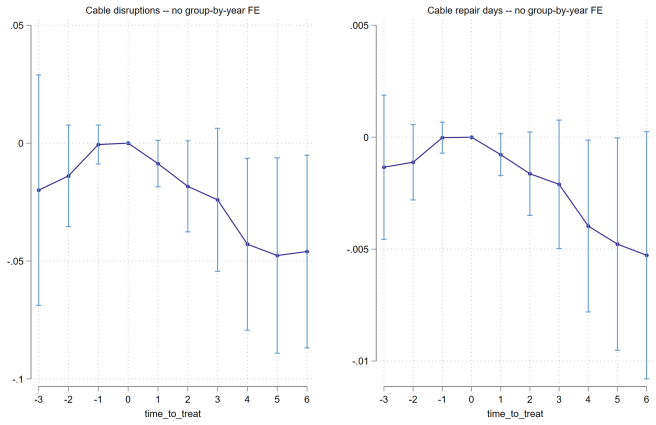
Therefore, countries entering the estimation period with more than one landed cable, more than two connected partners, at least one IXP, or more than two cable owners lose between 1.2% and 2.5% four years after a disruption, none of these estimates significantly different from zero. Deloitte (2016) attributes connectivity losses to the extent of a country's digitalisation, reflected by high internet penetration rates. Evidence does not support this assertion. Splitting the sample at the 2008 median of internet use (22.6%), GDP per capita falls 6.0% below trend four years after a disruption in below-median countries and 3.6% in above-median ones, and the two confidence intervals overlap across the

[27] An IXP is a facility where ISPs, CDNs and other autonomous systems interconnect to exchange traffic directly without transiting international routes. A larger number of IXPs lowers latency, reduces transit dependence and increases the number of rerouting options.

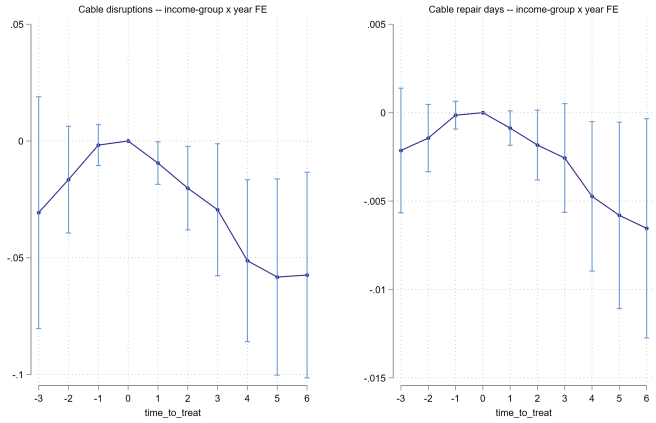
[28] Ownership diversity is distinct from route diversity. Two countries with the same number of independent routes can differ in whether these routes are held by one owner or by several.

[29] Online Appendixes OA.2 reports the lists of low-connectivity countries identified according to the four exposure factors and the detailed dynamic estimates.

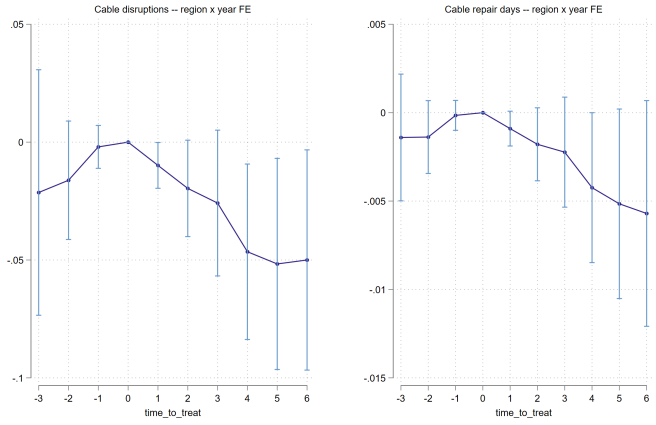
[30] This heterogeneous effect is robust to adding the number of cables as control, and therefore independent from cable density.



(a) No group-by-year fixed effects



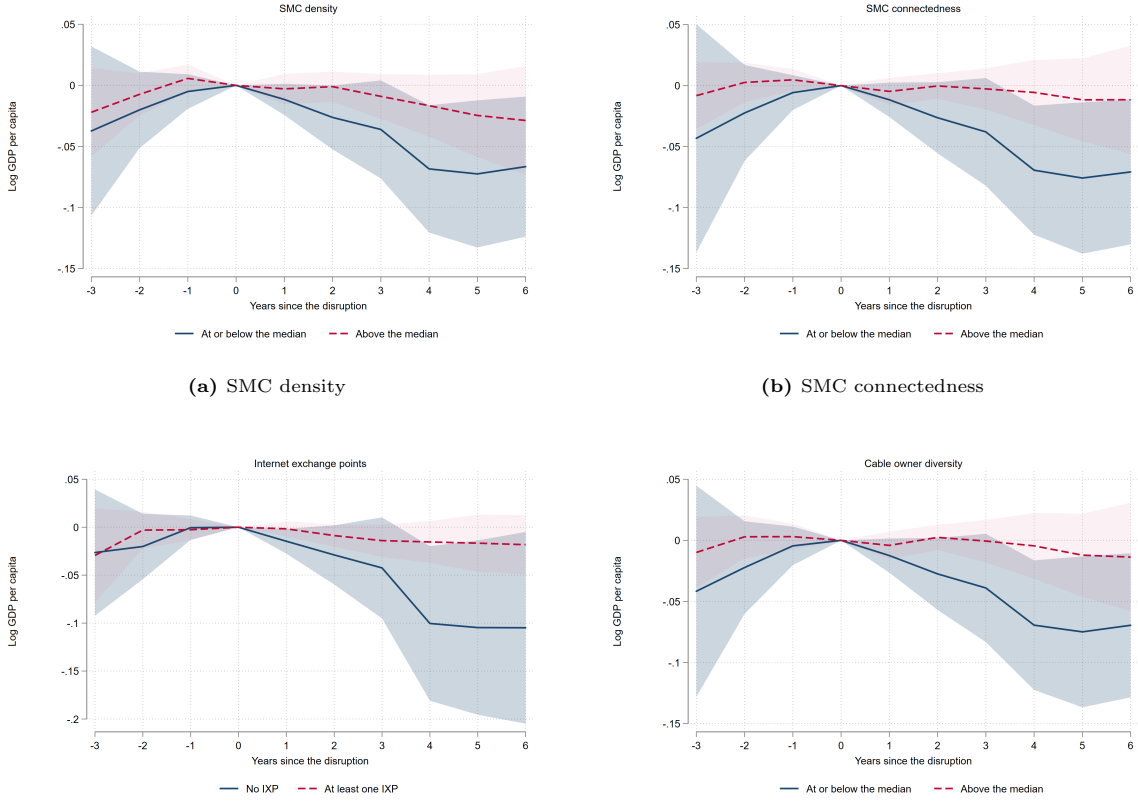
(b) Income-group by year fixed effects



(c) Region by year fixed effects

Fig. 5: Cable disruptions and GDP per capita (constant USD, ln): dynamic effects.

Notes: Within each panel the left graph uses the number of cable disruptions as treatment and the right graph the count of repair days. All graphs report 95% confidence bands. Specifications include country fixed effects, year fixed effects and linear country trends, with standauntry level. Panel (a) adds no group-by-year fixed effects, panel (b) addsincome-group-by-year fixed effects, and panel (c) region-by-year fixed effects. Year 6 corresponds to five years after the cable failure.



(a) SMC density

(b) SMC connectedness

(c) Internet Exchange Points

(d) Cable ownership diversity

Fig. 6: Policy margins: network redundancy and resilience to SMC failures.

Notes: Red hollow markers = above-median value at baseline; blue filled markers = below-median. Group assignments are fixed at the start of the sample. Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. Panels (a)–(b): SMC data drawn from Telegeography's SMC map. Panels (c): IXP and cable-ownership data drawn from Telegeography's IXP map, Packet Clearing House, and the PeeringDB databases; panel (d)'s ownership-diversity measure counts distinct owners of cables landed in a country.

event window (Appendix Figure A.2.5.1, panel (a)). The joint ITU–UNDRR–Sciences Po report (ITU et al. 2026) makes the same observation, noting that network architecture governs exposure to digital risk more than the level of digitalisation does, with some small coastal or island states relying on a single submarine cable. Income per capita does not account for the split either. Splitting at the 2008 median of GDP per capita, the below-median half falls 5.1% below trend at the same horizon and the above-median half 5.4%. These results are consistent with Porcellacchia et al. (2026), whose model of traffic routing makes a country's welfare loss from a cable cut increase as data flows concentrate on few routes and decrease with redundancy.

4.3 Regional spillovers

In the presence of interference the baseline GDP effects should be interpreted as *net* losses reflecting a divergence process, since part of the relative decline in exposed economies may reflect contemporaneous gains in proximate, non-disrupted countries. To probe the neighbour response directly, I construct two peer-disruption measures that aggregate, at the World Bank region level, the annual number of disrupted economies (extensive margins) and the annual number of cable-disruption episodes (intensive margins) occurring in a country's region, where one episode can affect several economies. I re-estimate the event study on the subsample of country-years with no recorded failure ($D_{it} = 0$), so

that treatment varies only through exposure to disruptions elsewhere in the same region, including country fixed effects, year fixed effects, and a linear country trend. Figure 7 shows the unaffected neighbours' GDP response to regional connectivity shocks (Appendix A.2.4 reports the detailed estimates).

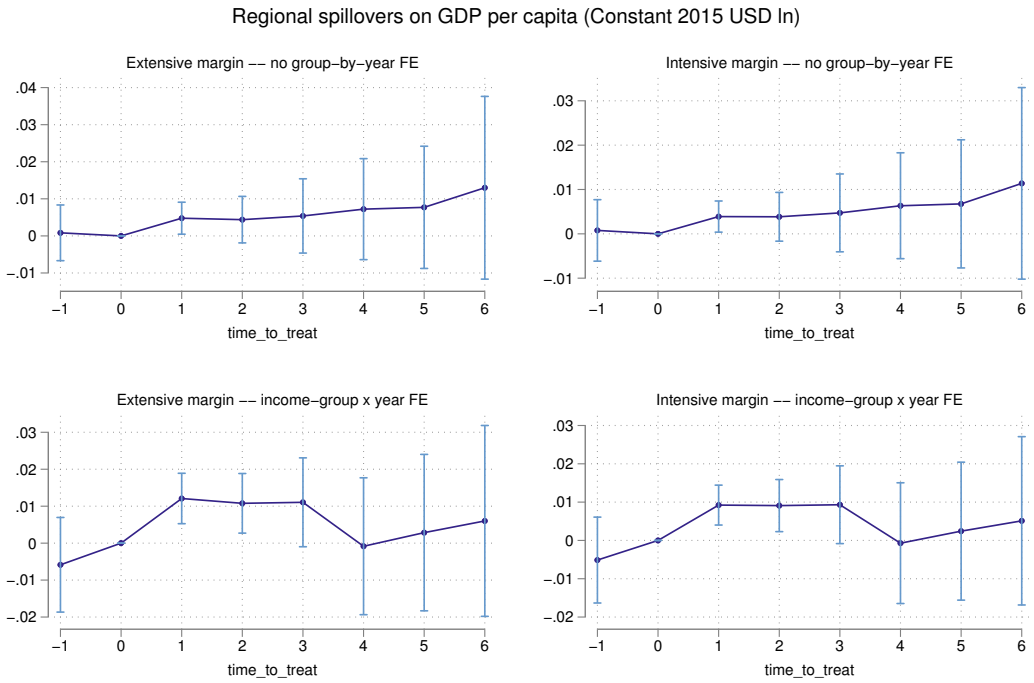


Fig. 7: Regional spillovers of cable disruptions on unaffected neighbours.

Notes: Event-study coefficients estimated on the subsample of country-years with no recorded cable failure ($D_{it} = 0$), where treatment varies through exposure to cable disruptions elsewhere in the same World Bank region. The outcome is log GDP per capita in all four panels. The left column measures regional exposure by the number of disrupted economies and the right column by the number of cable-disruption episodes. The top row includes country fixed effects, year fixed effects and linear country trends. The bottom row adds income-group-by-year fixed effects. Standard errors are clustered at the country level, with 95% confidence bands and one placebo lead. The joint test of post-treatment effects yields $p = 0.03$ in the top row and $p < 0.001$ in the bottom row. The estimates are identified from 85 switching groups (N=127) on impact falling to 80 (N=90) at five years in the top row, and from 56 (N=72) falling to 32 (N=38) in the bottom row, against only two never-exposed control countries. Year 6 corresponds to a period of five years after the cable failure occurrence.

Placebo leads are flat, and the post-treatment coefficients turn positive. In the year of a regional disruption, an unaffected neighbour's GDP per capita stands about 0.5% above trend on the extensive margin (the count of disrupted economies) and 0.4% on the intensive margin (the count of disruption episodes), both significantly different from zero (Figure 7 and Appendix Table A.2.4.1). Adding income-group-by-year fixed effects raises these gains to 1.2% and 0.9%, significant through the following year. The gains concentrate in the first year or two and fade thereafter. Estimates' magnitude matches the 0.4 percentage-point gap between the within-region baseline (Appendix Table A.2.2.4) and the broader counterfactual (Appendix Table A.2.2.2), stressing that part of the estimated within-region loss reflects neighbours' economic gains.

The mechanisms underlying GDP spillovers are further investigated in Section 6.3, which asks whether these gains come with a diversion of service exports towards unaffected neighbours, and Section 6.4, which tests whether foreign investment and productivity relocate with them.

5 Robustness

The baseline results show large and persistent output losses after submarine-cable failures. This section probes their robustness against alternative explanations and identification threats. Because the annual effects build up over several years, a modest undetected pre-trend could generate them on its own. I first turn to an infra-annual analysis of breaks on the Africa Coast to Europe cable, using monthly night-time lights as a proxy for economic activity. The monthly frequency provides a long pre-break window to test for diverging trends and leaves no multi-year window over which such a trend could build. I then address potential bias from non-random exposure to cable shocks through falsification tests in the spirit of Borusyak and Hull (2023). The main concern is that countries with weak telecommunications policy accumulate both more disruptions and worse macro trends. Next, I control for key confounding factors. Finally, I run sensitivity checks that sequentially exclude regions and countries, and verify that the magnitude of the losses is not driven by a few long-lasting disruptions.

5.1 Undetected pre-trend?

Because the baseline effects in Section 4 accumulate over several years, a pre-trend undetected by the CDH estimator could account for them. Since the specification already includes country-specific linear trends, the concern is a differential trend that departs from them. Table 2 reports the two bounds proposed by Rambachan and Roth (2023). The first, Δ^{SD}, limits how much the slope of the differential trend can change from one year to the next, and suits a secular trend that evolves smoothly. The second, Δ^{RM}, bounds the post-break differential trend by a multiple of the largest one observed before the break, and suits a shock that hits treated countries around the treatment date. Both concerns apply here. First, breaks are caused by fishing gear and anchors, and their timing is unrelated to macroeconomic conditions, but disruption-prone economies, often peripheral and poorly connected, may follow a growth path that slowly diverges from that of other countries. Second, several large disruptions also occurred in 2008, that is, during the financial crisis. While year and income-group-by-year fixed effects absorb the common part of the financial crisis, they do not absorb differences in its impact across countries.

The bounds are expressed in log units of GDP per capita.[31] Under Δ^{SD}, $M = 0$ allows any linear extension of the pre-break trend, while a larger M allows the growth gap between disrupted and control countries to change by up to M each year. At the 95% level, the estimates at three to five years remain significant under a linear extension, and for values of M up to 0.0020, 0.0014 and 0.0008 respectively. At four years, the growth gap can thus depart from its linear extension by 0.14 percentage points each year, which shifts GDP per capita by about 2% over the post-break years, a little more than a third of the estimated loss of 5.8%. The breakdown value falls with the horizon because the same departure builds up over more years. At the 90% level, all horizons remain significant under a linear extension, and the four-year bound rises to 0.18 percentage points per year, or about 2.7% of GDP per capita, close to half of the estimated loss.

Under Δ^{RM}, $\bar{M} = 1$ allows the post-break GDP gap to be as large as in the worst pre-break year. At four years, the breakdown value is 0.25 at the 95% level and 0.35 at the 90% level. The largest year-to-year change in the placebo estimates is about 1.5 percentage points, so the four-year loss remains significant as long as the post-break growth gap stays below about 0.4 percentage points per year. Two features of the data qualify this result. The placebo estimates are individually and jointly

[31] For the magnitudes involved, a difference of 0.01 in the level corresponds to about 1% of GDP per capita, so that a change of 0.001 in the slope corresponds to 0.1 percentage points of annual growth.

insignificant, and they come from countries first disrupted after 2008, since the sample starts that year. A growth gap of 1.5 percentage points per year would also exceed the rate at which the estimated loss deepens after the break, about 1.2% of GDP per capita per year over the first four years.

The medium-run losses are therefore robust to a linear extension of the pre-break trend and to moderate departures from it, but not to violations as large as the largest pre-break one. The next subsection complements this test by exploiting quasi-experimental disruptions on the Africa-Coast-to-Europe (ACE) cable over 2016-2019 with monthly night-time light data.

Table 2: Sensitivity of the baseline estimates to violations of parallel trends.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Smoothness (Δ^{SD})				Relative magnitudes (Δ^{RM})			
		95% level		90% level		95% level		90% level	
Horizon	Estimate	CI at $M=0$	M	CI at $M=0$	M	Original CI	$\bar{M}$	Original CI	$\bar{M}$
t	−0.009	[−0.025, 0.002]	0.0000	[−0.023, −0.001]	0.0014	[−0.019, −0.000]	0.05	[−0.017, −0.002]	0.20
$t+1$	−0.020	[−0.048, 0.003]	0.0000	[−0.044, −0.001]	0.0020	[−0.038, −0.002]	0.15	[−0.035, −0.005]	0.25
$t+2$	−0.029	[−0.070, 0.004]	0.0000	[−0.064, −0.002]	0.0014	[−0.058, −0.001]	0.10	[−0.053, −0.006]	0.20
$t+3$	−0.051	[−0.110, −0.017]	0.0020	[−0.102, −0.024]	0.0026	[−0.086, −0.017]	0.30	[−0.080, −0.022]	0.40
$t+4$	−0.058	[−0.132, −0.015]	0.0014	[−0.123, −0.024]	0.0018	[−0.100, −0.016]	0.25	[−0.094, −0.023]	0.35
$t+5$	−0.057	[−0.144, −0.008]	0.0008	[−0.133, −0.019]	0.0012	[−0.101, −0.013]	0.20	[−0.094, −0.021]	0.25

Notes: Sensitivity of the baseline event-study estimates to violations of parallel trends, following Rambachan and Roth (2023). Column (1) reports the estimates of Figure 5, panel (b), the income-group-by-year specification. Columns (3), (5), (7) and (9) report the breakdown value, the smallest M or $\bar{M}$ at which the robust confidence set includes zero. In columns (2) to (5), M bounds the change in the slope of the differential trend from one year to the next, in log units of GDP per capita. $M=0$ allows any linear extension of the pre-break trend, and columns (2) and (4) report the robust confidence set in that case. A breakdown value of zero means the estimate is not significant under a linear extension. The grid runs from 0 to 0.005 in steps of 0.0002. Columns (6) and (8) report the original confidence intervals. In columns (7) and (9), $\bar{M}$ bounds the change in the differential trend between two post-break years as a multiple of the largest such change over the three pre-break years. The grid runs from 0 to 1 in steps of 0.05. Following Rambachan and Roth (2023), robust confidence sets use fixed-length confidence intervals under Δ^{SD} and the conditional-least-favorable hybrid method under Δ^{RM}. All specifications include country-specific linear trends, so both bounds apply to departures from these trends.

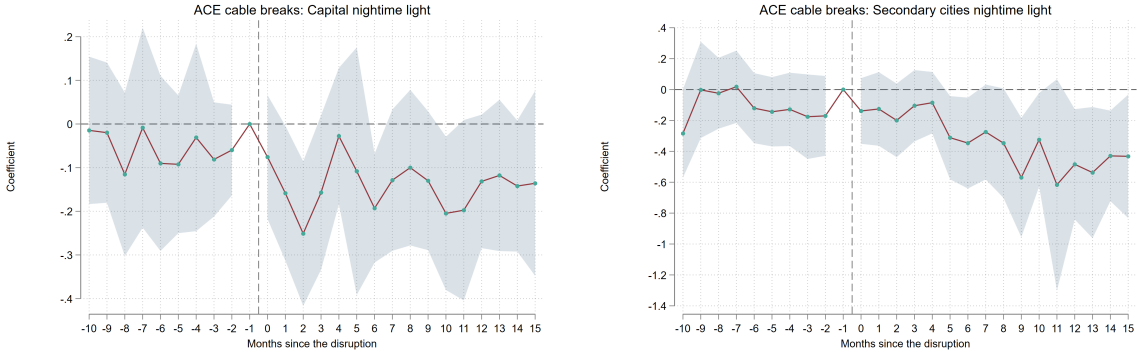
5.2 Monthly evidence from ACE breaks over 2016–2019

Monthly data on economic activity can address the concern of an undetected pre-trend from a different angle. National accounts being unavailable at this frequency, I measure economic activity with satellite night-time radiance, a source independent of the administrative statistics used in the annual analysis. Breaks on the Africa Coast to Europe (ACE) cable over 2016–2019 provide a suitable quasi-experiment, because each break cuts off only some of the countries landed on the cable, while the others remain connected and serve as a comparison group. The event study covers the ten months before and the fifteen months after each break. Over the pre-break months, a slow pre-existing trend would be visible in the data, and such a trend could not explain a sharp fall in activity in the months right after a break. These breaks also occur ten years after the 2008 financial crisis.[32]

The sample includes 19 countries observed monthly from January 2016 to December 2019. The treated group consists of ten ACE landing countries disrupted at least once over the period. The nine untreated countries comprise four ACE landing countries that were not disrupted, two landlocked countries connected to ACE through neighbouring landing countries, and three distant control countries.[33] Night-time radiance is measured separately for each country's economic capital and for its

[32] Exposure to the break may nonetheless not be random. Section 5.3 addresses this concern through placebo-permutation tests (Borusyak and Hull 2023).

[33] The treated countries are Mauritania, Senegal, the Gambia, Guinea-Bissau, Guinea, Sierra Leone, Liberia, Côte d'Ivoire, Benin, and Equatorial Guinea, all disrupted on 30 March 2018. The Gambia was disrupted again in July and November 2019,



(a) Night-time light, capital city

(b) Night-time light, remaining cities

Fig. 8: ACE cable breaks: monthly night-time radiance.

Notes: Local projections difference-in-differences estimates (Dube et al. 2023) of the effect of an ACE cable break on the logarithm of population-weighted night-time radiance, by month relative to the break. Panel (a) uses the economic capital of each country; panel (b) aggregates all sampled cities above 20,000 inhabitants outside the economic capital. Radiance is measured within ten kilometres of the city centre from the VIIRS stray-light-corrected monthly composite, with each city's own calendar-month mean removed. The sample is nineteen countries observed from January 2016 to December 2019; ten were disrupted by the break of 30 March 2018 or by the faults of July and November 2019. Specifications include one lag of the outcome, a clean-comparison window of six months, and standard errors clustered on the country. Bands are 95% confidence intervals. Month 0 is the month of the break, and the month before it is the reference period. The nightime light radiance variable's summary statistics are reported in Appendix A.2.3.1.

other cities of more than 20,000 inhabitants.[34] I use the local projections difference-in-differences estimator of Dube et al. (2023), with ten monthly leads, fifteen monthly lags, and standard errors clustered by country. Unlike the annual panel, where disruptions vary in timing and intensity across countries, treatment here is binary and hits all treated countries at the same dates. LP-DiD estimates the effect separately at each horizon against countries not disrupted over a six-month clean-comparison window, and includes one lag of the outcome to account for the strong persistence of monthly radiance series.[35]

Figure 8 reports the results. In Panel (a), radiance in capital cities falls over the fifteen months after the break, with a pooled post-break estimate of −15.5%, while the ten pre-break months are flat and jointly indistinguishable from zero. In Panel (b), radiance in secondary cities falls by 28.2%. Both series follow a similar path through month six, with local troughs at months two and six. The drop then holds steady in capitals, while radiance in secondary cities keeps falling until month 15. At that horizon, capitals stand 12.7% below the control group and secondary cities 35.1%.[36] The CDH estimator gives estimates similar in sign and dynamics, with average declines of 10% in capitals and 22% in secondary cities over the fifteen months (Appendix A.2.3.4). Capital effects are felt in the first four months and fade thereafter, while those in secondary cities deepen from the ninth month onward.[37]

The costs of cable disruptions are therefore visible at monthly frequency and in satellite radiance, in addition to national accounts. They are not confined to the economic centre. Capitals stabilise at

and Liberia and Sierra Leone in November 2019. The untreated landing countries are Ghana, Nigeria, Gabon, and São Tomé and Príncipe. The landlocked countries are Mali, connected via Senegal or Côte d'Ivoire, and Niger, connected via Benin. The distant control countries are South Africa, France, and Portugal. Appendix A.2.3 gives the full panel composition.

[34] Appendix A.2.3 details how radiance variables are built.

[35] Specification details and estimates are reported in Appendix A.2.3.

[36] Estimates are robust to dropping France and Portugal and re-estimating on the seventeen African countries (Appendix A.2.3).

[37] The two specifications differ. The local projections estimator includes one lag of the dependent variable, which absorbs the persistence of radiance, an option the CDH estimator does not offer. The CDH specification instead includes income-by-month fixed effects, which remove movements common to countries at the same income level in the same month. The two sets of estimates therefore agree in sign, timing, and ranking between capitals and other cities, but not in level.

their post-break level while losses keep deepening in secondary cities, suggesting that infrastructure redundancy is higher in capitals than in smaller, peripheral cities.

5.3 Non-random exposure to connectivity shocks

A country's exposure to disruption may not be independent of its growth path. Weak telecom policy can leave a country with both more disruptions and slower growth, which would lead the event study to overstate the losses. Operators may also choose where to lay cables. Caoui and Steck (2025) model that choice as a dynamic entry game in which buyers demand route diversification to hedge against disruption risk, making the number of routes serving a country, and hence its exposure, an equilibrium outcome. I test for such non-random exposure following Borusyak and Hull (2023), reassigning the timing of disruptions at random within countries and re-estimating the baseline specification 500 times over 2008–2019 on the same estimation sample.

Figure 9 reports the two distributions generated by these draws, at the reference horizon of four years and for the six post-disruption horizons taken jointly. Both distributions are centred around zero, so the same countries, disrupted at other dates, show no loss on average. In the data, GDP per capita falls by 5.84% at four years, with a permutation p-value of 0.048, and the joint test over the six horizons gives a p-value of 0.046. The observed losses exceed those obtained in about 95% of the draws. They are therefore unlikely to stem from the characteristics of disruption-prone countries, such as structurally weak telecom policy. The test leaves open a confounder timed with the breaks themselves, which Section 5.4.3 addresses by controlling for key confounding factors such as internet penetration, governance quality, shipping connectivity and natural disasters.

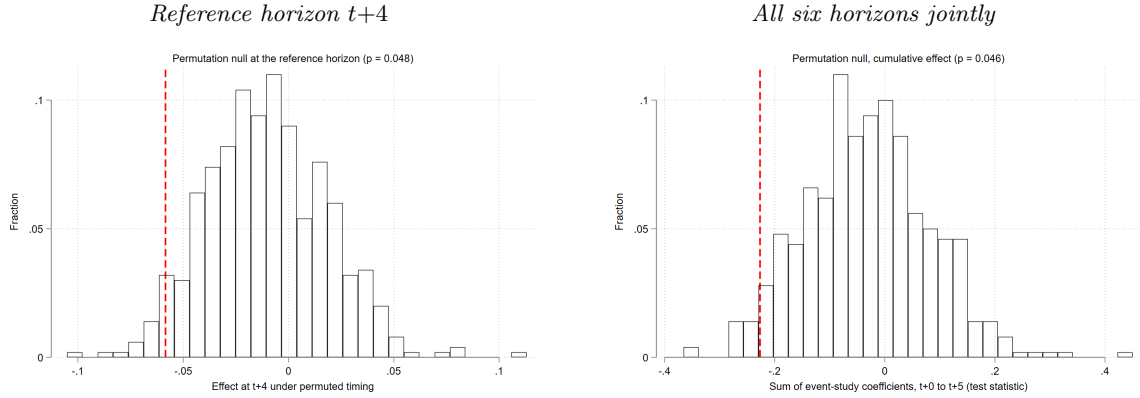


Fig. 9: Permutation test for non-random exposure: real vs. placebo (2008–2019).

Notes: Both panels are based on 500 random re-assignments of cable disruption timing within countries, over the 2008–2019 sample. The outcome is log GDP per capita. *Left panel*: histogram of the permutation null at the reference horizon t+4, against a real effect of -0.058 ($p = 0.048$). *Right panel*: histogram of the permutation null for the sum of event-study coefficients over t to t+5, against a real value of -0.227 ($p = 0.046$). In each panel the red dashed vertical line marks the actual test statistic, and p-values are the share of permutation statistics below it. Specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. Draws are reproducible from a fixed random seed.

While the permutation test concerns the timing of disruptions, Appendix Table A.3.1.1 examines whether countries with more disruptions over 2008–2020 differ in their pre-disruption growth, income, or network position. The number of disruptions is uncorrelated with GDP per capita growth before the first disruption and with the level of GDP per capita, in both the linear and the Poisson specifications.

It increases with population and, in the Poisson specification, with cable connectedness. Exposure thus appears to reflect a country's size and position in the network, not its growth path.[38]

5.4 Sensitivity Analysis

5.4.1 Excluding regions from the sample

A natural concern is that the baseline dynamics might be driven by a single geographic block such as SSA, which concentrates the largest number of disruptions. To probe these potential sources of sample bias, I re–estimate the dynamic specification sequentially excluding each World Bank region from the estimation sample.

Appendix Figure A.3.1.1 reports the resulting event–time coefficients for GDP per capita excluding regions one by one. Pre–trends remain flat in every exclusion, supporting the parallel–trends restriction independent of regional composition. The post–event contraction remains consistent across sub-samples. In fact, coefficients move below zero within two–to–three years after first exposure and continue to decline thereafter, cumulating to medium–run losses of roughly 6–10 percentage points by horizons $[t+5;\, t+6]$, well within the baseline confidence intervals. The trajectory obtained after dropping EAP, LAC, MENA, SA, and more importantly SSA, is very close to the baseline. Dropping SSA also flattens the pre–break path while keeping medium-run losses close to the baseline, which reduces the concern that the baseline estimates reflect an undetected trend. Dropping ECA, which represents about one third of the estimation sample and contains many never-treated countries that serve as controls (Figure 2 and Table 1), leaves the shape and magnitude of the dynamic effects essentially unchanged, but slightly widens confidence intervals.

5.4.2 Leave-one-out estimations

To assess whether the headline estimates are driven by any single influential switcher, I re-estimate each baseline event-study sequentially excluding each treated country from the estimation sample. The resulting distribution of point estimates at horizons t+4 and t+5 using the baseline specification is reported in Appendix Figure A.3.1.4, without group-by-year fixed effect in Appendix Figure A.3.1.5, and adding the excluded outliers (the year 2020 and Vietnam) in Appendix Figure OA.3.4. Across estimations, point estimates remain negative across the entire leave-one-out (LOO) distribution, and the baseline magnitudes are robust in sign and order of magnitude to any single-country exclusion. The most influential switcher are Armenia, Liberia, and Sierra Leone, but no single exclusion moves the point estimate outside the full-sample 95% confidence band.

5.4.3 Controlling for confounding factors.

The baseline dynamic is unchanged when I sequentially add controls for time-varying factors correlated with both disruptions and growth: disaster occurrence (EM-DAT), liner shipping connectivity, institutional quality, ICT regulatory quality, internet penetration, the number of submarine cables and cable path diversity, the real exchange rate, and the net barter terms of trade. Across all specifications the pre-treatment coefficients stay near zero and the post-disruption coefficients decline over the medium run, as in the baseline (Appendix Figure A.3.1.6).

[38] Better-connected countries record more disruptions, yet Section 4 finds their output losses indistinguishable from zero. If a characteristic that raises exposure were driving the estimated losses, the countries most exposed to disruptions should lose more. Among well-connected countries, the opposite holds.

6 Mechanisms

This section explores the mechanisms underlying the persistent growth effects of submarine cable failures.[39] I begin by exploring the trade channel, which is the most directly impacted by international connectivity disruptions, documenting adjustments in export and import components, exporter participation on both the export and import sides using customs transaction data, and showing that previous GDP spillovers are supported by trade diversion patterns toward nearby non-disrupted countries. Then, I test the direct and spillover effects of disruptions on foreign direct investment inflows and total factor productivity growth. Finally, I explore the role of financial frictions as a potential shock amplification force.

6.1 Direct effects on trade components

The most immediate economic adjustment to international connectivity disruptions passes through trade flows. Goods-and-services trade falls after a disruption, by 11% for exports and 17% for imports at five years, although these estimates are imprecise (Appendix Figure A.3.2.1). Interestingly, this trade dynamic hides heterogeneous responses across export and import's components. First, the trade decline is mainly reflected in merchandise exchanges. Goods exports decline strongly and monotonically after a disruption, reaching around 30% from three to five years after the disruption.[40] Goods imports drop earlier, by 14% one year after a disruption, and stand 18% below trend five years after, although less precisely estimated.

Second, service trade declines on both the export and import sides, which follow however distinct dynamics. While service imports and exports stand about 22% below trend five years after a disruption, the import decline is cleaner, setting in one year after the disruption and holding onwards. On the export side, services decline builds more slowly. Service estimates are less precise than for merchandise trade, but point to the same pattern, with an earlier adjustment of imports and a gradual erosion of exports.

Within commercial services, transport and financial services stand out, echoing the mechanisms of Section 2.3. Transport services, which cover freight, port operations and logistics, follow merchandise trade closely, with exports falling by 38% and imports by 18% after five years. Part of this co-movement is mechanical, as lower goods trade reduces demand for freight, but this evidence is consistent with disrupted logistics weighing on merchandise exchanges. Financial services fall by 51% for exports and 36% for imports after five years, in line with payment, settlement and trade-finance frictions. By contrast, ICT service exports appear unresponsive to cable disruptions.

6.2 Import and export extensive margins

This section asks whether the aggregate trade response has a firm- and partner-level counterpart, using the World Bank Exporter Dynamics Database (EDD).[41] I measure a country's exporter extensive margin from the domestic-exporter side, recording the entry, exit, survival, and stock of incumbent domestic exporters. To measure the import side, I exploit the same panel from the destination's perspective, treating firms that export *into* a market as its foreign exporters. This yields three proxies for a country's extensive margin in import capacity: the total number of foreign exporters serving it,

[39] Since cable breaks have been identified in Section 5 as the main drivers of growth responses, I use this cable disruption definition as the preferred (because cleaner) treatment variable across this section.

[40] For comparison, Steinwender (2018) estimates that permanently eliminating information frictions via the transatlantic telegraph raised bilateral cotton trade volumes by 37%.

[41] The EDD is an unbalanced panel of about 30 exporting countries over 2008–2014; coverage is uneven across countries and years, generating attrition.

the number of foreign exporters per observed origin market, and the number of distinct origin countries with at least one observed exporter. These variables do not capture the full universe of foreign exporters and origin countries serving each destination, and so, do not capture potential changes in the structure of the suppliers basket. However, in addition to providing an import capacity perspective, they offer broader country and time coverage than the panel's origin-exporter side. Estimates of the direct and spillover effects of cable disruptions on import and export extensive margins are reported in Appendix Table A.3.2.1.

Starting with the import side, estimates shows that disruptions narrow the foreign supplier base (Panel A1). The average number of foreign exporters per partner falls by 35.8 on impact (11% of the mean) and by 84.6 at t+3 (27%). In levels the total count falls by 15,642 at t+3, and a destination loses 4.4 distinct exporter countries. The aggregate import capacity decline thus reflects the withdrawal of foreign exporters and a narrowing supplier base, not merely lower volumes per existing partner, consistent with information frictions shaping the set of trading partners (Akerman et al. 2022) and with importers withdrawing attention from markets that become costlier to monitor (Dasgupta and Mondria 2018). On the export side, the disrupted country's own exporter base contracts in parallel, with the exit rate rising by 15 pp at t+1 and surviving entrants falling by 170 on impact (Panel A2).

Non-disrupted regional peers show the mirror adjustment (Panels B1–B2). Incumbents expand, exits decline, the domestic exporter pool rises by 366–733 at t+2, and peers gain 2–5 distinct exporter countries on impact. From the firm perspective, this corroborates the aggregate service trade diversion documented in the following section.

6.3 Service trade diversion

I next test whether the growth spillovers come with trade diversion, testing whether cable breaks raise exports in unaffected neighbours (Appendix Figures A.3.2.4 and A.3.2.5). Spillovers operate almost only through services. Merchandise exports of unaffected countries rise briefly and then turn insignificant, whereas total and commercial service exports rise and settle about 5% above their pre-event level after four years. The response concentrates in financial and insurance services, the most bandwidth-intensive category, which roughly double over the medium run. Transport and ICT services rise by about +15% and +25% over the medium run.

Together with the large losses in directly exposed countries, this evidence is consistent with regional reallocation of digitally intensive activity, with unaffected neighbors capturing part of the displaced demand for high-bandwidth services. The next subsections turn to competitiveness and financial frictions as the mechanisms behind both sides of this reallocation.

6.4 Productivity and FDI: direct and spillover effects

Additional evidence support that some economic activity moves to affected neighbours. Appendix Figure A.3.2.6 reports the effects of cable disruptions on FDI inflows per capita and TFP growth for disrupted countries, and the spillovers on their non-disrupted neighbours.[42] In disrupted countries both series fall, though not in the reliable confidence level. TFP growth is 3.9 percentage points below trend at five years and FDI inflows decline over the same horizon. In neighbouring countries both rise significantly. TFP growth stands 1.1 percentage points above trend in the year of a regional disruption and 3.0 points at five years, jointly significant at the 2% level. FDI inflows rise as well, the coefficient reaching 0.4 in the year of the disruption and 2.1 at five years.

[42] Variables are drawn from the World Bank Global Productivity Database (Dieppe 2021). I report 90% confidence intervals given the imprecision of factor-productivity measures, which may not fully capture informal activity in lower-income economies.

6.5 Financial channels

Digital infrastructure underpins payment systems and information flows within and across borders (D'Andrea and Limodio 2024; D'Andrea et al. 2025), so connectivity losses that tighten domestic and cross-border finance depress competitiveness and productivity through the channels discussed in Section 2. On the domestic side, credit to non-financial agents falls after a disruption and the contraction deepens over the following years (Appendix Figure A.3.3.1).[43] The decline is driven by credit to private non-financial actors, while credit to general government follows a smoother path. Pre-trends are flat, and leave-one-out exclusions show that no single economy drives the result (Appendix Figure A.3.3.2).

On the external side, cross-border banking positions decline gradually after a disruption, with wider confidence intervals at longer horizons (Appendix Figure A.3.3.3).[44] The two results together indicate that financial frictions operate through both domestic and cross-border intermediation.

7 Conclusion

Submarine-cable disruptions impose large and persistent output losses. Within four to five years, GDP per capita falls 4–6% below the pre-disruption path, depending on the fixed-effect specification, while non-disrupted regional neighbours gain 0.5 to about 1% from the diversion of trade, investment, and bandwidth-intensive services. The losses concentrate in low-connectivity economies, that is, countries with few connected partners, a sparse cable network, held by few owners, and with no Internet exchange point. Economies with re-routing capacity and diversified ownership absorb the same shock with effects close to zero. Averaged over the disrupted economies, whose mean income is $11,757, the loss reaches $118 per head on impact and $564 to $694 at the four-to-five-year trough. In the typical disrupted country, at the median income of $3,512, it reaches $35 and $169 to $207.

The persistence and magnitude of these losses point to a set of reinforcing mechanisms that extend well beyond the immediate interruption of connectivity. First, the channel moving earliest and farthest is merchandise trade. Goods imports drop soon after a disruption and keep declining thereafter, as foreign suppliers withdraw, while the fall in goods exports sets in three years after the disruption, as the domestic exporter base contracts. Transport and financial services, which provide the logistics and financing of merchandise trade, decline alongside, with the largest losses among service exports. Second, cable disruptions undermine competitiveness, pulling down productivity growth and foreign investment in disrupted countries and shifting them toward their neighbours. These supply-side effects weaken firms' ability to compete in international markets and are accompanied by the diversion of trade toward unaffected neighbouring countries. Third, shocks are amplified by a tightening of domestic credit to non-financial actors, including in relatively large and financially developed economies, as well as a gradual weakening of cross-border banking relationships.

Taken together, these results support a policy agenda centered on reducing exposure to SMC failures and accelerating repairs when they occur. First, building *redundancy* through more diverse landing sites, partner routes, and cable ownership, and avoiding shared choke points, reduces the probability that a single break isolates a country. Second, densifying *IXPs* and domestic peering localizes traffic and facilitates rapid fail-over when international links are impaired. Third, shortening *repair queues* through adequate maintenance agreements and fleet renewal reduces outage duration. Because

[43] BIS data, within-country z-scores winsorized at the 90th percentile. I use z-scores rather than logs because credit stocks are far higher in the six switchers than in non-switchers (Online Appendix Figure A.3.3.2); the variable distributions are in Online Appendix Figure A.3.3.4.

[44] BIS Locational Banking Statistics, non-bank loans and deposits per capita.

the losses concentrate in economies with thin route diversity, few IXPs, and concentrated cable ownership, whatever their income level, these investments are a question of development finance as much as one of national and global security, and fall within the remit of the World Bank, Regional Economic Communities, and national regulators. This matters all the more as new capacity is increasingly built as single-owner cables by a few large content providers, and as economies turn to single-owned satellite networks such as Starlink for backup. Private owners optimizing their own traffic need not internalize the value of route diversity for the economies they serve, so that redundancy is likely underprovided, a gap that public coordination can correct and that future research should investigate.

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Appendix

A.1 Sample statistics and additional estimations

A.1.1 Sample statistics

Table A.1.1.1: Estimation sample countries by World Bank region

East Asia and Pacific					
AUS	FJI	IDN	JPN	KHM	KIR
MMR	MNG	NZL	PLW	PNG	SLB
THA	TON	TUV	VUT	WSM	
Europe and Central Asia					
ALB	AND	ARM	AUT	AZE	BEL
BGR	BLR	CHE	CYP	CZE	DEU
DNK	ESP	EST	FIN	FRA	GBR
GEO	GRC	GRL	HRV	HUN	IRL
ISL	ITA	KAZ	KGZ	LTU	LUX
LVA	MCO	MDA	MKD	NLD	NOR
POL	PRT	RUS	SMR	SVK	SVN
SWE	TJK	TKM	TUR	UKR	UZB
Latin America and Caribbean					
ABW	ARG	ATG	BHS	BLZ	BRA
BRB	CHL	COL	CRI	CUB	CYM
DMA	ECU	GRD	GTM	GUY	HND
JAM	KNA	LCA	MEX	NIC	PAN
PER	PRI	PRY	SLV	SUR	TTO
URY	VCT				
Middle East and North Africa					
DZA	IRQ	JOR	LBY	MAR	TUN
North America					
CAN	USA				
South Asia					
BTN	MDV				
Sub-Saharan Africa					
AGO	BDI	BFA	BWA	CAF	CIV
CMR	COG	COM	CPV	GAB	GHA
GIN	GMB	GNB	GNQ	KEN	LSO
MDG	MLI	MOZ	MRT	MUS	MWI
NAM	RWA	SDN	SEN	SLE	SOM
STP	SWZ	SYC	TCD	TZA	UGA
ZAF	ZWE				

Notes: Regions follow World Bank's region classification. Sample corresponds to estimates reported in Figure 5.

Table A.1.1.2: **Summary statistics of cable failure variables**

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Panel A: Overall sample statistics (187 countries)						
Cable disruptions (breaks + outages)	0.074	0.304	0.268	0	3	2,244
Cable breaks	0.057	0.261	0.233	0	3	2,244
Cable outages	0.016	0.144	0.134	0	2	2,244
Repair days	0.629	4.296	3.974	0	80	2,237
Panel B: Treated countries, sample statistics (71 countries)						
Cable disruptions	1.137	0.460	0.342	0.085	3	146
Cable breaks	0.884	0.566	0.402	0	3	146
Cable outages	0.253	0.509	0.349	0	2	146
Repair days	10.130	14.216	8.367	1	80	139
Panel C: Annual sums, 2008–2020						
Cable disruptions	20.08	16.52	–	8	69	13
Cable breaks	14.92	15.44	–	6	65	13
Cable outages	3.15	4.26	–	0	16	13
Repair days	202.23	191.89	–	67	754	13
Panel D: Annual sums, 2008–2019						
Cable disruptions	16.00	7.87	–	8	37	12
Cable breaks	10.75	3.62	–	6	17	12
Cable outages	3.08	4.44	–	0	16	12
Repair days	156.25	100.93	–	67	432	12

Notes: Cable disruptions sums the number of cable breaks and cable outages. Cables breaks refer to physical breaks, while outages refer to cable disability due to technical faults, such as repeater failures. The average repair days per disruption (total repair days ÷ total disruptions, whole sample): 8.482.

Table A.1.1.3: **Summary statistics of cable failure variables, estimation sample.**

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Panel A: Overall panel statistics (country–year)						
Cable disruptions	0.045	0.245	0.228	0	3	1,804
Cable breaks	0.032	0.201	0.191	0	3	1,804
Cable outages	0.014	0.135	0.126	0	2	1,804
Repair days	0.375	3.546	3.385	0	80	1,802
Panel size: Countries (n) = 164; years per country = 11						
Panel B: Annual sums, estimation sample						
Cable disruptions	7.73	6.72	–	1	26	11
Cable breaks	5.45	3.80	–	1	13	11
Cable outages	2.27	3.74	–	0	13	11
Repair days	61.55	45.37	–	0	135	11

Notes: Estimation sample, 2008–2019 excluding Vietnam. Annual sums are totals across countries within each year. Cable disruptions sums the number of cable breaks and cable outages. Cables breaks refer to physical breaks, while outages refer to cable disability due to technical faults, such as repeater failures. The average repair days per disruption (total repair days ÷ total disruptions): 8.232.

Table A.1.1.4: Repair days by incidence of cable disruptions

Disrupt. count	Mean	Median	Std. Dev.	Freq.
		All countries		
1	9.23	5.5	13.12	116
2	13.43	11.0	17.41	21
3	27.50	27.5	33.23	2
Total	10.13	6.0	14.22	139
		Estimation sample		
1	9.89	2.0	16.75	55
2	10.75	10.5	9.40	12
3	4.00	4.0	–	1
Total	9.96	2.0	15.53	68

Notes: Repair days are SMC repair days per country-year, among country-years with at least one cable disruption. The thirteen countries whose December 2008 event is split across 2008 and 2009 carry fractional disruption counts; they are grouped with one disruption rather than rounded to zero.

Table A.1.1.5: Macroeconomic outcomes, treatments and regional peer measures: summary statistics, estimation sample 2008–2019.

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
GDP per capita growth (%)	1.661	4.245	3.735	-48.43	24.64	1,944
GDP per capita (USD, const 2015)	15,165	22,719	2,043	253	185,583	1,945
GDP (USD, const 2015)	3.15e11	1.47e12	1.12e11	3.05e7	2.02e13	1,945
Private consumption (USD pc)	7,697	9,184	654.7	198	44,875	1,568
Government consumption (USD pc)	2,812	4,017	206.0	25.2	21,775	1,569
Gross fixed capital formation (USD pc)	3,368	4,573	1,076	21.6	38,835	1,548
Exports of goods and services (USD pc)	8,035	20,103	2,796	25.6	214,390	1,583
Imports of goods and services (USD pc)	7,324	16,209	2,420	75.1	179,347	1,583
Log GDP per capita (const 2015 USD)	8.738	1.386	0.103	5.53	12.13	1,957
Cable disruptions (count)	0.043	0.237	0.222	0	3	1,980
Cable repair days	0.342	3.387	3.247	0	80	1,977
Regional peers disrupted (count)	1.961	2.652	2.043	0	11	1,980
Regional peer disruption episodes	2.484	3.514	2.806	0	14	1,980

Panel size: Countries (n) = 165; years per country = 12 (2008–2019)

Table A.1.1.6: Mechanism outcomes: summary statistics, estimation sample.

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Private consumption pc (const USD)	7,701	9,185	629	198	44,875	1,445
Government consumption pc (const USD)	2,813	4,021	188	30	21,775	1,446
Gross fixed capital formation pc (const USD)	3,352	4,565	1,092	22	38,835	1,426
Exports of goods and services pc (const USD)	8,111	20,262	2,741	26	214,390	1,457
Imports of goods and services pc (const USD)	7,380	16,355	2,434	75	179,347	1,457
Net services balance (const USD billions)	1.06	52.4	9.44	-593.0	262.0	1,546
Merchandise exports (const USD billions)	81.1	201.0	23.5	0.00	1,600.0	1,419
BoP services exports (const USD billions)	29.4	83.3	14.6	0.01	870.0	1,364
Commercial services exports (const USD billions)	28.9	81.6	14.6	0.00	848.0	1,364
Financial services exports (const USD billions)	4.02	16.70	2.15	0.00	156.00	1,270
Transport services exports (const USD billions)	4.95	11.70	1.64	0.00	89.60	1,341
ICT services exports (const USD billions)	2.72	8.57	3.31	0.00	136.00	1,317
Merchandise imports (const USD billions)	86.7	240.0	30.4	0.06	2,580.0	1,419
BoP services imports (const USD billions)	25.7	64.0	13.7	0.02	603.0	1,364
Commercial services imports (const USD billions)	25.3	62.4	13.7	0.02	579.0	1,364
Financial services imports (const USD billions)	2.42	8.90	1.19	0.00	96.00	1,342
Transport services imports (const USD billions)	5.22	12.10	1.61	0.02	113.00	1,351
FDI inflows pc (const USD)†	7,195	106,746	90,958	-2,280,293	2,144,355	1,715
TFP growth (%)	0.076	4.094	3.753	-44.108	61.404	1,056
Labor productivity (real, const USD)	6,008	26,621	1,308	4	244,363	774
Labor reallocation contribution	1.216	3.757	3.281	-36.573	20.932	1,360

Notes: Estimation sample (realized CDH sample), 2008–2019. †FDI inflows pc range reflects a small number of tiny-population financial-center economies with very large absolute FDI flows; the regression uses the inverse hyperbolic sine of this variable to accommodate the skew and sign changes.

A.2 Detailed estimation tables

A.2.1 First-stage estimation tables

This section reports the tables underlying the first-stage check of Section 3.4: the verified-onset episodes used in the daily event study (Table A.2.1.1) and the country-month panel estimates (Table A.2.1.2).

Table A.2.1.1: Verified-onset disruption episodes used in the first-stage event study

Onset date	Cable system(s)	Countries (n)	Examples
23 Jan 2008	FALCON	2	Iran, India
30 Jan 2008	SEA-ME-WE 4, FLAG	9	Egypt, Pakistan, India, UAE
01 Feb 2008	FALCON	2	Oman, UAE
19 Dec 2008	SMW-3/SMW-4/FLAG/GO-1	15	Egypt, Saudi Arabia, India, Taiwan
10 Jul 2009	SAT-3	4	Nigeria, Benin, Togo, Niger
17 Aug 2009	APCN-2	6	China, Hong Kong, Taiwan, Philippines
14 Apr 2010	SEA-ME-WE 4	7	Egypt, India, UAE, France, Italy
22 Mar 2013	SEACOM	2	Mozambique, Tanzania
27 Mar 2013	SEA-ME-WE 4	5	Egypt, Saudi Arabia, India, Pakistan
23 Apr 2015	AAG	3	Vietnam, Hong Kong, Singapore
18 Jun 2017	MainOne	2	Nigeria, Cameroon
03 Dec 2017	SEA-ME-WE 3	3	Australia, Singapore, Vietnam
30 Mar 2018	ACE	10	Mauritania, Sierra Leone, Liberia, Senegal
09 Jan 2020	FALCON, FLAG-FEA	7	Yemen, Sudan, Tanzania, Saudi Arabia
16 Jan 2020	SAT-3, WACS	16	Nigeria, South Africa, Angola, Gabon
20 Feb 2020	ECFS	6	Antigua, Dominica, Grenada, St Kitts
14 Mar 2020	EASSy	13	Egypt, Kenya, Tanzania, Djibouti
Total (RIPE RIS coverage)		**107 country-events**	62 distinct countries

Notes: Documented multi-country submarine-cable disruptions whose onset date is web-verified from contemporaneous operator and press sources, and used in the daily event study of Section 3.4. A country enters once per episode that affects it. "Countries (n)" is the coded scope. A few country-events without RIPE RIS coverage (Ethiopia, Vietnam in 2015 and 2017, Anguilla, Montserrat) are dropped, leaving 107 country-events across 62 countries.

Table A.2.1.2: First stage: TWFE estimates, country-month panel

Outcome (country-month)	Two-way FE coef. (SE)
Outage days	0.792^{***} (0.234)
Outage episodes	0.130^{***} (0.031)
Severity ($\sum$ |drawdown|)	32.82^{***} (9.86)
Worst-month drawdown (pp)	-6.29^{***} (1.43)
Treatment	SubTel cable disruptions (count, per country-month)
Fixed effects	country, year-month
Economies / Obs.	172 / 26,288

Notes: All outcomes are within-country measures of routed-prefix degradation (RIPE RIS), where a day's *drawdown* is the percentage shortfall of routed prefixes relative to their trailing 60-day baseline. *Outage days* = number of days in the month with drawdown below -10%. *Outage episodes* = number of distinct runs of such days (gaps $\leq$ 3 days merged). *Severity* = sum of |drawdown| over the month's outage days (depth $\times$ duration). *Worst-month drawdown* = the month's deepest daily drawdown (percentage points, negative). The column is a two-way fixed-effects regression on the SubTel cable-disruption count, SE clustered by country. The heterogeneity-robust dynamic effects of de Chaisemartin and d'Haultfoeuille (2024) on the same panel are plotted in Figure 4(c)–(d) and reported in full, with placebos, in Table A.2.1.3. A complementary Poisson (PPML) specification for the episode count gives a coefficient of 1.04 on the disruption count ($z = 6.8$). The monthly mean routed level shows no statistically significant change at any horizon, consistent with routed connectivity returning to trend after the acute withdrawal. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.2.1.3: First stage: CDH dynamic estimates, country-month panel

Event-time	Worst-month drawdown (pp)	Outage episodes	Outage days	Severity ($\sum$ |drawdown|)
Effect_1	-10.03^{***} (2.73)	0.163^{***} (0.056)	0.85^{*} (0.50)	53.09^{**} (25.34)
Effect_2	-3.46 (2.15)	0.039 (0.053)	1.87^{**} (0.88)	52.86^{**} (25.60)
Effect_3	-1.80 (1.59)	0.057 (0.048)	0.44 (0.52)	8.42 (11.82)
Effect_4	0.41 (1.44)	-0.01 (0.049)	0.16 (0.52)	0.25 (10.71)
Effect_5	0.27 (1.14)	0.019 (0.041)	-0.13 (0.36)	-2.86 (6.80)
Effect_6	—	-0.063 (0.042)	-0.44 (0.35)	-8.93 (8.00)
Placebo_1	0.38 (1.75)	0.010 (0.063)	-0.12 (0.41)	-1.52 (9.49)
Placebo_2	-0.76 (1.46)	-0.011 (0.038)	0.39 (0.57)	7.26 (12.26)
Placebo_3	-0.58 (1.37)	0.013 (0.049)	0.55 (0.57)	11.12 (14.19)
Joint p (effects)	0.004	<0.001	0.002	0.028
Joint p (placebos)	0.894	0.944	0.454	0.806

Notes: Dynamic effects from the CDH estimator, using a 172-country country-month panel, with treatment the SubTel cable-disruption count, and standard errors clustered by country in parentheses. Effect_ℓ is the normalized effect ℓ months into a disruption and Placebo_ℓ the corresponding pre-trend lead; the worst-month drawdown is estimated over five post-periods and the other outcomes over six. Outcomes are defined as in Table A.2.1.2. 77 switchers contribute to the effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.2.2 Baseline estimations

The baseline sample runs 2008–2019 and excludes the pandemic year 2020, and Vietnam, which records an exceptionally large number of cable failures and repair days. The year 2020 coincides with the COVID-19 traffic surge, while Vietnam is repeatedly exposed to cable disruption due to recurrent faults on the AAG cable system. Estimates that add 2020 and/or Vietnam back, add Vietnam back are reported in Appendix OA.3. This appendix reports the detailed event-study and placebo estimates behind the baseline, the average effects over alternative horizon windows, and the same estimates without group-by-year fixed effects. Leave-one-out stability for this specification is reported together with the other leave-one-out checks in Appendix Section A.3.1 and in Online Appendix Figure OA.3.4.

Table A.2.2.1: Treatment paths at the estimation window $[t-1; t+5]$, baseline specification.

	#Groups	%Groups	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$	$\ell = 6$
Path 1	12	54.55	0	1	0	0	0	0	0
Path 2	3	13.64	0	1	0	0	0	1	2
Path 3	3	13.64	0	2	0	0	0	0	0
Path 4	1	4.55	0	1	0	0	0	0	2
Path 5	1	4.55	0	1	0	0	1	1	0
Path 6	1	4.55	0	1	2	0	0	0	1
Path 7	1	4.55	0	1	2	1	0	0	1

Notes: Treatment paths detected by the CDH estimator's path-detection diagnostic for the baseline specification: log GDP per capita on the cable-disruption count, income-group-by-year fixed effects, 2008–2019 excluding Vietnam. $\ell = k$ is the treatment value k periods after the first switch. The seven paths cover 100% of the 22 switching groups with a fully observed six-period window; path 1 alone covers 54.6%. Panels for repair days and for dropping 2020 are no longer reported: the baseline already excludes 2020, and treatment paths are detected for the disruption count only.

Table A.2.2.2: **Cable failures and GDP per capita (const USD, ln) — event-study estimates and placebo tests, no group × year FEs, 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects						
Effect_1	-0.0086666	0.0050241	-0.0185137	0.0011804	1444	49
Effect_2	-0.0183263	0.0098445	-0.0376211	0.0009685	1272	37
Effect_3	-0.0240353	0.0154781	-0.0543719	0.0063012	1110	29
Effect_4	-0.0428873	0.0186043	-0.0793511	-0.0064235	954	24
Effect_5	-0.0476756	0.0211534	-0.0891355	-0.0062157	807	23
Effect_6	-0.0459741	0.0208434	-0.0868265	-0.0051217	666	22
Sup-t test of joint nullity of the effects: p-value = 0.0635						
Placebo leads						
Placebo_1	-0.0005918	0.0042219	-0.0088665	0.0076829	1281	46
Placebo_2	-0.0139090	0.0109957	-0.0354602	0.0076421	958	28
Placebo_3	-0.0199292	0.0249508	-0.0688318	0.0289734	662	18
Sup-t test of joint nullity of the placebos: p-value = 0.3926						
Panel B: Cable repair days						
Event-study effects						
Effect_1	-0.0007800	0.0004773	-0.0017156	0.0001556	1308	47
Effect_2	-0.0016329	0.0009505	-0.0034958	0.0002301	1142	36
Effect_3	-0.0021055	0.0014643	-0.0049754	0.0007645	988	28
Effect_4	-0.0039710	0.0019585	-0.0078095	-0.0001325	844	23
Effect_5	-0.0047783	0.0024198	-0.0095210	-0.0000356	811	23
Effect_6	-0.0052769	0.0028170	-0.0107982	0.0002443	670	22
Sup-t test of joint nullity of the effects: p-value = 0.1047						
Placebo leads						
Placebo_1	-0.0000217	0.0003534	-0.0007144	0.0006709	1145	44
Placebo_2	-0.0011182	0.0008602	-0.0028041	0.0005677	828	27
Placebo_3	-0.0013413	0.0016451	-0.0045657	0.0018830	540	17
Sup-t test of joint nullity of the placebos: p-value = 0.3688						

Notes: Estimator: the CDH (2024) dynamic DID estimator, normalized for the non-binary treatment, six event-study effects and three placebo leads, on the 2008–2019 sample excluding Vietnam, with no group-by-year fixed effects at all (neither income-group-by-year nor region-by-year). Effect_k is the effect at $t + (k - 1)$. Specifications include country and year fixed effects and country-specific linear trends; standard errors are clustered at the country level. The joint test is the conservative sup-t test, which draws $\max |t|$ from a multivariate t with $G - 1$ degrees of freedom and never inverts the covariance matrix; it replaces the package's F-test, which is flagged as unreliable when the variance of the effects is near-singular. The six-horizon specification is used throughout; the seven-horizon version for GDP per capita in level is not reported, since its F-test is unreliable for the same reason.

Table A.2.2.3: **Cable failures and GDP per capita (const USD, ln) — event-study estimates and placebo tests, income-group × year FEs, 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects						
Effect_1	-0.0094485	0.0046550	-0.0185721	-0.0003250	899	49
Effect_2	-0.0202079	0.0091207	-0.0380842	-0.0023316	783	37
Effect_3	-0.0294453	0.0144160	-0.0577002	-0.0011904	633	29
Effect_4	-0.0512806	0.0176916	-0.0859554	-0.0166057	537	24
Effect_5	-0.0582885	0.0214418	-0.1003136	-0.0162633	499	23
Effect_6	-0.0574203	0.0224736	-0.1014676	-0.0133729	413	22
Sup-t test of joint nullity of the effects: p-value = 0.0181						
Placebo leads						
Placebo_1	-0.0017559	0.0044690	-0.0105151	0.0070033	853	46
Placebo_2	-0.0165349	0.0116792	-0.0394257	0.0063558	623	28
Placebo_3	-0.0306726	0.0253189	-0.0802968	0.0189516	454	18
Sup-t test of joint nullity of the placebos: p-value = 0.3076						
Panel B: Cable repair days						
Event-study effects						
Effect_1	-0.0008696	0.0004971	-0.0018438	0.0001046	881	47
Effect_2	-0.0018321	0.0010071	-0.0038061	0.0001418	766	36
Effect_3	-0.0025630	0.0015682	-0.0056367	0.0005107	620	28
Effect_4	-0.0047316	0.0021561	-0.0089575	-0.0005058	527	23
Effect_5	-0.0058076	0.0026893	-0.0110785	-0.0005367	502	23
Effect_6	-0.0065420	0.0031634	-0.0127421	-0.0003419	417	22
Sup-t test of joint nullity of the effects: p-value = 0.0764						
Placebo leads						
Placebo_1	-0.0001419	0.0004011	-0.0009281	0.0006443	835	44
Placebo_2	-0.0014357	0.0009706	-0.0033380	0.0004667	606	27
Placebo_3	-0.0021383	0.0017981	-0.0056625	0.0013860	441	17
Sup-t test of joint nullity of the placebos: p-value = 0.2757						

Notes: Estimator: the CDH (2024) dynamic DID estimator, normalized for the non-binary treatment, six event-study effects and three placebo leads, on the 2008–2019 sample excluding Vietnam. Effect$_k$ is the effect at $t+(k-1)$. Specifications include country and year fixed effects and country-specific linear trends; standard errors are clustered at the country level. The joint test is the conservative sup-t test, which draws max $|t|$ from a multivariate t with $G-1$ degrees of freedom and never inverts the covariance matrix; it replaces the package's F-test, which the command itself flags as unreliable when the variance of the effects is near-singular.

Table A.2.2.4: **Cable failures and GDP per capita (const USD, ln) — event-study estimates and placebo tests, region × year FEs, 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects						
Effect_1	-0.0098528	0.0049588	-0.0195719	-0.0001337	570	49
Effect_2	-0.0196173	0.0104360	-0.0400716	0.0008370	419	37
Effect_3	-0.0258410	0.0157765	-0.0567623	0.0050804	363	29
Effect_4	-0.0464985	0.0189839	-0.0837064	-0.0092907	315	24
Effect_5	-0.0516688	0.0228490	-0.0964521	-0.0068856	287	23
Effect_6	-0.0499916	0.0238346	-0.0967067	-0.0032766	262	22
Sup-t test of joint nullity of the effects: p-value = 0.0503						
Placebo leads						
Placebo_1	-0.0020119	0.0046409	-0.0111079	0.0070840	518	46
Placebo_2	-0.0161712	0.0128164	-0.0412908	0.0089484	284	28
Placebo_3	-0.0213653	0.0265733	-0.0734480	0.0307173	200	18
Sup-t test of joint nullity of the placebos: p-value = 0.3994						
Panel B: Cable repair days						
Event-study effects						
Effect_1	-0.0009005	0.0005020	-0.0018844	0.0000833	544	47
Effect_2	-0.0017891	0.0010534	-0.0038536	0.0002754	399	36
Effect_3	-0.0022330	0.0015877	-0.0053449	0.0008789	350	28
Effect_4	-0.0042415	0.0021635	-0.0084819	-0.0000010	305	23
Effect_5	-0.0051544	0.0027353	-0.0105156	0.0002067	290	23
Effect_6	-0.0057007	0.0032570	-0.0120843	0.0006830	266	22
Sup-t test of joint nullity of the effects: p-value = 0.1237						
Placebo leads						
Placebo_1	-0.0001522	0.0004318	-0.0009985	0.0006940	492	44
Placebo_2	-0.0013775	0.0010510	-0.0034375	0.0006824	264	27
Placebo_3	-0.0014030	0.0018290	-0.0049877	0.0021817	187	17
Sup-t test of joint nullity of the placebos: p-value = 0.3715						

Notes: Estimator: the CDH (2024) dynamic DID estimator, normalized for the non-binary treatment, six event-study effects and three placebo leads, on the 2008–2019 sample excluding Vietnam. Effect$_k$ is the effect at $t+(k-1)$. Specifications include country and year fixed effects, country-specific linear trends, and region-by-year fixed effects (no income-group-by-year fixed effects), with standard errors clustered at the country level. The joint test is the conservative sup-t test, which draws $\max|t|$ from a multivariate t with $G-1$ degrees of freedom and never inverts the covariance matrix; it replaces the package's F-test, which the command itself flags as unreliable when the variance of the effects is near-singular.

Table A.2.2.5: **Cable failures and GDP per capita (const USD, ln), income-group-by-region-by-year fixed effects: event-study estimates and placebo tests, 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects						
Effect_1	-0.0098088	0.0053548	-0.0203040	0.0006865	369	46
Effect_2	-0.0181098	0.0112242	-0.0401089	0.0038893	274	35
Effect_3	-0.0254702	0.0165085	-0.0578263	0.0068859	236	28
Effect_4	-0.0501709	0.0208019	-0.0909418	-0.0094000	196	23
Effect_5	-0.0588331	0.0265186	-0.1108087	-0.0068575	180	22
Effect_6	-0.0596226	0.0296285	-0.1176934	-0.0015517	163	21
Sup-t test of joint nullity of the effects: p-value = 0.0547						
Placebo leads						
Placebo_1	0.0017957	0.0059778	-0.0099206	0.0135120	340	43
Placebo_2	-0.0081694	0.0153540	-0.0382628	0.0219240	190	26
Placebo_3	-0.0222304	0.0301470	-0.0813173	0.0368566	137	17
Sup-t test of joint nullity of the placebos: p-value = 0.7279						
Panel B: Cable repair days						
Event-study effects						
Effect_1	-0.0008101	0.0005349	-0.0018584	0.0002383	357	44
Effect_2	-0.0016128	0.0011703	-0.0039065	0.0006808	264	34
Effect_3	-0.0021229	0.0017546	-0.0055619	0.0013161	230	27
Effect_4	-0.0044678	0.0024542	-0.0092780	0.0003424	192	22
Effect_5	-0.0056288	0.0031494	-0.0118016	0.0005439	183	22
Effect_6	-0.0064978	0.0038298	-0.0140040	0.0010085	167	21
Sup-t test of joint nullity of the effects: p-value = 0.1561						
Placebo leads						
Placebo_1	0.0001233	0.0005046	-0.0008657	0.0011123	328	41
Placebo_2	-0.0008250	0.0012152	-0.0032067	0.0015568	180	25
Placebo_3	-0.0015191	0.0020758	-0.0055875	0.0025494	131	16
Sup-t test of joint nullity of the placebos: p-value = 0.7296						

Notes: Estimator: the CDH (2024) dynamic DID estimator, normalized for the non-binary treatment, six event-study effects and three placebo leads, on the 2008–2019 sample excluding Vietnam. Effect$_k$ is the effect at $t+(k-1)$. Specifications include country and year fixed effects, country-specific linear trends, and the interaction of income-group-by-year and region-by-year fixed effects (17 income×region cells; two cells reduce to a single switching country, Seychelles and Haiti, which therefore drop from the estimation), with standard errors clustered at the country level. The joint test is the conservative sup-t test, which draws $\max|t|$ from a multivariate t with $G-1$ degrees of freedom and never inverts the covariance matrix; it replaces the package's F-test, which the command itself flags as unreliable when the variance of the effects is near-singular.

A.2.3 Monthly evidence from the ACE cable

A.2.3.1 Construction of the night-time-light outcome

The outcome is built from the VIIRS Day/Night Band monthly composite (NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG), a stray-light-corrected night-time radiance product, extracted through Google Earth Engine. Night-time radiance is measured separately for each country's economic capital and for its other cities of more than 20,000 inhabitants. For each country, I construct two monthly radiance series, one for the economic capital and one for the other cities, both log-transformed and deseasonalized. Each country's capital is represented by one city, its largest by population when administrative capital is not the demographic center (so Abidjan stands for Côte d'Ivoire and Lagos for Nigeria rather than Yamoussoukro or Abuja). For the capital, radiance is averaged over a ten-kilometre disc around the city centroid, weighting each pixel by the population living there. For the other cities, radiance is averaged over the same disc without weights, and the city averages are then combined using city population as weights. Each series is deseasonalized by subtracting its calendar-month mean and adding back its overall mean, which preserves its level.

Treatment timing and intensity follow the same SubTel Forum coding as the annual database of Online Appendix OA.1.1, read at monthly frequency. Cloud-free observation counts and aerosol optical depth are extracted alongside radiance and used as placebo outcomes, to check that the estimated response is not an artifact of weather-driven changes in satellite visibility.

A.2.3.2 Summary statistics

Table A.2.3.1 reports summary statistics for the outcome used in Section 5.2, log population-weighted radiance density within ten kilometres of the city centre, over the nineteen-country ACE panel (January 2016–December 2019).

Table A.2.3.1: Summary statistics: night-time radiance, ACE panel (10 km ring).

	Mean	Std. Dev.	Min	Max	Obs.
Overall (19 countries)	2.456	1.109	-0.692	4.709	912
Treated (disrupted, lands ACE)	1.965	1.031	-0.692	3.461	480
Comparison, lands ACE, not disrupted	2.453	0.553	1.465	3.496	192
Comparison, landlocked, untreated	2.406	0.347	1.717	2.840	96
Comparison, far placebo	4.132	0.362	3.555	4.709	144

Notes: The outcome is the logarithm of population-weighted radiance density, VIIRS stray-light-corrected monthly composite, within ten kilometres of each sampled city's centre. Panel of 19 countries observed monthly from January 2016 to December 2019 (48 months), one city per country. "Treated" countries land the ACE cable and are disrupted by the break of 30 March 2018, three of them again by the 2019 faults (10 countries); the three comparison groups are countries that land ACE but were not disrupted, untreated landlocked countries, and a far-placebo group of three cities, two of them outside Africa. See Section 5.2 for the estimation sample and specification.

A.2.3.3 Excluding the non-African far-placebo countries

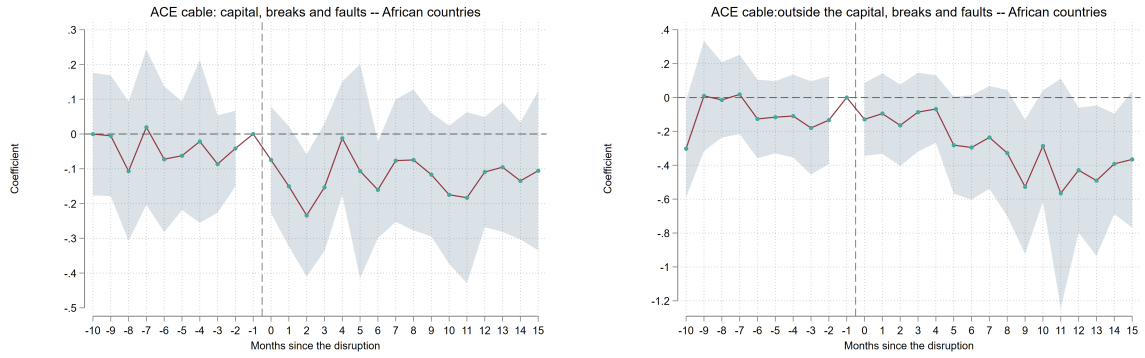
Two of the three far-placebo cities in Section 5.2's design, Paris and Lisbon, sit outside Africa entirely, included to widen the untreated comparison pool beyond the continent. Table A.2.3.2 re-estimates the three specifications of Figure 8 excluding France and Portugal, leaving seventeen countries with Johannesburg as the sole far-placebo.

Table A.2.3.2: Pooled post-break estimates: full sample vs. African countries only.

	Full sample (19 countries)		African countries only (17)	
	Estimate	p-value	Estimate	p-value
Economic capital	-0.169	0.013	-0.152	0.038
Outside the capital	-0.332	0.029	-0.303	0.051
Whole country	-0.244	0.040	-0.228	0.066

Notes: Pooled post-break coefficient ("Post") from the local projections difference-in-differences estimator of Dube et al. (2023), ten monthly leads, fifteen monthly lags, standard errors clustered on the country. Full sample: fourteen countries landing ACE, two landlocked, and three far-placebo controls (Johannesburg, Paris, Lisbon). African countries only: the same fourteen and two, with Johannesburg as the sole far-placebo, excluding France and Portugal.

Dropping Paris and Lisbon shrinks all three estimates modestly. The economic-capital result stays significant at the five percent level; the other two specifications remain significant only at ten percent. The sign and pre-trend pattern are unchanged throughout.



(a) Night-time light, capital city

(b) Night-time light, secondary cities

Fig. A.2.3.1: ACE cable breaks, African countries only: monthly night-time radiance.

Notes: Same specification as Figure 8, excluding France and Portugal from the far-placebo group. Seventeen countries, Johannesburg the sole far-placebo control. Bands are 95% confidence intervals.

A.2.3.4 CDH estimations

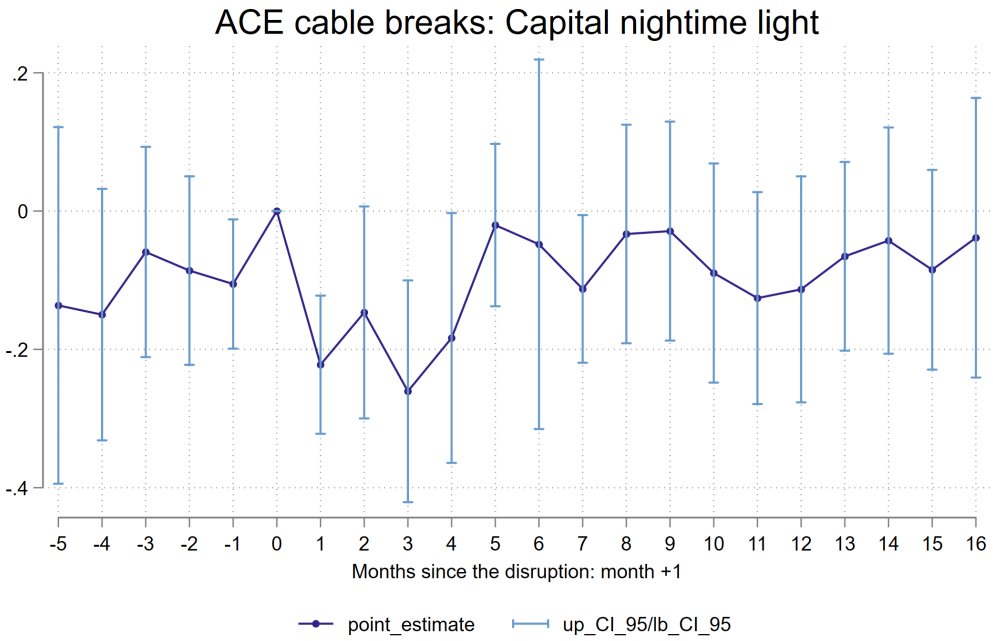


Fig. A.2.3.2: ACE cable breaks and capital-city night-time radiance, CDH estimator.

Notes: Event-study estimates from the de Chaisemartin and d'Haultfoeuille (2024) estimator on the monthly ACE panel, with country, year and income group-by-year fixed effects, with standard errors clustered at the country level. Ten countries switch treatment status. Point k on the horizontal axis reports the effect in the k-th month counted from the break, so point 1 is the month of the break itself. Bands are 95% confidence intervals.

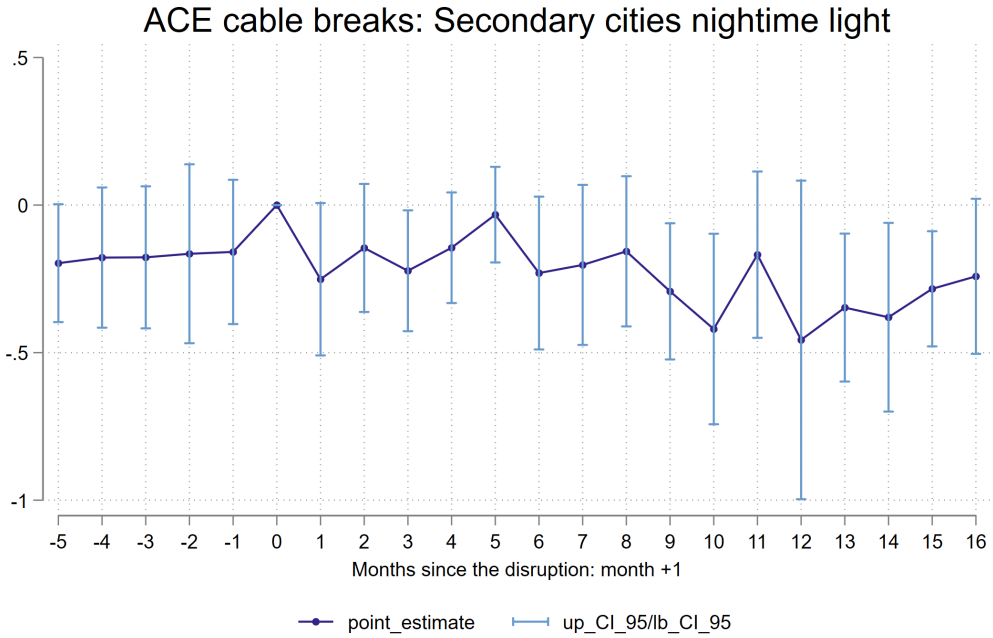


Fig. A.2.3.3: ACE cable breaks and night-time radiance outside the capital, CDH estimator.

Notes: Event-study estimates from the de Chaisemartin and d'Haultfoeuille (2024) estimator on the monthly ACE panel, with country, year and income group-by-year fixed effects, with standard errors clustered at the country level. Radiance measured over each country's remaining sampled cities outside the capital recording more the 20,000 inhabitants.

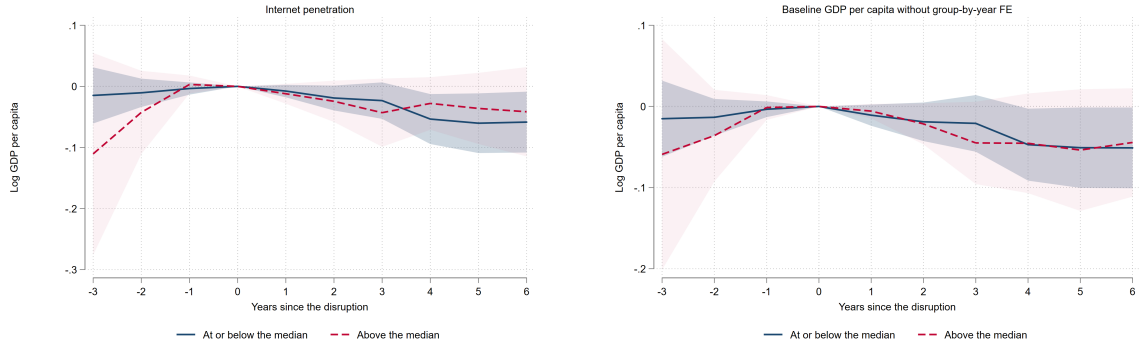
A.2.4 Regional spillovers

Table A.2.4.1: **Regional peer disruptions and GDP per capita (const USD, ln): event-study estimates and placebo tests, 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Regional number of disrupted economies – income-group×year FE						
Event-study effects						
Effect_1	0.0120833	0.0034797	0.0052633	0.0189033	72	56
Effect_2	0.0107780	0.0041142	0.0027143	0.0188416	67	53
Effect_3	0.0110511	0.0061391	-0.0009812	0.0230835	68	53
Effect_4	-0.0008413	0.0094520	-0.0193669	0.0176842	37	32
Effect_5	0.0028480	0.0107997	-0.0183190	0.0240150	37	32
Effect_6	0.0060120	0.0131767	-0.0198137	0.0318378	38	32
Sup-t test of joint nullity of the effects: p-value = 0.0042						
Placebo lead						
Placebo_1	-0.0058564	0.0065347	-0.0186642	0.0069513	11	7
Sup-t test of joint nullity of the placebos: p-value = 0.3712						
Panel B: Regional number of disrupted economies – no group-by-year FE						
Event-study effects						
Effect_1	0.0047778	0.0021984	0.0004690	0.0090866	127	85
Effect_2	0.0043840	0.0032001	-0.0018881	0.0106562	120	80
Effect_3	0.0053886	0.0051098	-0.0046264	0.0154036	123	80
Effect_4	0.0072253	0.0069482	-0.0063929	0.0208434	87	80
Effect_5	0.0077079	0.0084082	-0.0087720	0.0241877	87	80
Effect_6	0.0129828	0.0125715	-0.0116569	0.0376224	90	80
Sup-t test of joint nullity of the effects: p-value = 0.0751						
Placebo lead						
Placebo_1	0.0008364	0.0038320	-0.0066742	0.0083471	40	36
Sup-t test of joint nullity of the placebos: p-value = 0.8303						
Panel C: Regional number of disruption episodes – income-group×year FE						
Event-study effects						
Effect_1	0.0092273	0.0026572	0.0040193	0.0144353	72	56
Effect_2	0.0090894	0.0034696	0.0022891	0.0158898	67	53
Effect_3	0.0093198	0.0051773	-0.0008275	0.0194671	68	53
Effect_4	-0.0007156	0.0080396	-0.0164730	0.0150417	37	32
Effect_5	0.0024224	0.0091859	-0.0155817	0.0204265	37	32
Effect_6	0.0051137	0.0112077	-0.0168531	0.0270804	38	32
Sup-t test of joint nullity of the effects: p-value = 0.0042						
Placebo lead						
Placebo_1	-0.0051244	0.0057179	-0.0163312	0.0060824	11	7
Sup-t test of joint nullity of the placebos: p-value = 0.3712						
Panel D: Regional number of disruption episodes – no group-by-year FE						
Event-study effects						
Effect_1	0.0038924	0.0017910	0.0003821	0.0074028	127	85
Effect_2	0.0038432	0.0028054	-0.0016552	0.0093416	120	80
Effect_3	0.0047239	0.0044794	-0.0040557	0.0135034	123	80
Effect_4	0.0063340	0.0060910	-0.0056042	0.0182722	87	80
Effect_5	0.0067570	0.0073710	-0.0076899	0.0212039	87	80
Effect_6	0.0113812	0.0110207	-0.0102189	0.0329813	90	80
Sup-t test of joint nullity of the effects: p-value = 0.0751						
Placebo lead						
Placebo_1	0.0007721	0.0035373	-0.0061608	0.0077050	40	36
Sup-t test of joint nullity of the placebos: p-value = 0.8303						

Notes: Estimated on country-years with no domestic cable failure ($D_{it} = 0$), so treatment varies only through disruptions elsewhere in the same World Bank region. The extensive margin counts other disrupted economies in the region-year; the intensive margin counts their disruption episodes. ONE placebo lead only: pre-treatment periods empty out fast here, the second lead resting on six switchers against the eighteen-switcher floor applied elsewhere in the paper. Six effects on the 2008–2019 sample excluding Vietnam; Effect$_k$ is the effect at $t + (k - 1)$. Standard errors clustered at the country level; joint tests are the conservative sup-t test.

A.2.5 Exposure factors: Internet penetration and income group



(a) Internet penetration

(b) Income level

Fig. A.2.5.1: Digitalisation, income, and exposure to SMC failures.

Notes: Panel (a) splits countries at the 2008 median of Internet penetration, 22.6% of the population. Panel (b) splits them at the 2008 median of GDP per capita, US$5,049 in constant prices. Both variables are drawn from the World Bank's World Development Indicators, and both assignments are fixed at the country's first sample year. The solid navy line reports the at-or-below-median group and the dashed red line the above-median group, with 95% confidence bands. Both panels include country fixed effects, year fixed effects and linear country trends, with standard errors clustered at the country level. Panel (a) adds income-group-by-year fixed effects. Panel (b) omits them, since they absorb part of the income variation the split is built on. Point k on the horizontal axis reports the estimate for $k-1$ years after the disruption, so point 1 is the year of the disruption and point 6 is five years after. Point 0 is the last year before the disruption, to which the estimates are normalised, and negative points report the placebo estimates. In panel (b) the at-or-below-median estimates rest on 16 switchers four years after the disruption and the above-median estimates on seven, which are significant at no horizon.

A.3 Robustness checks and mechanisms

A.3.1 Robustness checks

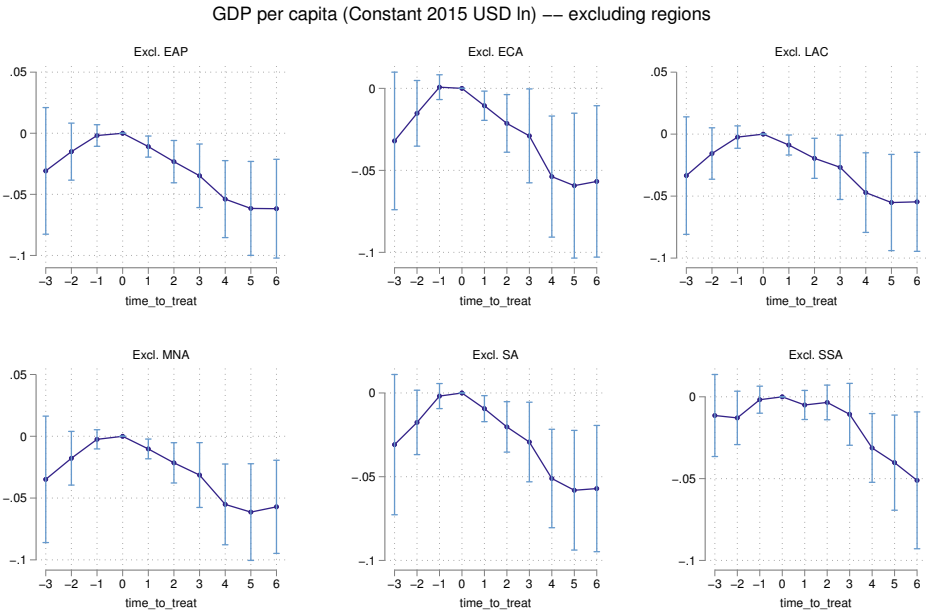


Fig. A.3.1.1: Cable disruptions and GDP per capita (const USD, ln) – Estimations sequentially removing regions.

Notes: Event–study coefficients for log GDP per capita after sequentially dropping the indicated region. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and $income - group \times year$ fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence.

Table A.3.1.1: Determinants of disruption exposure: cross-sectional balance.

	OLS			Poisson (IRR)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dep. var:	**# cable disruptions over 2008–2020**					
Pre-treatment GDP per cap. growth	−0.026	0.005	−0.063	0.845	0.855	0.861
	(0.217)	(0.227)	(0.218)	(0.119)	(0.115)	(0.128)
Log GDP per capita	−0.090	−0.114	0.064	0.745	0.726	0.913
	(0.215)	(0.225)	(0.205)	(0.162)	(0.199)	(0.194)
Log population	0.429**	0.417**	0.661**	1.401*	1.375*	1.900***
	(0.191)	(0.184)	(0.314)	(0.249)	(0.234)	(0.440)
Cable connectedness (relations)	0.641	0.521*		1.883**	1.804***	
	(0.430)	(0.305)		(0.517)	(0.371)	
Number of landed cables	−0.281		0.013	0.847		1.146
	(0.327)		(0.161)	(0.437)		(0.313)
Internet exchange points	−0.156	−0.264	−0.174	0.805	0.771	0.763
	(0.167)	(0.178)	(0.164)	(0.226)	(0.173)	(0.193)
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Countries	172	172	172	172	172	172
R^2 / Pseudo R^2	0.21	0.21	0.19	0.34	0.34	0.31

Notes: The dependent variable is a country's total number of SMC-induced connectivity disruptions over 2008–2020. Of the 172 countries in the sample, 72 are disrupted at least once, for 193 events in total. Pre-treatment growth is each country's average annual GDP-per-capita growth rate over the years strictly before its first recorded disruption. For the 100 countries never disrupted, the pretreatment growth is the average growth over 2008–2020. Cable connectedness is the number of distinct cable relations. The five regressors in level are measured at baseline, in each country's first year of non-missing data. All six regressors are standardized to mean zero and unit variance. Column (1) is OLS with region fixed effects and heteroskedasticity-robust (HC1) standard errors in parentheses, so each coefficient is the change in the disruption count associated with a one-standard-deviation increase in the regressor. Column (2) reports incidence-rate ratios from a Poisson count model on the same specification and sample, with robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

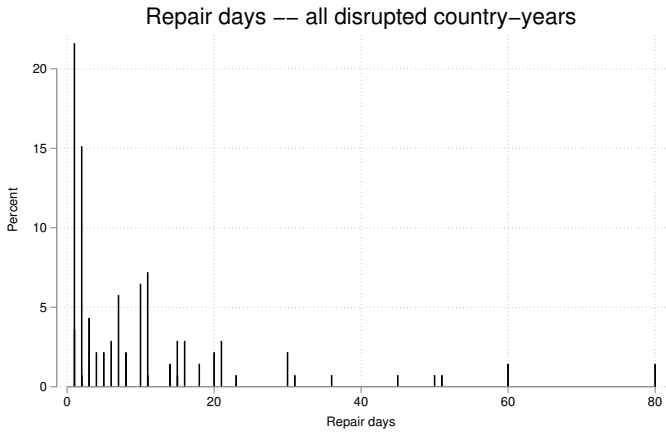


Fig. A.3.1.2: Distribution of repair days, disrupted country-years.

Notes: Repair days in every disrupted country-year of the estimation sample, including recurring disruptions in later years for the same switcher (139 country-years). Median 6 days; this is the median used to split countries in Figure A.3.1.3.

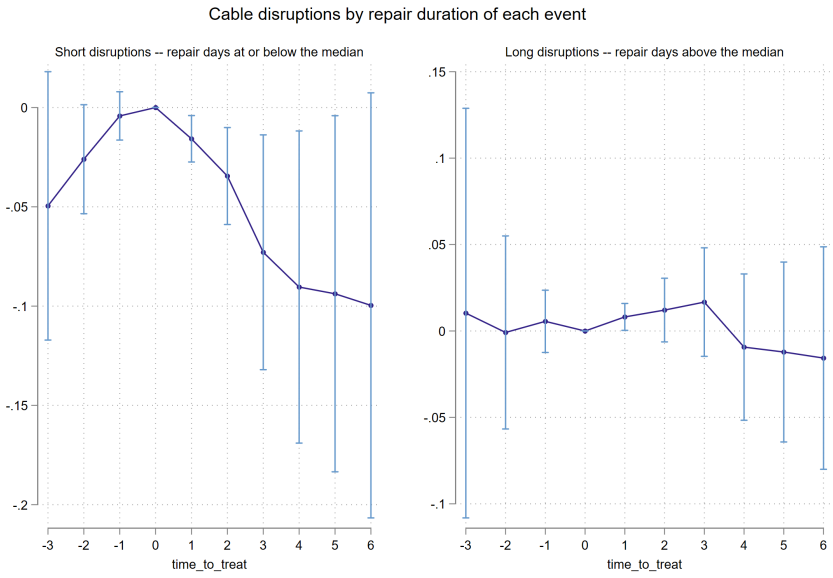


Fig. A.3.1.3: Sensitivity to the repair duration of the disruption.

Notes: The treatment splits into two counts. Disruptions whose annual repair days are at or below 6, the median across the 139 disrupted country-years of the estimation sample (Figure A.3.1.2), enter the left panel, and those above 6 the right panel. The two samples are disjoint. A country whose disruption lasted more than six days is coded zero in the short-disruption count for that year. Left in the short-disruption sample, it would serve as an untreated control while its GDP is falling, which understates the estimated loss. Every country recording a disruption above the median therefore leaves the short-disruption sample, and every country recording one at or below the median leaves the long-disruption sample. The 20 countries recording both over the period leave both, which leaves 27 treated countries in the left panel and 22 in the right. Countries never disrupted remain in both as controls. Specifications include country and year fixed effects, country-specific linear trends, and $income - group \times year$ fixed effects, with standard errors clustered at the country level.

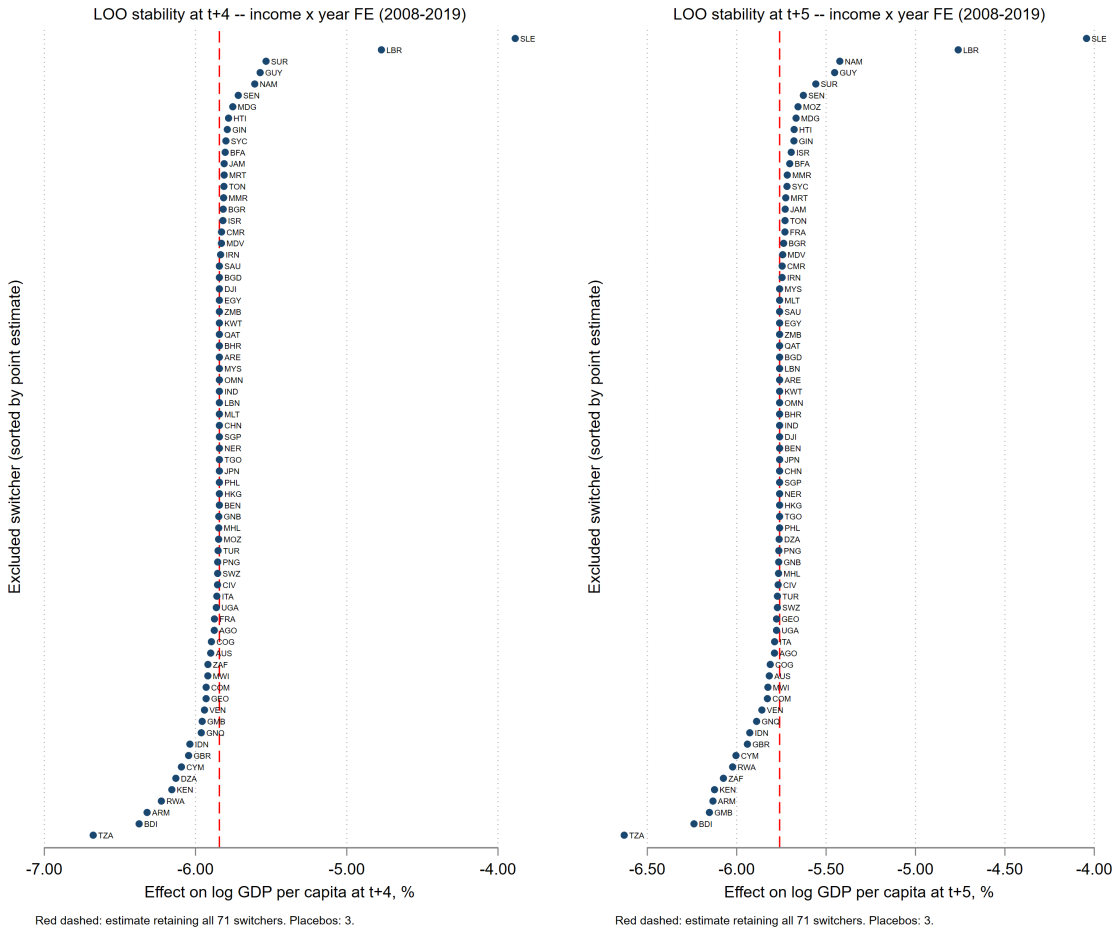


Fig. A.3.1.4: Leave-one-out stability: log GDP per capita response, income-group × year FEs.

Notes: Each row reports the estimated cumulative deviation in log GDP per capita at t+4 (left panel) or t+5 (right panel) from a leave-one-out re-estimation of the baseline event-study in which the labeled country is excluded from the sample. Red dashed line: full-sample point estimate; solid vertical line: zero. Specification follows Figure 5, treatment: the cable-disruption count.

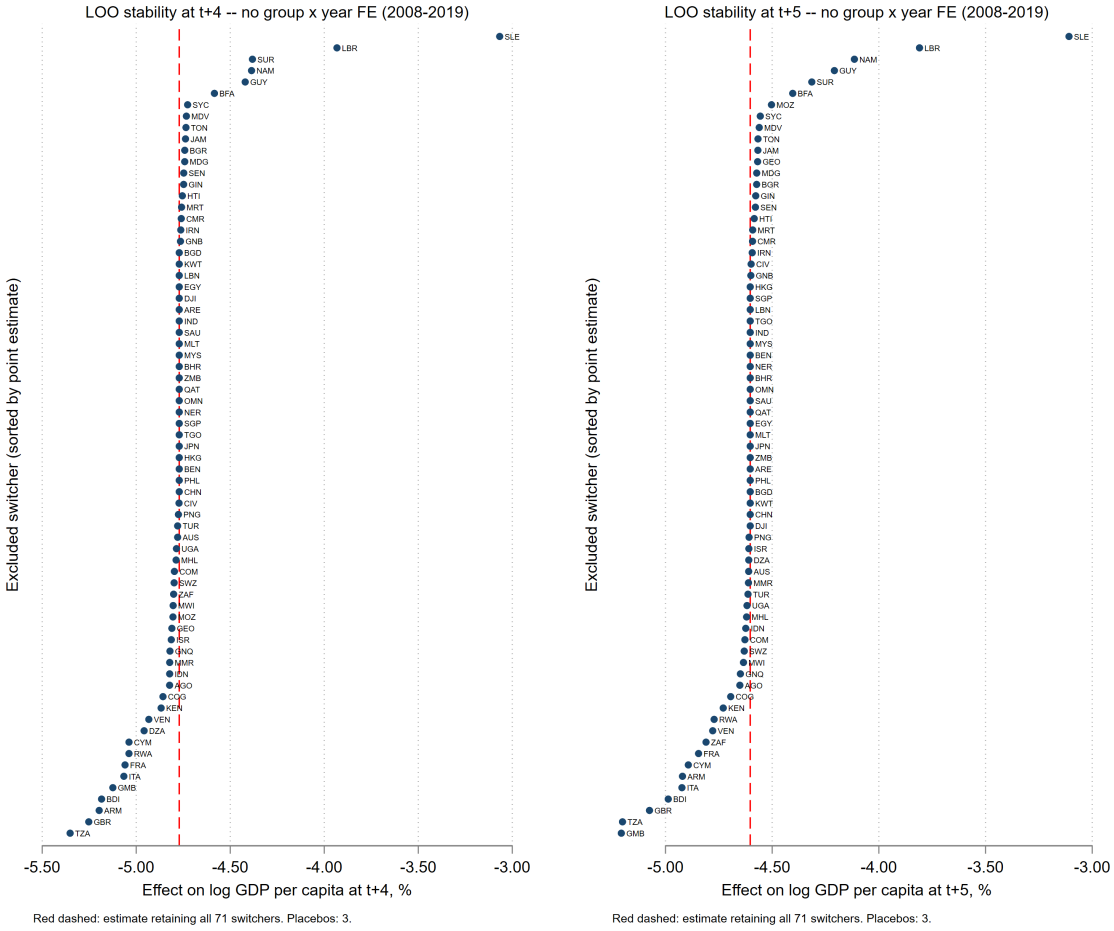


Fig. A.3.1.5: Leave-one-out stability: log GDP per capita response, no group-by-year FE).

Notes: Each row reports the estimated cumulative deviation in log GDP per capita at $t+4$ (left panel) or $t+5$ (right panel) from a leave-one-out re-estimation of the event-study in which the labeled country is excluded from the sample. Red dashed line: full-sample point estimate; solid vertical line: zero. Specification follows Figure 5, treatment: the cable-disruption count.

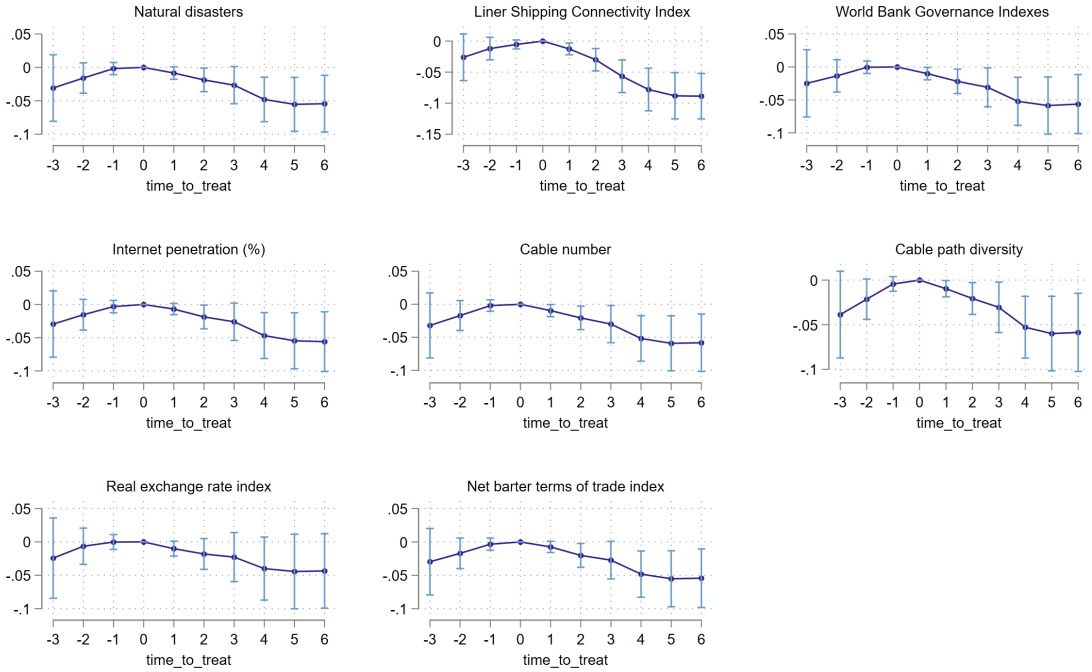


Fig. A.3.1.6: Cable disruptions and GDP per cap. (const. USD, ln), controlling for key confounding factors.

Notes: Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. *Disaster occurrence* controls include the annual number of biological, climatological, geophysical, hydrological, and meteorological disaster events, as reported in the EM-DAT database. To avoid sample attrition, missing value are replaced by 0. Results remain robust to replacing missing values by region×income group×year averages or maximums, and can be provided on request. The *Liner Shipping Connectivity Index*, drawn from the UNCTAD, measures the level of integration of a country or port into the global maritime transport network, by assessing factors such as the number of ships, their capacity, the size of the ships, the number of services and the number of shipping companies. The *ICT Regulatory quality index* is measured by International Telecommunications Union's ICT Regulatory tracker. *World Bank Governance Indexes* include Voice and Accountability, Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption indexes. *SMC connectedness* is the number of distinct cable-connected partners.

A.3.2 Mechanisms

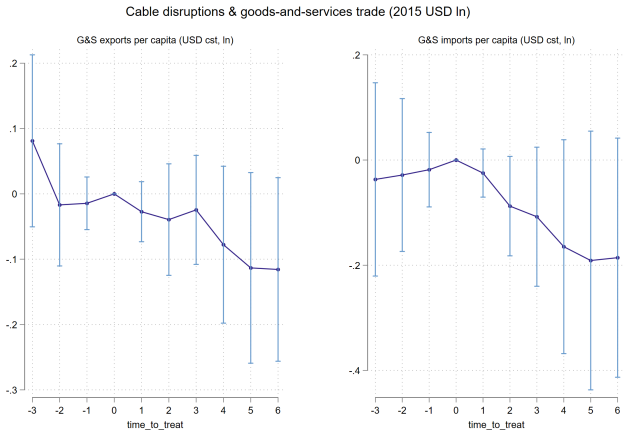


Fig. A.3.2.1: Cable disruptions and goods-and-services trade (USD, ln).

Markers show point estimates; bars denote 95% confidence intervals. The left panel reports exports of goods and services per capita and the right panel imports, both in constant 2015 USD and logged. Specifications include country fixed effects, year fixed effects, linear country trends, and $income - group \times year$ fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence.

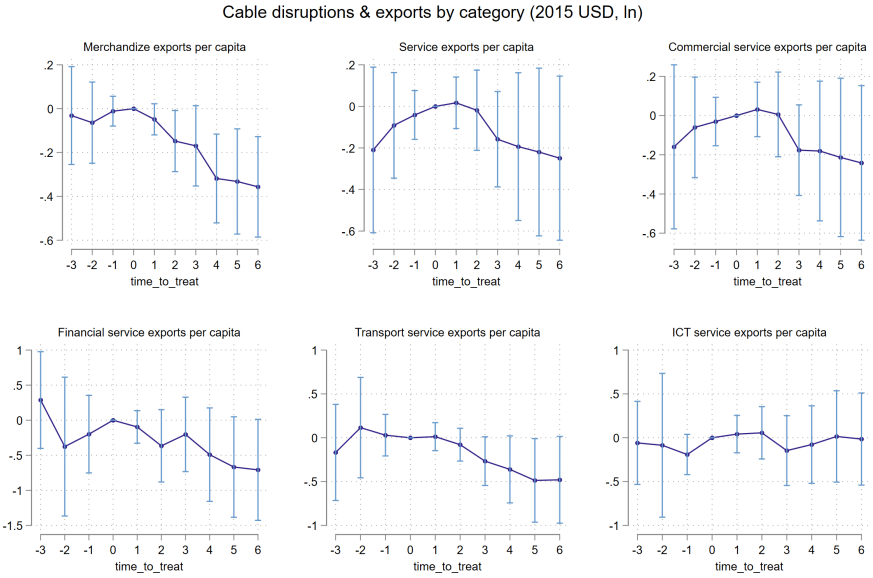


Fig. A.3.2.2: Cable disruptions and exports (USD, ln), by category.

Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and $income - group \times year$ fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence.

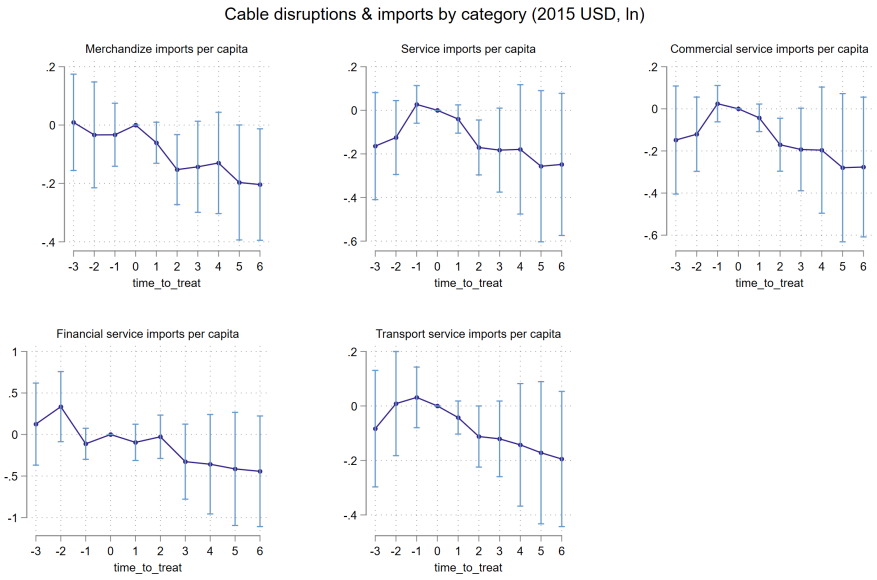


Fig. A.3.2.3: Cable disruptions and imports (USD, ln), by category.

Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and $income - group \times year$ fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence.

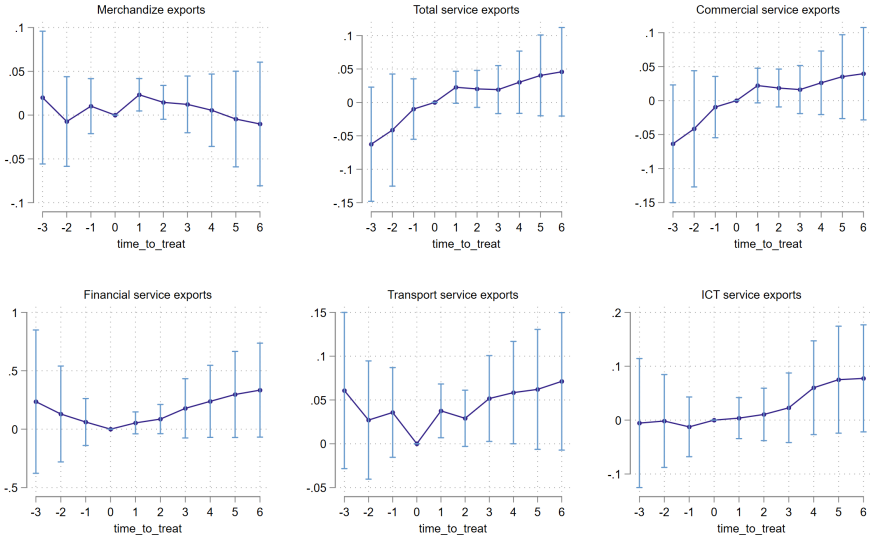


Fig. A.3.2.4: Regional spillovers on exports (USD, ln) - regional number of affected countries.

Notes: Figures report event–time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, and linear country trends, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. The treatment is the number of affected economies recorded in the region of the non-affected country.

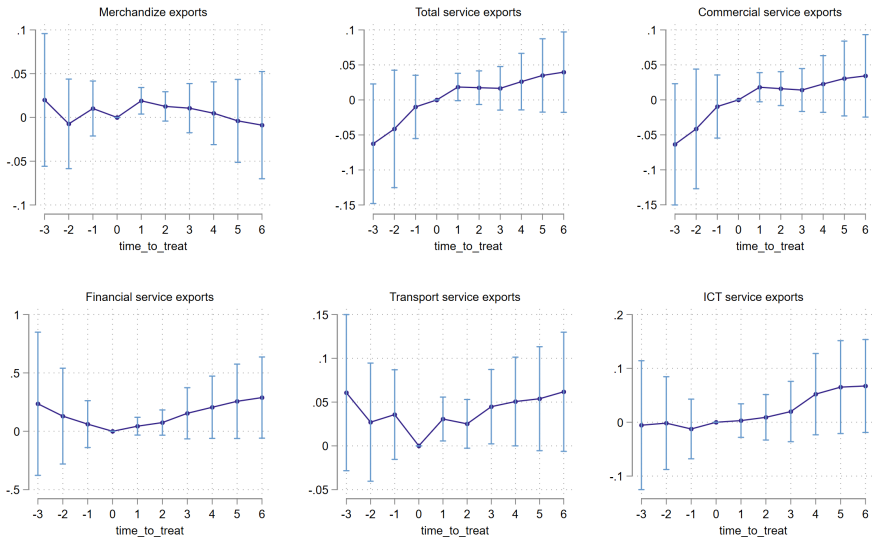


Fig. A.3.2.5: Regional spillovers on exports (USD, ln) - regional number of disruption episodes.

Notes: Figures report event–time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, and linear country trends, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. The treatment is the number of cable disruptions recorded in the region of the non-affected country.

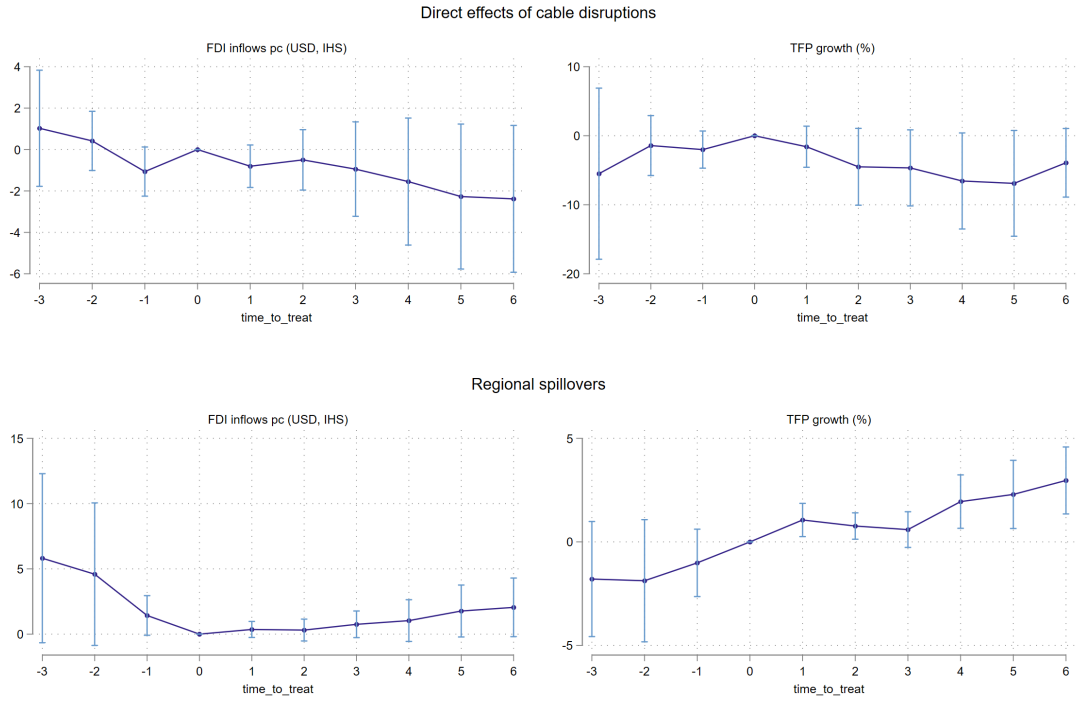


Fig. A.3.2.6: Cable disruptions and competitiveness, direct effects and regional spillovers.

Markers show point estimates; bars denote 90% confidence intervals. The top row reports direct effects on disrupted countries, treated by the number of cable disruptions. The bottom row reports regional spillovers, estimated on country-years with no recorded disruption and treated by the number of disruption episodes recorded in the same region. All four panels include country fixed effects, year fixed effects, linear country trends, and $income - group \times year$ fixed effects, with standard errors clustered at the country level. FDI inflows per capita are transformed using the inverse hyperbolic sine. TFP is a human-capital-adjusted TFP growth metric, calculated as the residual of labour productivity growth after subtracting the contributions of human capital and capital deepening. TFP growth is drawn from the World Bank Global Productivity Database (Dieppe 2021) and FDI inflows from the World Development Indicators. Year 6 corresponds to a period of five years after the cable failure occurrence.

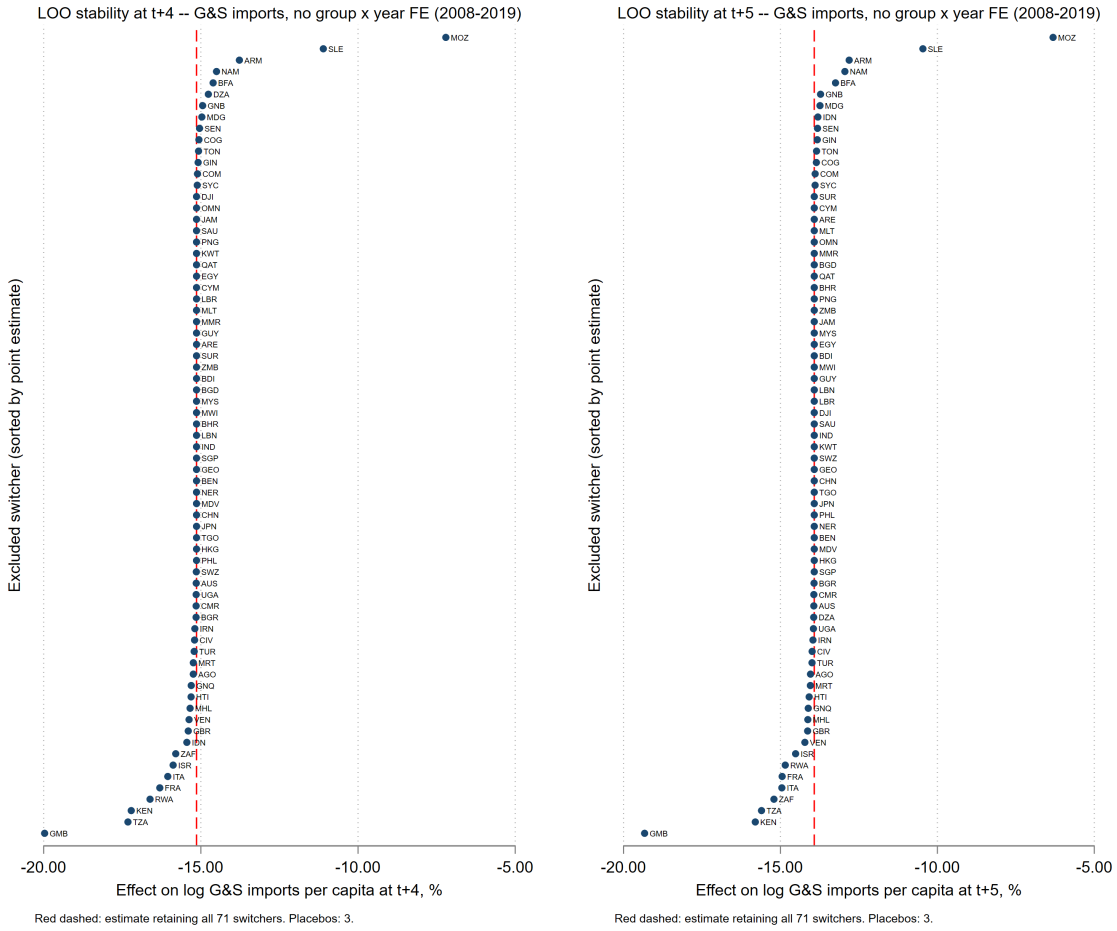


Fig. A.3.2.7: Leave-one-out stability: goods-and-services import response, income-group × year FEs.

Notes: Each row reports the estimated cumulative deviation in log goods-and-services imports per capita at t+4 (left panel) or t+5 (right panel) from a leave-one-out re-estimation in which the labeled country is excluded from the sample. Treatment: the cable-disruption count. Specifications include country and year fixed effects and linear country trends, without group-by-year fixed effects. Red dashed line: full-sample point estimate; solid vertical line: zero.

Table A.3.2.1: Import extensive margin and domestic export participation: direct effects and regional spillovers

	t	$t+1$	$t+2$	$t+3$	$p_{\text{eff.}}$	$p_{\text{plac.}}$
Panel A1: Direct effects – Import extensive margin (Treatment: cable disruptions)						
Avg. foreign exporters per partner	-35.8^{**}	-64.6^{**}	-64.8	-84.6	0.066	0.162
Sample mean = 309.2	(15.4)	(23.1)	(39.9)	(57.4)		
N	305	240	157	123		
Sw.	22	16	11	9		
Total foreign exporters	$-1{,}601^{*}$	$-3{,}550^{**}$	$-6{,}156^{**}$	$-15{,}642^{**}$	0.036	0.302
Sample mean = 10,599	(885)	(1,604)	(2,919)	(6,614)		
N	305	240	157	123		
Sw.	22	16	11	9		
Distinct foreign exporter countries	0.54	-0.34	-1.80^{*}	-4.41^{***}	0.022	
Sample mean = 29.6	(0.78)	(0.98)	(0.98)	(1.49)		
N	305	240	157	123		
Sw.	22	16	11	9		
Panel A2: Direct effects – Domestic export participation (Treatment: cable disruptions)						
Domestic exporter exit rate	0.041	0.152^{**}				
Sample mean = 0.383	(0.043)	(0.068)				
N	28	17				
Sw.	5	4				
Surviving entrants	-170^{**}					
Sample mean = 2,398	(76)					
N	18					
Sw.	4					
Panel B1: Spillover effects – Domestic export participation (Treatment: disrupted economies in region)						
Distinct foreign exporter countries	2.04^{***}	1.36^{***}	-0.10		<0.001	
Sample mean = 31.5	(0.19)	(0.18)	(0.40)			
N	112	110	78			
Sw.	75	73	43			
Total domestic exporters	-21	75	366^{*}		0.046	
Sample mean = 17,620	(216)	(201)	(175)			
N	24	22	20			
Sw.	13	11	9			
Incumbent domestic exporters	128^{*}	145^{*}	376^{**}		0.194	
Sample mean = 11,314	(75)	(85)	(177)			
N	23	21	18			
Sw.	13	11	9			
Domestic exiters	-179^{*}	-205^{*}	-502^{*}		0.113	
Sample mean = 5,071	(93)	(107)	(293)			
N	23	21	18			
Sw.	13	11	9			
Panel B2: Spillover effects – Domestic export participation (Treatment: disruption episodes in region)						
Distinct foreign exporter countries	4.89^{***}	3.07^{***}	-0.19		<0.001	
Sample mean = 31.7	(0.46)	(0.41)	(0.80)			
N	112	110	78			
Sw.	75	73	43			
Total domestic exporters	-42	149	733^{*}		0.046	
Sample mean = 17,620	(431)	(403)	(349)			
N	24	22	20			
Sw.	13	11	9			
Incumbent domestic exporters	255^{*}	289^{*}	752^{*}		0.194	
Sample mean = 11,314	(150)	(170)	(355)			
N	23	21	18			
Sw.	13	11	9			
Domestic exiters	-359^{*}	-410^{*}	$-1{,}004^{*}$		0.113	
Sample mean = 5,071	(186)	(214)	(585)			
N	23	21	18			
Sw.	13	11	9			

Notes: World Bank EDD data. Panel A: CDH estimator with country and year FEs, country-specific linear trends, and region×year FEs. Panels B: country and year FEs and country-specific linear trends; region nonparametric trends omitted to preserve cross-regional identifying variation. Panel B treatment: count of disrupted economies (*peer_break*); Panel C: count of disruption episodes (*peer_maxbreak*). SEs clustered by country. *Distinct exporter countries*: origin countries with ≥ 1 observed exporting firm at destination (sample max. 57). N, Sw., and means inline; sample contracts at longer horizons. $p_{\text{eff.}}/p_{\text{plac.}}$: joint-test p-values for effects/placebos; blank = not estimated. Panel A2 results rest on 4–5 switchers (directional). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.3.2.2: **Exporter Dynamics Database variables: summary statistics, estimation sample.**

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Surviving entrants (exporters)	1,654	3,268	325	1	18,759	146
Exporter exit rate	0.358	0.101	0.035	0.063	0.675	203
Foreign exporters (count)	10,539	16,786	7,329	2	121,011	975
Foreign exporters per origin country	310.5	406.6	121.0	0.2	3,008.5	975
Origin countries observed (count)	28.61	15.43	12.56	0	57	978
Total exporters	12,100	23,179	1,992	18	119,399	219
Incumbent exporters	7,876	15,111	1,035	10	95,921	204
Exiting exporters	3,795	7,801	995	1	52,322	203

Notes: World Bank Exporter Dynamics Database. Estimation sample (realized CDH sample), 2008–2019 excluding Vietnam.

A.3.3 Financial-channel

Table A.3.3.1: Financial-channel variables: summary statistics, estimation sample.

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Non-bank loan/deposits pc (BIS LBS, USD millions)	0.100	1.128	0.193	0	19.462	1,760
Non-bank loan/deposits pc growth (%)	0.063	0.767	0.647	-4.821	13.960	1,760
Credit to non-financial sector (Z-score)	-0.032	0.758	0.750	-2.226	1.906	374
Credit to general government (Z-score)	-0.031	0.758	0.751	-1.741	2.420	374
Credit to private non-financial sector, all lenders (Z-score)	-0.017	0.816	0.810	-2.400	1.920	352
Credit to private non-financial sector, banks (Z-score)	-0.014	0.830	0.823	-2.362	1.952	352

Notes: Estimation sample (realized CDH sample), 2008–2019 excluding Vietnam. Credit variables are within-country Z-scores, BIS data.

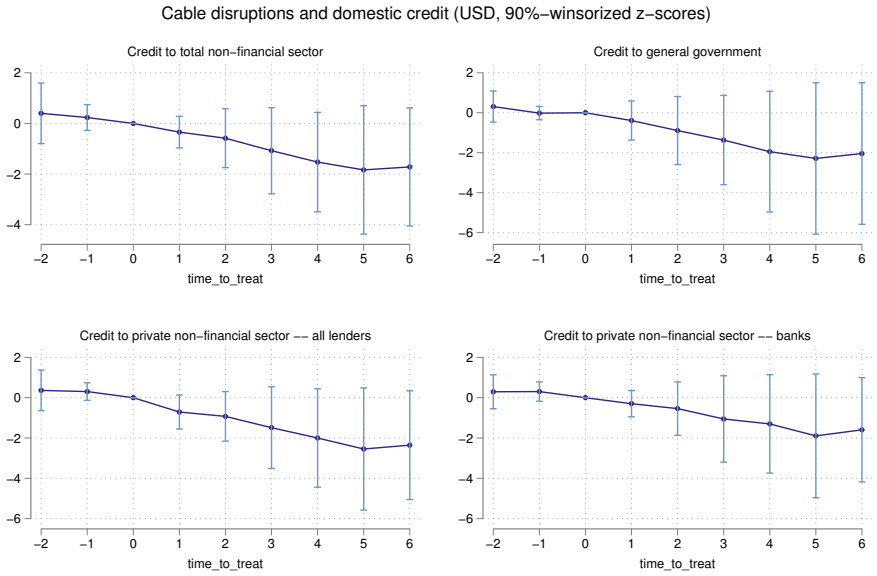


Fig. A.3.3.1: Cable disruptions and credit to non-financial domestic agents (USD, Z-scores).

Data from the Bank of International Settlements (BIS). Variables are winsorized at 90% and standardized afterwards. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. The estimation sample includes 27 high-income and emerging economies (271 observations), with six switchers (76 obs.): Australia, France, United Kingdom, Indonesia, Italy, and Turkey.

A.3.4 Economic cost of disruptions in low connectivity settings

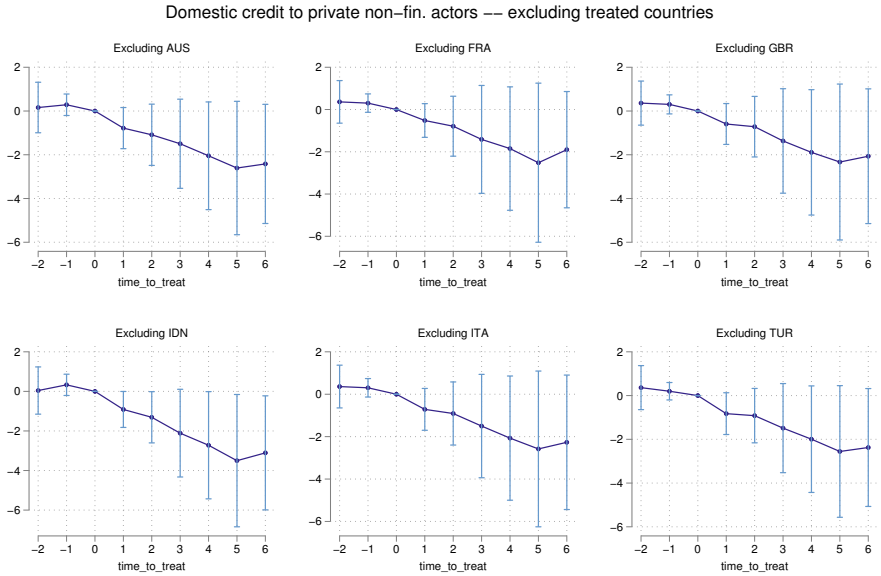


Fig. A.3.3.2: Cable disruptions and credit to non-financial domestic agents (USD, Z-scores), sequentially excluding switchers.

Data drawn from the Bank of International Settlements (BIS). Variables are winsorized at 90% and standardized afterwards. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence.

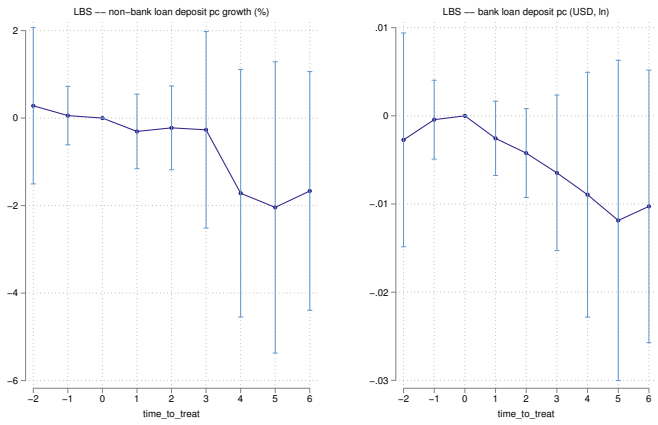


Fig. A.3.3.3: Non-bank loan deposits per capita (USD millions, ln) - Location Banking Statistics.

Markers show point estimates; bars denote 90% confidence intervals. Data drawn from the Locational Banking Statistics (LBS) from the Bank of International Settlements (BIS). Specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. The estimation sample includes 137 countries (1,781 observations), including 50 switchers (650 obs.). High-income and middle-income economies respectively represent 33.6% and 54% of the estimation sample.

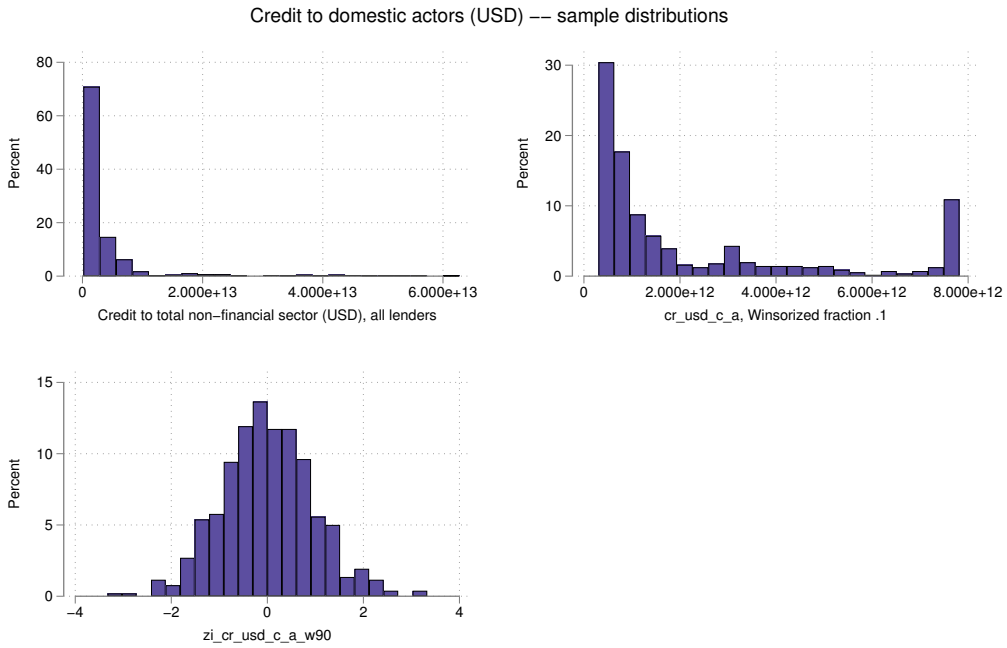


Fig. A.3.3.4: Domestic credit to non-financial domestic actors (USD), variable distributions.

Table A.3.4.1: Dynamic effects and relative costs (net of spillovers) of cable disruptions on GDP per capita (log), under low connectivity settings.

Effect	Cost (USD pc)	Estimate	SE	LB CI	UB CI	N	Switchers
Low connectedness: Cable disruptions							
Effect 1	-130.11	-0.0116273	0.0071882	-0.0257159	0.0024613	381	30
Effect 2	-294.96	-0.0263578	0.0148768	-0.0555157	0.0028001	317	23
Effect 3	-424.58	-0.0379430	0.0225302	-0.0821015	0.0062154	271	19
Effect 4	-775.95	-0.0693418	0.0269811	-0.1222238	-0.0164599	212	15
Effect 5	-847.77	-0.0757609	0.0316319	-0.1377582	-0.0137636	192	14
Effect 6	-791.87	-0.0707655	0.0303009	-0.1301542	-0.0113768	179	14
Sample average GDP per capita: 11190.15 USD							
Low connectedness: Repair days							
Effect 1	-12.58	-0.0011244	0.0008706	-0.0028309	0.0005820	366	28
Effect 2	-32.70	-0.0029216	0.0020049	-0.0068511	0.0010079	303	22
Effect 3	-40.44	-0.0036133	0.0026751	-0.0088565	0.0016299	260	18
Effect 4	-84.40	-0.0075424	0.0038842	-0.0151552	0.0000704	203	14
Effect 5	-101.51	-0.0090714	0.0047876	-0.0184549	0.0003121	195	14
Effect 6	-115.72	-0.0103414	0.0054623	-0.0210474	0.0003646	183	14
Sample average GDP per capita: 11190.15 USD							
No IXP: Cable disruptions							
Effect 1	-169.44	-0.0146245	0.0065787	-0.0275186	-0.0017305	535	35
Effect 2	-334.58	-0.0288787	0.0157368	-0.0597222	0.0019649	452	24
Effect 3	-491.41	-0.0424146	0.0269047	-0.0951468	0.0103176	361	18
Effect 4	-1162.80	-0.1003622	0.0410900	-0.1808971	-0.0198273	297	13
Effect 5	-1212.36	-0.1046398	0.0463961	-0.1955745	-0.0137051	276	13
Effect 6	-1214.75	-0.1048466	0.0509957	-0.2047962	-0.0048969	215	12
Sample average GDP per capita: 11585.77 USD							
No IXP: Repair days							
Effect 1	-15.70	-0.0013547	0.0007289	-0.0027834	0.0000740	534	34
Effect 2	-31.75	-0.0027410	0.0017866	-0.0062426	0.0007606	452	24
Effect 3	-38.07	-0.0032856	0.0025086	-0.0082025	0.0016312	361	18
Effect 4	-89.02	-0.0076832	0.0040227	-0.0155676	0.0002011	297	13
Effect 5	-107.14	-0.0092472	0.0050607	-0.0191660	0.0006715	276	13
Effect 6	-124.88	-0.0107786	0.0063233	-0.0231721	0.0016149	215	12
Sample average GDP per capita: 11585.77 USD							

Online Appendix

OA.1 Disruption data: coding protocol and maintenance areas

OA.1.1 Internet disruption database: coding protocol

Sources.

The database is built from systematic monitoring of connectivity disruption reports published by *SubTel Forum*, the submarine cable industry's primary technical platform, which documents national-level disruptions caused by cable infrastructure failures in real time. The *SubTel Forum* cable faults and maintenance desk is complemented by two secondary sources: the *Internet Disruption Report* (internetdisruption.report), a periodic compilation of global connectivity incidents cross-referenced against operator announcements; and Akamai's *State of the Internet Connectivity* reports for earlier years. Each event is cross-checked against targeted web searches combining the country name, the cable system name, and the approximate date. The database covers 2008–2020; the estimation sample is restricted to 2008–2020.

Unit of observation.

The raw database records connectivity disruption events at the country–event level. A single physical cable fault can impair connectivity in multiple countries simultaneously: each affected country is recorded as a separate observation, reflecting the principle that effects are measured from the perspective of what each country experiences, not from proximity to the physical break. For instance, the December 2008 failure of the SEA-ME-WE 4 and FLAG systems, caused by a vessel anchor in the Mediterranean, generated 14 country-level disruption events across Asia, the Middle East, and Southern Europe.

Attribution criterion.

A country is coded as disrupted only when operators or contemporary sources explicitly report a measurable loss of international connectivity—degraded throughput, partial or full outage, or confirmed rerouting with service impact—attributable to a cable infrastructure failure. Countries that are topologically connected to the faulted cable but for which traffic is successfully rerouted without reported degradation are coded as not treated. This *material-impact* criterion ensures that treatment captures experienced connectivity loss.

Event taxonomy.

Each raw event is assigned one of the following categories on the basis of source descriptions. *Cable breaks* are disruptions explicitly linked to a physical cut or break requiring dispatch of a repair vessel; these constitute the primary treatment events. *Cable outages* are service-affecting faults without confirmed physical damage—repeater failures, power-feed faults, branching-unit malfunctions—and are retained as secondary treatment events. *Network disruptions* cover connectivity losses attributable to routing issues, DNS problems, or other technical failures not involving cable infrastructure damage. *Other-cause* events include government-ordered shutdowns, sabotage of terrestrial or submarine fiber, power outages, natural disasters (cyclones, earthquakes, floods), cyber attacks, and scheduled maintenance operations.

Treatment variable construction.

The estimation sample retains only cable breaks and cable outages as treatment-generating events, and constructs two country-year variables: (i) *cable disruptions*: the annual count of SMC-induced connectivity losses; and (ii) *repair days*: the annual sum of connectivity-loss days associated with these events. Government-ordered shutdowns, sabotage, power failures-caused disruptions, routing issues, and events of unknown cause are coded separately and excluded from the treatment variable. As robustness checks, I drop events associated with suspected or confirmed sabotage and events associated with natural disasters that could independently affect economic outcomes (Section 5).

OA.1.2 SMC maintenance areas



Fig. OA.1.1: SMC maintenance areas.
Source: Global Marine Systems Ltd..

OA.2 Exposure factors: detailed estimates and country lists

Table OA.2.1: **Exposure factor: cable density – dependent variable: GDP per capita (const USD, ln), 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Event-study effects, countries at or below the median number of landed cables (1)						
Effect_1	-0.0115114	0.0065267	-0.0243035	0.0012807	456	31
Effect_2	-0.0262537	0.0133497	-0.0524187	-0.0000888	382	25
Effect_3	-0.0360869	0.0204854	-0.0762375	0.0040638	297	20
Effect_4	-0.0683830	0.0265838	-0.1204862	-0.0162797	231	15
Effect_5	-0.0724753	0.0307196	-0.1326846	-0.0122659	208	14
Effect_6	-0.0665315	0.0292553	-0.1238708	-0.0091922	195	14
Placebo leads						
Placebo_1	-0.0050236	0.0072230	-0.0191805	0.0091332	456	31
Placebo_2	-0.0200332	0.0159361	-0.0512674	0.0112011	305	21
Placebo_3	-0.0373184	0.0353096	-0.1065240	0.0318872	203	14
Event-study effects, countries above the median number of landed cables (1)						
Effect_1	-0.0028819	0.0062981	-0.0152260	0.0094622	275	17
Effect_2	-0.0010251	0.0062394	-0.0132542	0.0112039	236	11
Effect_3	-0.0089986	0.0093398	-0.0273043	0.0093071	184	9
Effect_4	-0.0166117	0.0129142	-0.0419231	0.0086997	181	9
Effect_5	-0.0246248	0.0172561	-0.0584462	0.0091966	172	9
Effect_6	-0.0287255	0.0226987	-0.0732141	0.0157632	140	8
Placebo leads						
Placebo_1	0.0057088	0.0057616	-0.0055837	0.0170012	244	14
Placebo_2	-0.0073326	0.0087060	-0.0243961	0.0097309	170	6
Placebo_3	-0.0220286	0.0184003	-0.0580925	0.0140353	118	4

Notes: Dependent variable: log GDP per capita (constant 2015 USD). Treatment: the cable-disruption count; the repair-days panel of earlier versions is not reported, the subgroup estimations being run on the disruption count only. Six event-study effects and three placebo leads on the 2008–2019 sample excluding Vietnam; Effect_k is the effect at $t + (k - 1)$. Groups are fixed at the country's first sample year, split at the 2008 cross-sectional median. Specifications include country and year fixed effects, country-specific linear trends and income-group-by-year fixed effects, with standard errors clustered at the country level. No joint sup-t test is reported: each subgroup is estimated separately, which does not leave the full parameter vector the test requires.

Table OA.2.2: **Exposure factor: connected partners – dependent variable: GDP per capita (const USD, ln), 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Event-study effects, countries at or below the median number of connected partners (2)						
Effect_1	-0.0116273	0.0071882	-0.0257159	0.0024613	381	30
Effect_2	-0.0263578	0.0148768	-0.0555157	0.0028001	317	23
Effect_3	-0.0379430	0.0225302	-0.0821015	0.0062154	271	19
Effect_4	-0.0693418	0.0269811	-0.1222238	-0.0164599	212	15
Effect_5	-0.0757609	0.0316319	-0.1377582	-0.0137636	192	14
Effect_6	-0.0707655	0.0303009	-0.1301542	-0.0113768	179	14
Placebo leads						
Placebo_1	-0.0057423	0.0072742	-0.0199994	0.0085148	381	30
Placebo_2	-0.0224080	0.0200349	-0.0616756	0.0168597	246	18
Placebo_3	-0.0432002	0.0476742	-0.1366400	0.0502395	183	12
Event-study effects, countries above the median number of connected partners (2)						
Effect_1	-0.0047630	0.0056603	-0.0158569	0.0063309	339	19
Effect_2	-0.0003173	0.0053145	-0.0107335	0.0100988	295	14
Effect_3	-0.0027097	0.0085476	-0.0194628	0.0140433	231	10
Effect_4	-0.0056076	0.0135656	-0.0321957	0.0209806	196	9
Effect_5	-0.0116709	0.0173577	-0.0456912	0.0223495	187	9
Effect_6	-0.0115716	0.0225485	-0.0557658	0.0326226	151	8
Placebo leads						
Placebo_1	0.0047160	0.0045595	-0.0042205	0.0136525	308	16
Placebo_2	0.0026004	0.0082769	-0.0136221	0.0188229	225	10
Placebo_3	-0.0082463	0.0139050	-0.0354996	0.0190070	161	6

Notes: Dependent variable: log GDP per capita (constant 2015 USD). Treatment: the cable-disruption count; the repair-days panel of earlier versions is not reported, the subgroup estimations being run on the disruption count only. Six event-study effects and three placebo leads on the 2008–2019 sample excluding Vietnam; Effect_k is the effect at $t + (k - 1)$. Groups are fixed at the country's first sample year, split at the 2008 cross-sectional median. Specifications include country and year fixed effects, country-specific linear trends and income-group-by-year fixed effects, with standard errors clustered at the country level. No joint sup-t test is reported: each subgroup is estimated separately, which does not leave the full parameter vector the test requires.

Table OA.2.3: **Exposure factor: Internet exchange points – dependent variable: GDP per capita (const USD, ln), 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Event-study effects, countries with no Internet exchange point						
Effect_1	-0.0146245	0.0065787	-0.0275186	-0.0017305	535	35
Effect_2	-0.0288787	0.0157368	-0.0597222	0.0019649	452	24
Effect_3	-0.0424146	0.0269047	-0.0951468	0.0103176	361	18
Effect_4	-0.1003622	0.0410900	-0.1808971	-0.0198273	297	13
Effect_5	-0.1046398	0.0463961	-0.1955745	-0.0137051	276	13
Effect_6	-0.1048466	0.0509957	-0.2047962	-0.0048969	215	12
Placebo leads						
Placebo_1	-0.0004512	0.0064829	-0.0131575	0.0122551	535	35
Placebo_2	-0.0200765	0.0173824	-0.0541454	0.0139923	371	21
Placebo_3	-0.0264038	0.0335585	-0.0921774	0.0393697	267	14
Event-study effects, countries with at least one Internet exchange point						
Effect_1	-0.0016578	0.0040484	-0.0095925	0.0062769	188	14
Effect_2	-0.0087099	0.0058525	-0.0201806	0.0027608	181	13
Effect_3	-0.0139450	0.0085793	-0.0307602	0.0028701	132	11
Effect_4	-0.0153581	0.0111468	-0.0372054	0.0064893	129	11
Effect_5	-0.0166056	0.0151680	-0.0463344	0.0131232	120	10
Effect_6	-0.0181870	0.0157408	-0.0490384	0.0126644	119	10
Placebo leads						
Placebo_1	-0.0025830	0.0055085	-0.0133794	0.0082134	158	11
Placebo_2	-0.0029662	0.0098279	-0.0222286	0.0162961	118	7
Placebo_3	-0.0294027	0.0248499	-0.0781077	0.0193022	63	4

Notes: Dependent variable: log GDP per capita (constant 2015 USD). Treatment: the cable-disruption count; the repair-days panel of earlier versions is not reported, the subgroup estimations being run on the disruption count only. Six event-study effects and three placebo leads on the 2008–2019 sample excluding Vietnam; Effect_k is the effect at $t + (k - 1)$. Groups are fixed at the country's first sample year, split at the 2008 cross-sectional median. Specifications include country and year fixed effects, country-specific linear trends and income-group-by-year fixed effects, with standard errors clustered at the country level. No joint sup-t test is reported: each subgroup is estimated separately, which does not leave the full parameter vector the test requires.

Table OA.2.4: **Exposure factor: cable ownership diversity – dependent variable: GDP per capita (const USD, ln), 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Event-study effects, countries at or below the median number of distinct cable owners (2)						
Effect_1	-0.0124370	0.0071445	-0.0264399	0.0015659	393	30
Effect_2	-0.0274540	0.0150395	-0.0569310	0.0020229	330	23
Effect_3	-0.0389533	0.0225989	-0.0832463	0.0053398	283	19
Effect_4	-0.0693983	0.0270508	-0.1224170	-0.0163797	225	15
Effect_5	-0.0748912	0.0314964	-0.1366231	-0.0131594	204	14
Effect_6	-0.0694985	0.0301025	-0.1284984	-0.0104986	191	14
Placebo leads						
Placebo_1	-0.0044921	0.0080689	-0.0203069	0.0113227	393	30
Placebo_2	-0.0222347	0.0193206	-0.0601024	0.0156329	261	19
Placebo_3	-0.0416649	0.0440922	-0.1280839	0.0447542	197	13
Event-study effects, countries above the median number of distinct cable owners (2)						
Effect_1	-0.0041206	0.0057452	-0.0153810	0.0071398	314	19
Effect_2	0.0024382	0.0052210	-0.0077947	0.0126711	270	14
Effect_3	-0.0005733	0.0087995	-0.0178201	0.0166734	206	10
Effect_4	-0.0044812	0.0137148	-0.0313617	0.0223993	170	9
Effect_5	-0.0120852	0.0173335	-0.0460581	0.0218878	163	9
Effect_6	-0.0136696	0.0226741	-0.0581101	0.0307709	126	8
Placebo leads						
Placebo_1	0.0030575	0.0054199	-0.0075653	0.0136803	285	16
Placebo_2	0.0029320	0.0089032	-0.0145179	0.0203820	200	9
Placebo_3	-0.0098879	0.0147081	-0.0387153	0.0189394	136	5

Notes: Dependent variable: log GDP per capita (constant 2015 USD). Treatment: the cable-disruption count; the repair-days panel of earlier versions is not reported, the subgroup estimations being run on the disruption count only. Six event-study effects and three placebo leads on the 2008–2019 sample excluding Vietnam; Effect_k is the effect at $t + (k - 1)$. Groups are fixed at the country's first sample year, split at the 2008 cross-sectional median. Specifications include country and year fixed effects, country-specific linear trends and income-group-by-year fixed effects, with standard errors clustered at the country level. No joint sup-t test is reported: each subgroup is estimated separately, which does not leave the full parameter vector the test requires.

Table OA.2.5: **Exposure factor: Internet penetration – dependent variable: GDP per capita (const USD, ln), 2008–2019.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Event-study effects, countries at or below the median Internet penetration (22.6%)						
Effect_1	-0.0076897	0.0051962	-0.0178740	0.0024946	466	36
Effect_2	-0.0190521	0.0103899	-0.0394159	0.0013117	392	29
Effect_3	-0.0233098	0.0152455	-0.0531903	0.0065707	321	23
Effect_4	-0.0535250	0.0208594	-0.0944087	-0.0126413	260	19
Effect_5	-0.0602932	0.0249331	-0.1091611	-0.0114253	232	18
Effect_6	-0.0584893	0.0253961	-0.1082647	-0.0087138	185	17
Placebo leads						
Placebo_1	-0.0034998	0.0050922	-0.0134802	0.0064807	466	36
Placebo_2	-0.0105154	0.0118277	-0.0336972	0.0126664	316	23
Placebo_3	-0.0148199	0.0234294	-0.0607406	0.0311008	227	15
Event-study effects, countries above the median Internet penetration (22.6%)						
Effect_1	-0.0120625	0.0082757	-0.0282826	0.0041577	233	12
Effect_2	-0.0244423	0.0171991	-0.0581519	0.0092672	198	7
Effect_3	-0.0430447	0.0284458	-0.0987974	0.0127080	154	6
Effect_4	-0.0279499	0.0219481	-0.0709675	0.0150677	123	5
Effect_5	-0.0361116	0.0296654	-0.0942547	0.0220314	121	5
Effect_6	-0.0415488	0.0372557	-0.1145685	0.0314710	120	5
Placebo leads						
Placebo_1	0.0031103	0.0074350	-0.0114620	0.0176827	189	9
Placebo_2	-0.0425820	0.0346896	-0.1105724	0.0254084	154	4
Placebo_3	-0.1105229	0.0842413	-0.2756328	0.0545871	110	3

Notes: Dependent variable: log GDP per capita (constant 2015 USD). Treatment: the cable-disruption count; the repair-days panel of earlier versions is not reported, the subgroup estimations being run on the disruption count only. Six event-study effects and three placebo leads on the 2008–2019 sample excluding Vietnam; Effect_k is the effect at $t + (k - 1)$. Groups are fixed at the country's first sample year, split at the 2008 cross-sectional median. Specifications include country and year fixed effects, country-specific linear trends and income-group-by-year fixed effects, with standard errors clustered at the country level. No joint sup-t test is reported: each subgroup is estimated separately, which does not leave the full parameter vector the test requires.

Table OA.2.6: **Control variables and exposure factors: summary statistics, estimation sample.**

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Submarine-cable connectivity index (LSCI)	85.83	95.55	8.25	4.45	489.93	1,105
Internet users (% of population)	45.87	29.03	10.72	0.22	99.55	1,740
Real exchange rate index	135.12	116.93	83.05	3.39	2230.76	1,317
Terms of trade index	102.45	14.23	10.45	54.40	199.10	1,715
Control of corruption (WGI score)	48.54	19.99	3.27	9.68	95.06	1,780
Government effectiveness (WGI score)	52.69	18.00	3.33	9.76	95.54	1,779
Political stability (WGI score)	68.08	14.68	3.47	23.14	95.95	1,791
Rule of law (WGI score)	57.55	16.85	2.06	16.85	93.26	1,791
Regulatory quality (WGI score)	55.55	14.91	2.22	20.60	87.66	1,779
Voice and accountability (WGI score)	59.36	16.48	2.48	15.52	91.11	1,791
Number of landed cables	3.45	6.72	1.17	0	60	1,804
Connected partners (relations)	10.52	13.30	3.03	0	62	1,804
Number of cable owners	12.31	16.59	3.76	0	116	1,804
Internet exchange points (count)	2.58	7.75	2.59	0	120	1,804

Notes: Estimation sample (realized CDH sample), 2008–2019 excluding Vietnam. WGI = World Governance Indicators. The lower panel reports the exposure factors underlying the heterogeneity splits reported above and in Section 4.

Table OA.2.7: List of low-connectedness countries – criteria: connected partners $\leq$ 2 (2008 cross-sectional median), 96 countries.

AND	ARM	AUT	AZE	BDI	BFA
BGR	BIH	BLR	BOL	BTN	BWA
CAF	CHE	COG	COM	CUB	CZE
ERI	EST	FSM	GEO	GIN	GMB
GNB	GNQ	GRL	GUY	HUN	KAZ
KEN	KGZ	KHM	KIR	LAO	LBN
LBR	LIE	LSO	LTU	LUX	LVA
MAC	MCO	MDA	MDG	MDV	MHL
MKD	MLI	MLT	MNE	MNG	MOZ
MRT	MWI	NAM	NER	NPL	NRU
PLW	PNG	POL	PRK	PRY	PSE
ROU	RWA	SLB	SLE	SLV	SMR
SRB	SSD	STP	SUR	SVK	SVN
SWZ	SYC	TCD	TGO	TJK	TKM
TLS	TON	TUV	TZA	UGA	URY
UZB	VUT	WSM	XKX	ZMB	ZWE

Notes: ISO-3 codes, alphabetical. Median split of the number of connected partners in the 2008 cross-section, the same split used in Figure 6b and Table OA.2.2.

Table OA.2.8: List of no-IXP countries – criteria: no Internet exchange point, 112 countries.

ABW	ALB	AND	ARE	ARM	ATG
AZE	BDI	BEN	BFA	BGR	BHS
BIH	BLR	BLZ	BOL	BRB	BRN
BTN	CAF	CIV	CMR	COG	COM
CPV	CRI	CYM	CYP	DJI	DMA
DZA	ERI	FJI	FSM	GAB	GEO
GIN	GMB	GNB	GNQ	GRD	GRL
GTM	GUY	HKG	HND	HTI	IRN
JAM	JOR	KGZ	KIR	KNA	KWT
LAO	LBR	LCA	LIE	LSO	MAC
MAR	MCO	MDA	MDG	MDV	MEX
MHL	MKD	MLI	MMR	MNE	MRT
NAM	NER	NRU	OMN	PLW	PNG
PRI	PRK	PSE	QAT	SAU	SEN
SLB	SLE	SLV	SMR	SRB	SSD
STP	SUR	SWZ	SYC	TCD	TGO
TJK	TKM	TLS	TON	TTO	TUN
TUR	TUV	URY	VCT	VEN	VUT
WSM	XKX	ZMB	ZWE		

Notes: ISO-3 codes, alphabetical. Countries with no Internet exchange point, the same split used in Figure 6c and Table OA.2.3.

Table OA.2.9: List of low-ownership-diversity countries – criteria: cable owners $\leq$ 2 (2008 cross-sectional median), 94 countries.

AND	ARM	AUT	AZE	BDI	BFA
BGR	BIH	BLR	BOL	BTN	BWA
CAF	CAN	CHE	COG	COM	CUB
CZE	ERI	FSM	GIN	GMB	GNB
GNQ	GRL	GUY	HUN	ISL	ISR
JAM	JOR	KAZ	KEN	KGZ	KHM
KIR	LAO	LBR	LIE	LSO	LTU
LUX	MAC	MCO	MDA	MDG	MHL
MKD	MLI	MNE	MNG	MOZ	MRT
MWI	NAM	NER	NPL	NRU	PLW
PNG	PRK	PRY	PSE	ROU	RWA
SLB	SLE	SLV	SMR	SRB	SSD
STP	SUR	SVK	SVN	SWZ	SYC
TCD	TGO	TJK	TKM	TLS	TON
TUV	TZA	UGA	URY	UZB	VUT
WSM	XKX	ZMB	ZWE		

Notes: ISO-3 codes, alphabetical. Median split of the number of distinct cable owners in the 2008 cross-section, the same split used in Figure 6d.

OA.3 Outlier inclusion: 2020 and Vietnam

The baseline sample excludes the pandemic year 2020 and Vietnam. Vietnam records 26 cable disruptions and 467 repair days over 2013–2019, of which 11 events and 259 days fall in 2017 alone, making it the most intensely disrupted economy in the data by a wide margin. This appendix adds each back in turn, and then both together.

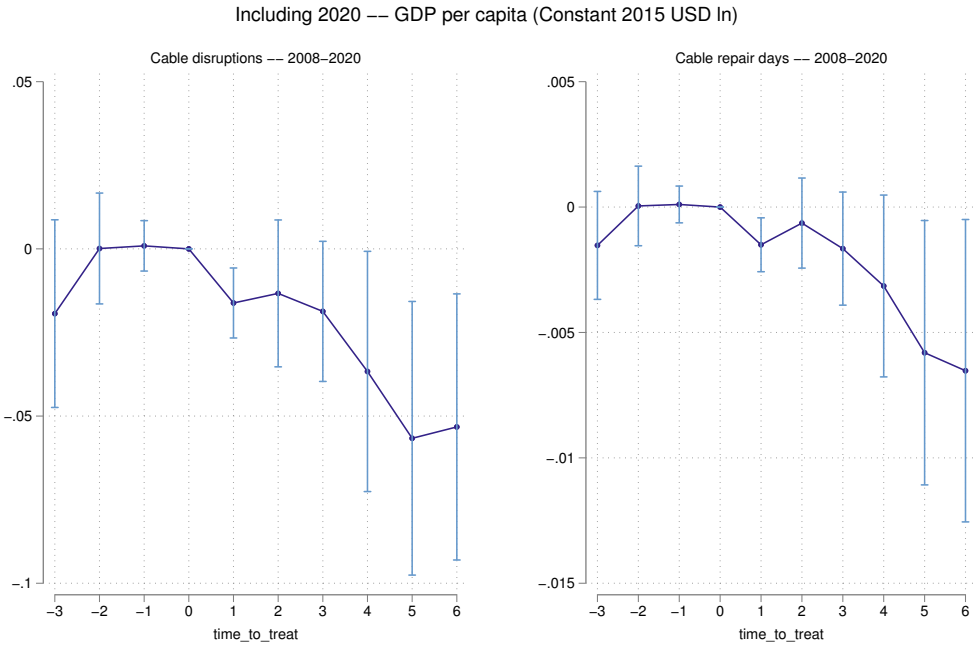


Fig. OA.3.1: Adding the 2020 pandemic year.

Notes: Baseline specification estimated on 2008–2020, Vietnam still excluded, so the panel isolates the effect of the pandemic year. Left: cable disruptions. Right: cable repair days. Specification as in Figure 5: country and year fixed effects, country-specific linear trends, income-group-by-year fixed effects, standard errors clustered at the country level, six event-study effects and three placebo leads.

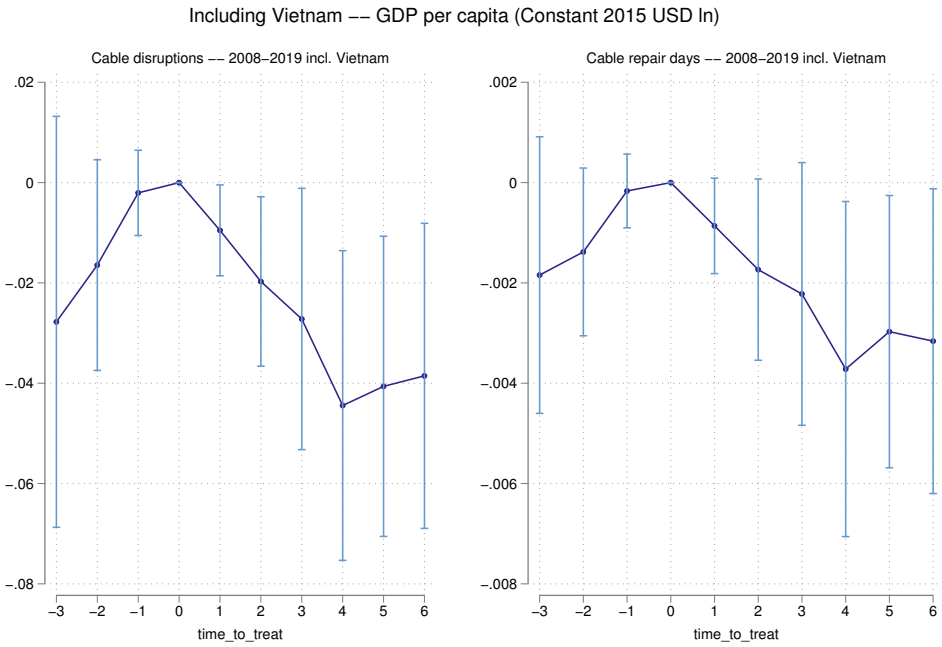


Fig. OA.3.2: Adding Vietnam.

Notes: Baseline specification estimated on 2008–2019 with Vietnam included, so the panel isolates that single economy. Specification otherwise as in Figure 5.

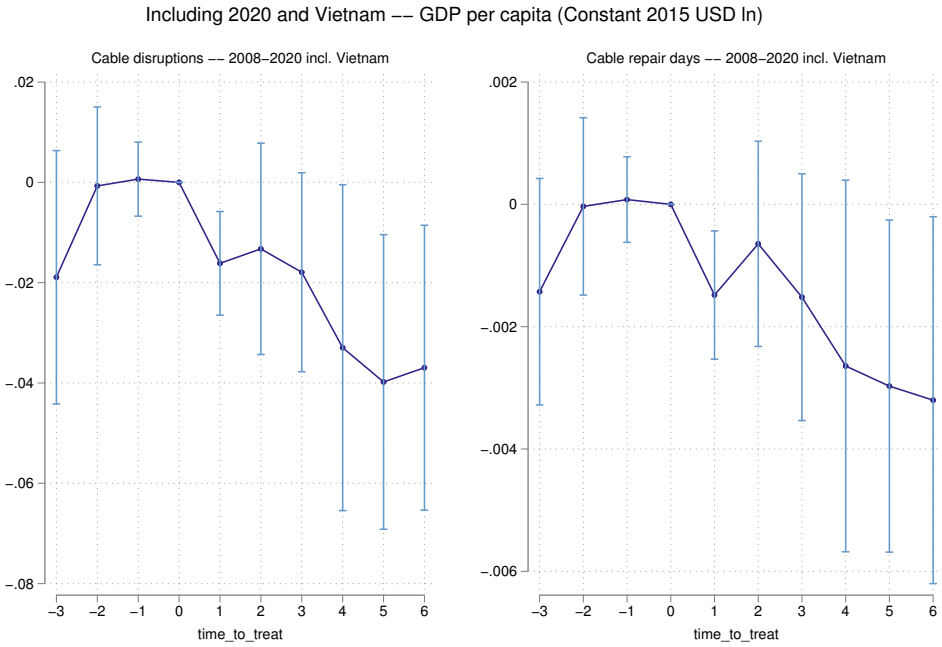


Fig. OA.3.3: Adding both 2020 and Vietnam.

Notes: Baseline specification estimated on the unrestricted 2008–2020 sample including Vietnam. Specification otherwise as in Figure 5.

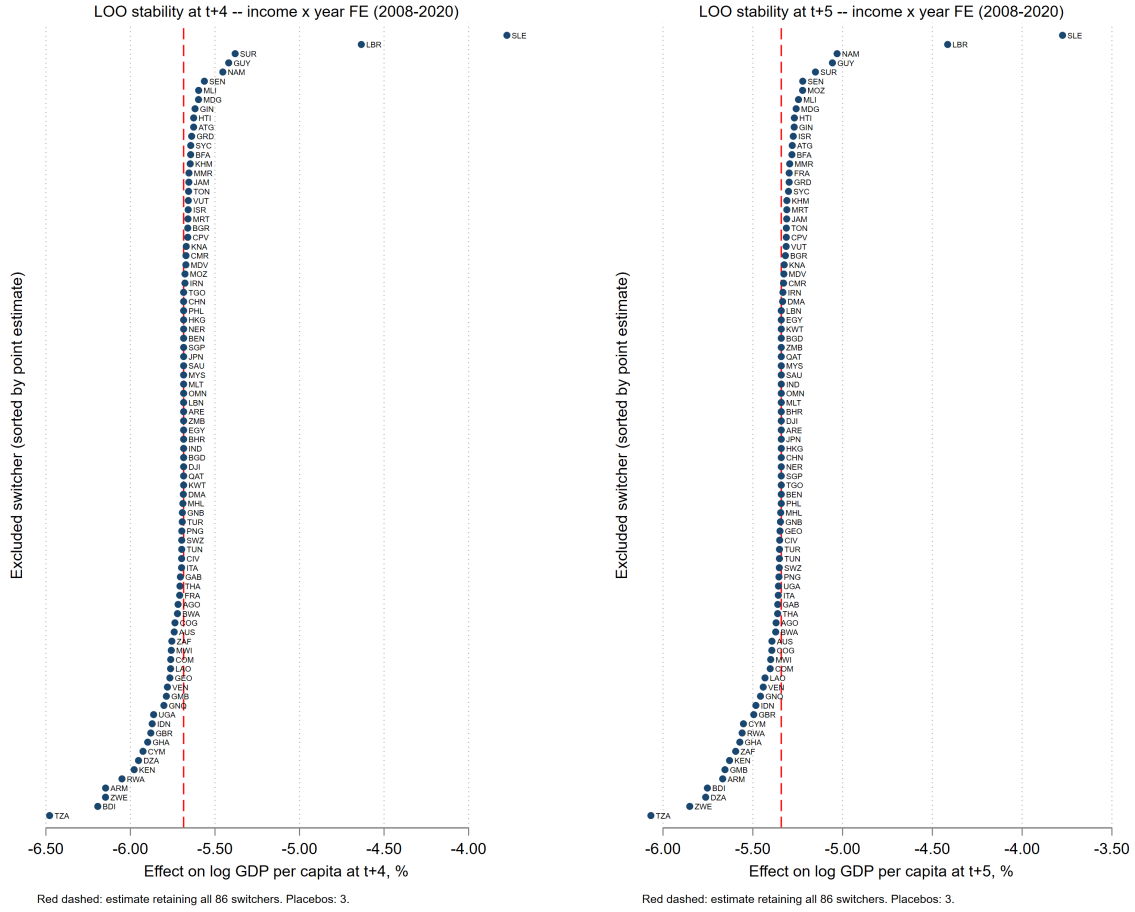


Fig. OA.3.4: Leave-one-out stability: log GDP per capita response, income-group × year FEs, including 2020 and Vietnam.

Notes: Each row drops one switcher and re-estimates the baseline on the unrestricted sample, which adds back both the 2020 pandemic year and Vietnam, the most heavily disrupted economy in the data. Specifications include country and year fixed effects, linear country trends, and income-group-by-year fixed effects. Left: effect at t+4. Right: effect at t+5. Red dashed line: full-sample point estimate; horizontal bars: 95% confidence intervals. Compare to Appendix Figure A.3.1.4, the excl-2020 baseline.